Lecture 1: Linear Algebra

Math for Machine Learning

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Introduction

Machine learning works with data, and data are almost always stored as lists of numbers: an image is a list of pixel intensities, a customer is a list of features such as age and income, and a word is represented by an embedding, a list of a few hundred numbers. Linear algebra is the mathematics of such lists, which we call vectors, and of the maps between them that are compatible with adding and scaling, the linear transformations. It is the language of the whole subject:

- a data set of N examples with n features each is an $N \times n$ matrix, and a linear model predicts a number from a feature vector $\mathbf{x}$ by $w_1 x_1 + \cdots + w_n x_n$; data with more indices, such as a batch of color images, are stored as tensors;

- a layer of a neural network multiplies its input vector by a weight matrix, so a network is built from matrix multiplications and simple nonlinear functions;
- fitting a linear model to data leads to a system of linear equations, and whether its solution is unique depends on the rank of the data matrix;
- eigenvalues and eigenvectors describe what happens when a linear map is applied again and again, as in Markov chains and the PageRank algorithm, and they are the basis of principal component analysis (Lecture 2).

In this lecture:

- we start from fields, the number systems in which we can add, subtract, multiply and divide, and define vector spaces over them, including spaces of functions such as the continuous and the smooth functions, with bases, dimension and subspaces;
- we study linear transformations, including differentiation and integration, and the matrices that represent them: matrix operations, row reduction for solving linear systems and computing inverses, special types of matrices, the trace and the determinant, which measures how a linear map scales areas and volumes, and block matrices, with the LU decomposition behind every linear solver and the Woodbury identity for low-rank updates;
- we relate the null space, the column space and the rank of a matrix by the rank–nullity theorem;
- we introduce dual spaces, which explain what the transpose of a matrix is, and quotient spaces with the isomorphism theorems;
- we find eigenvalues and eigenvectors, diagonalize matrices, prove the Cayley–Hamilton theorem, characterize diagonalizable matrices by their minimal polynomials, use powers of matrices to analyze Fibonacci numbers, Markov chains and power iteration, and show that the discrete Fourier transform diagonalizes convolutions, the linear maps behind convolutional networks;
- we construct the tensor product of vector spaces, define tensors as its elements, and derive their description by arrays (the tensors of deep learning libraries), multilinear forms, the trace as a contraction, tensor rank, and Einstein notation.

Lecture 2 adds geometry to vector spaces: lengths, angles and orthogonality. The later lectures use the language of this lecture throughout: derivatives are linear maps (Lecture 3), covariance matrices describe random vectors (Lecture 4), and the eigenvalues of a matrix decide whether the solutions of a linear differential equation grow or decay (Lecture 3).

Notation. Vectors of $\mathbb{R}^n$ are written in bold, $\mathbf{x} = (x_1, \ldots, x_n)$, with their entries in normal font; so are the zero vector $\mathbf{0}$ and the standard basis vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$. Elements of general vector spaces, such as polynomials and functions, are denoted by lowercase letters in normal font, such as u, v and f. Scalars are denoted by letters such as a, b and λ, and matrices by capital letters such as A. The same conventions are used in all lectures.

1 Number Sets and Fields

Remark 1.1. When we want to say that a number x is in a set S, we write $x \in S$. When we want to say that a number x is not in a set S, we write $x \notin S$.

Definition 1.2 (Number sets)**.** $\mathbb{Z}$ is the set of all integers. Meaning,

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

$\mathbb{Q}$ is the set of all rational numbers. Meaning,

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

$\mathbb{R}$ is the set of all real numbers. Meaning, $\mathbb{R}$ is obtained from $\mathbb{Q}$ by "filling in the gaps": $\mathbb{R} \supset \mathbb{Q}$, every Cauchy sequence in $\mathbb{Q}$ has a limit in $\mathbb{R}$ (completeness), and every real number is the limit of some Cauchy sequence in $\mathbb{Q}$. In other words, $\mathbb{R}$ is the completion of $\mathbb{Q}$.

$\mathbb{C}$ is the set of all complex numbers. Meaning,

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\},$$

where i is a number satisfying $i^2 = -1$.

Example 1.3 (Number sets)**.** 2 is a real number, and also an integer and a rational number.
$\sqrt{2}$ is a real number, but not a rational number.
π is a real number, but not a rational number.
e is a real number, but not a rational number.
$2 + 5i$ is a complex number, but not a real number.

Definition 1.4 (Field)**.** A field is a set $\mathbb{F}$ with two operations, an addition and a multiplication, which assign to all $a, b \in \mathbb{F}$ elements $a + b \in \mathbb{F}$ and $a \cdot b = ab \in \mathbb{F}$, and with two elements $0, 1 \in \mathbb{F}$, such that for all $a, b, c \in \mathbb{F}$:

1. (associativity) $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$;
2. (commutativity) $a + b = b + a$ and $ab = ba$;
3. (identities) $a + 0 = a$ and $a \cdot 1 = a$;
4. (inverses) there is an element $-a \in \mathbb{F}$ with $a + (-a) = 0$, and if $a \neq 0$, there is an element $a^{-1} \in \mathbb{F}$ with $aa^{-1} = 1$;
5. (distributivity) $a(b + c) = ab + ac$;
6. (nontriviality) $0 \neq 1$.

The last axiom says that a field has at least two elements. The set $\{0\}$, with $0 + 0 = 0$ and $0 \cdot 0 = 0$, satisfies axioms 1 to 5, with the element 0 as the identity for both operations (so $1 = 0$ there); the inverse axiom for nonzero elements holds because there are none. Axiom 6 excludes it: $\{0\}$ is not a field. In fact, $\{0\}$ is the only set that satisfies axioms 1 to 5 with $0 = 1$: the distributive law gives $a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$, so $a \cdot 0 = 0$ after adding $-(a \cdot 0)$, and if $0 = 1$, then $a = a \cdot 1 = a \cdot 0 = 0$ for every a.

Example 1.5. The real numbers $\mathbb{R}$, with the usual addition and multiplication, form a field: axioms 1, 2, 3 and 5 are the familiar rules of arithmetic, the inverses of a are $-a$ and, for $a \neq 0$, $\frac{1}{a}$, and $0 \neq 1$.

Example 1.6. The set of integers $\mathbb{Z}$ with the usual addition and multiplication operations does not form a field, since not every non-zero integer has a multiplicative inverse in $\mathbb{Z}$. For example, the integer 2 does not have a multiplicative inverse in $\mathbb{Z}$: there is no integer b with $2 \cdot b = 1$, because the only solution of this equation, $b = \frac{1}{2}$, is not an integer.

Exercise 1.7. Verify that the set of complex numbers $\mathbb{C}$ and the set of rational numbers $\mathbb{Q}$ with the usual addition and multiplication operations form fields.

2 Vector Spaces

Definition 2.1 (Vector space)**.** A vector space over a field $\mathbb{F}$ is a set V, whose elements are called vectors, with a vector addition, which assigns to $u, v \in V$ a vector $u + v \in V$, and a scalar multiplication, which assigns to $a \in \mathbb{F}$ and $v \in V$ a vector $a \cdot v = av \in V$, such that for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

1. $(u + v) + w = u + (v + w)$ and $u + v = v + u$;
2. there is a vector $0 \in V$, the zero vector (also written 0_V), with $v + 0 = v$;
3. there is a vector $-v \in V$ with $v + (-v) = 0$;
4. $a(bv) = (ab)v$ and $1v = v$;
5. $a(u + v) = au + av$ and $(a + b)v = av + bv$.

Axioms 1 to 3 are the rules for the addition of a field, now for vectors. Axioms 4 and 5 connect the two operations: in $(ab)v$, the scalars are multiplied in $\mathbb{F}$ before the vector is scaled, and in $(a + b)v = av + bv$, the left side adds scalars in $\mathbb{F}$, while the right side adds vectors in V.

Example 2.2. Every field $\mathbb{F}$ is a vector space over itself, with the field addition as vector addition and the field multiplication as scalar multiplication.

Proof. Since the operations are those of $\mathbb{F}$, every vector space axiom is a field axiom: axioms 1 to 3 are the field axioms for the addition, $a(bv) = (ab)v$ is the associativity of the multiplication, $1v = v$ is the identity axiom, and axiom 5 is the distributive law, which gives $(a+b)v = v(a+b) = va + vb = av + bv$ together with commutativity. □

Corollary 2.3. $\mathbb{Q}, \mathbb{R}$, *and* $\mathbb{C}$ *are vector spaces over themselves.*

Definition 2.4 (Product of vector spaces)**.** Let V and W be vector spaces over the same field $\mathbb{F}$. Their product is the set

$$V \times W = \{(v, w) : v \in V,\ w \in W\}$$

of all ordered pairs (v, w) with $v \in V$ and $w \in W$. The operations of vector addition and scalar multiplication are defined component-wise:

$$\begin{aligned}(v_1, w_1) + (v_2, w_2) &= (v_1 + v_2, w_1 + w_2),\\ a \cdot (v, w) &= (a \cdot v, a \cdot w), \qquad a \in \mathbb{F}.\end{aligned}$$

Proposition 2.5. *If V and W are vector spaces over the same field $\mathbb{F}$, then $V \times W$ is also a vector space over $\mathbb{F}$.*

Proof. The operations act on each component separately, so each axiom for $V \times W$ is the same axiom in V for the first components and in W for the second components. For example,

$$(a+b)(v,w) = ((a+b)v, (a+b)w) = (av+bv, aw+bw) = a(v,w) + b(v,w).$$

The zero vector is $(0_V, 0_W)$, and the additive inverse of (v,w) is $(-v,-w)$. □

Remark 2.6. For two (or finitely many) vector spaces, the product is also known as the (external) direct sum of the vector spaces, and it is often denoted by $V \oplus W$.

Definition 2.7 (The space $\mathbb{R}^n$)**.** Let n be a positive integer. The set $\mathbb{R}^n$ consists of all ordered n-tuples of real numbers,

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n\},$$

and the operations of vector addition and scalar multiplication are defined component-wise:

$$\begin{aligned}(x_1, \ldots, x_n) + (y_1, \ldots, y_n) &= (x_1 + y_1, \ldots, x_n + y_n),\\ a \cdot (x_1, \ldots, x_n) &= (ax_1, \ldots, ax_n), \qquad a \in \mathbb{R}.\end{aligned}$$

Corollary 2.8. *The set $\mathbb{R}^n$ is a vector space over the field $\mathbb{R}$ for every positive integer n.*

Proof. We use induction on n. For $n = 1$, $\mathbb{R}^1 = \mathbb{R}$ is a vector space over itself by Corollary 2.3.

Now let $n \geq 2$ and assume that $\mathbb{R}^{n-1}$ is a vector space over $\mathbb{R}$. Identify each n-tuple $(x_1, \ldots, x_n)$ with the pair $((x_1, \ldots, x_{n-1}), x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$. Under this identification, the component-wise operations on $\mathbb{R}^n$ are exactly the operations of the product $\mathbb{R}^{n-1} \times \mathbb{R}$. By Proposition 2.5, $\mathbb{R}^{n-1} \times \mathbb{R}$ is a vector space over $\mathbb{R}$, and hence so is $\mathbb{R}^n$. □

Vectors need not be lists of numbers. Some of the most important vector spaces consist of functions.

Definition 2.9 (The space of functions on a set)**.** Let S be a nonempty set. We write $\mathcal{F}(S)$ for the set of all functions $f : S \to \mathbb{R}$, with the pointwise operations

$$(f+g)(s) = f(s) + g(s), \qquad (a \cdot f)(s) = a\, f(s) \qquad (s \in S,\ a \in \mathbb{R}).$$

In words: to add two functions, we add their values at every point. Two functions $f, g \in \mathcal{F}(S)$ are equal if and only if $f(s) = g(s)$ for every $s \in S$.

Proposition 2.10. *For every nonempty set S, $\mathcal{F}(S)$ is a vector space over $\mathbb{R}$.*

Proof. Each axiom is an equation between functions, so it suffices to check it at every point $s \in S$, where it becomes the same rule of arithmetic in $\mathbb{R}$. For example,

$$((a+b)f)(s) = (a+b)f(s) = af(s) + bf(s) = (af + bf)(s).$$

The zero vector is the zero function, $0(s) = 0$ for all s, and the additive inverse of f is the function $-f$ with $(-f)(s) = -f(s)$. □

For $S = \{1, 2, \ldots, n\}$, a function $x : S \to \mathbb{R}$ is the same thing as the list of its values $(x(1), \ldots, x(n))$, and the pointwise operations are the componentwise operations of $\mathbb{R}^n$: so $\mathbb{R}^n$ is $\mathcal{F}(\{1, \ldots, n\})$ in disguise. In the same way, the real sequences $(a_0, a_1, a_2, \ldots)$ are the functions $\{0, 1, 2, \ldots\} \to \mathbb{R}$, so they form a vector space.

In calculus and in machine learning, we mostly work with functions that are continuous or differentiable. Let $I \subseteq \mathbb{R}$ be an open interval, such as $\mathbb{R}$ itself or $(0, 1)$. We write $f^{(k)}$ for the k-th derivative of a function f, with $f^{(0)} = f$.

Definition 2.11 (Continuous, differentiable and smooth functions)**.** We define the following sets of functions on the open interval I.

- $C(I)$, also written $C^0(I)$, is the set of all continuous functions $f : I \to \mathbb{R}$.
- For an integer $n \geq 1$, $C^n(I)$ is the set of all functions $f : I \to \mathbb{R}$ that are n times differentiable and whose n-th derivative $f^{(n)}$ is continuous. Such functions are called n times continuously differentiable.
- $C^\infty(I)$ is the set of all functions $f : I \to \mathbb{R}$ that have derivatives of all orders, i.e., $C^\infty(I) = C^1(I) \cap C^2(I) \cap \cdots$. Such functions are called smooth.

For a closed interval $[a, b]$, we write $C[a, b]$ for the set of continuous functions $[a, b] \to \mathbb{R}$; the proof of Proposition 2.13 below shows that it is a vector space as well.

A differentiable function is continuous, so a function in $C^n(I)$ also has the continuous derivatives $f, f', \ldots, f^{(n-1)}$ of all lower orders, and

$$C^\infty(I) \subseteq \cdots \subseteq C^3(I) \subseteq C^2(I) \subseteq C^1(I) \subseteq C(I) \subseteq \mathcal{F}(I).$$

Example 2.12.

1. Polynomials, e^x, $\sin x$ and $\cos x$ are smooth on $\mathbb{R}$: their derivatives are again functions of the same kind, so they can be differentiated again and again.
2. $f(x) = |x|$ is continuous on $\mathbb{R}$, but not differentiable at 0. So $f \in C(\mathbb{R})$, but $f \notin C^1(\mathbb{R})$.
3. $g(x) = x|x|$ has the derivative $g'(x) = 2|x|$ (check this separately for $x > 0$, for $x < 0$, and, with the definition of the derivative, for $x = 0$). So $g \in C^1(\mathbb{R})$, but $g \notin C^2(\mathbb{R})$. In the same way, $x^n|x| \in C^n(\mathbb{R})$, but $x^n|x| \notin C^{n+1}(\mathbb{R})$ (Exercise 36), so all the inclusions above are strict.
4. $h(x) = \frac{1}{x}$ is smooth on the interval $(0, \infty)$, but it is not even defined at 0, so it is not an element of $C(\mathbb{R})$. Whether a function belongs to one of these sets depends on the interval.

Proposition 2.13. *With the pointwise operations of $\mathcal{F}(I)$, the sets $C(I)$, $C^n(I)$ (for every $n \geq 1$) and $C^\infty(I)$ are vector spaces over $\mathbb{R}$.*

Proof. We first show that the operations do not lead outside these sets. If f and g are continuous and $a \in \mathbb{R}$, then $f + g$ and af are continuous, by the rules for limits. If f and g are differentiable, then so are $f + g$ and af, with

$$(f + g)' = f' + g' \qquad \text{and} \qquad (af)' = af',$$

by the sum rule and the constant factor rule of calculus. Applying these rules k times gives $(f + g)^{(k)} = f^{(k)} + g^{(k)}$ and $(af)^{(k)} = af^{(k)}$. So if $f, g \in C^n(I)$, then $f + g$ and af are n times differentiable, and their n-th derivatives $f^{(n)} + g^{(n)}$ and $af^{(n)}$ are continuous; hence $f + g, af \in C^n(I)$. Since this holds for every n, it also holds for $C^\infty(I)$.

The axioms of associativity, commutativity and distributivity, and $1 \cdot f = f$, hold because they hold in $\mathcal{F}(I)$ (Proposition 2.10), and the operations are the same. The zero function is smooth, so it lies in all of these sets. Finally, the additive inverse of a function f in one of these sets is $-f = (-1) \cdot f$, which lies in the same set, since the set is closed under scalar multiplication. $\square$

In Section 5, we will see that this argument is a general principle: $C(I)$, $C^n(I)$ and $C^\infty(I)$ are subspaces of $\mathcal{F}(I)$.

Remark 2.14. The set of all functions that are n times differentiable, without the condition that $f^{(n)}$ is continuous, is also a vector space, by the same argument. It is slightly larger than $C^n(I)$. For example, let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, and $f(0) = 0$. Then $f'(0) = \lim_{h \to 0} h \sin(1/h) = 0$, and $f'(x) = 2x \sin(1/x) - \cos(1/x)$ for $x \neq 0$, which has no limit as $x \to 0$. So f is differentiable on $\mathbb{R}$, but $f \notin C^1(\mathbb{R})$. The spaces $C^n(I)$ are used more often, because continuous derivatives are easier to work with.

Remark 2.15 (Function spaces in machine learning). A trained model is a function, for example from images to class probabilities, and learning means searching for a good function in a large set of candidates. Choosing the model fixes this set: linear models give the linear functions, polynomial regression (Programming Exercise P9) the polynomials of degree at most d, and neural networks with smooth activation functions such as tanh give smooth functions, while networks with the ReLU activation $\max(0, x)$ give continuous functions with kinks, like $|x|$. Kernel methods and Gaussian processes (Lecture 4) work with vector spaces of functions directly.

3 Bases and Dimension

Throughout this section, V is a vector space over a field $\mathbb{F}$.

Definition 3.1 (Linear combination and span)**.** Let $v_1, v_2, \ldots, v_k \in V$. A linear combination of $v_1, \ldots, v_k$ is a vector of the form

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k, \qquad a_1, \ldots, a_k \in \mathbb{F}.$$

The span of $v_1, \ldots, v_k$ is the set of all their linear combinations:

$$\operatorname{span}\{v_1, \ldots, v_k\} = \{a_1 v_1 + \cdots + a_k v_k : a_1, \ldots, a_k \in \mathbb{F}\}.$$

We say that $v_1, \ldots, v_k$ span V if $\operatorname{span}\{v_1, \ldots, v_k\} = V$, i.e., if every vector in V is a linear combination of $v_1, \ldots, v_k$.

Definition 3.2 (Linear independence)**.** The vectors $v_1, \ldots, v_k \in V$ are linearly independent if the only way to write the zero vector as a linear combination of them is with all coefficients equal to zero:

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k = 0 \implies a_1 = a_2 = \cdots = a_k = 0.$$

Otherwise, they are called linearly dependent.

For an infinite set $S \subseteq V$, we say that S is linearly independent if every finite subset of S is linearly independent, and that S spans V if every vector in V is a linear combination of finitely many vectors from S.

Example 3.3. In $\mathbb{R}^2$, the vectors $(1, 2)$ and $(2, 4)$ are linearly dependent, since

$$2 \cdot (1, 2) + (-1) \cdot (2, 4) = (0, 0).$$

Their span is the line $\operatorname{span}\{(1, 2), (2, 4)\} = \{(t, 2t) : t \in \mathbb{R}\}$, so they do not span $\mathbb{R}^2$. For example, $(1, 0)$ is not in their span.

Definition 3.4 (Basis)**.** A basis of a vector space V is a set of vectors in V that is linearly independent and spans V.

Proposition 3.5. *Let $\{v_1, v_2, \ldots, v_n\}$ be a basis of V. Then every vector in V can be expressed as a linear combination of the basis vectors in exactly one way.*

Proof. Since the basis spans V, every $v \in V$ can be written as $v = a_1v_1 + \cdots + a_nv_n$ for some $a_1, \ldots, a_n \in \mathbb{F}$. Suppose that also $v = b_1v_1 + \cdots + b_nv_n$. Subtracting the two expressions gives

$$(a_1 - b_1)v_1 + \cdots + (a_n - b_n)v_n = 0.$$

Since $v_1, \ldots, v_n$ are linearly independent, $a_i - b_i = 0$, i.e., $a_i = b_i$ for every i. So the two expressions are the same. □

The number of vectors in a basis is the same for all bases. This follows from the next lemma, which says that a spanning set always has at least as many vectors as a linearly independent set.

Lemma 3.6 (Exchange lemma)**.** *If $v_1, \ldots, v_m$ span V, and $u_1, \ldots, u_k \in V$ are linearly independent, then $k \leq m$.*

Proof. We replace the vectors v_i by the vectors u_j, one at a time, keeping a spanning list of m vectors. We show by induction on $j = 0, 1, \ldots, k$ that $j \leq m$ and that, after renumbering the v_i, the vectors

$$u_1, \ldots, u_j, v_{j+1}, \ldots, v_m$$

span V. For $j = 0$, this is the assumption. Suppose that it holds for some $j < k$. Then

$$u_{j+1} = a_1u_1 + \cdots + a_ju_j + b_{j+1}v_{j+1} + \cdots + b_mv_m$$

for some scalars a_i and b_i. If all b_i were 0 (in particular, if $j = m$, so that there are no b_i), then u_{j+1} would be a linear combination of $u_1, \ldots, u_j$, which is impossible since $u_1, \ldots, u_{j+1}$ are linearly independent. So $j + 1 \leq m$, and after renumbering, $b_{j+1} \neq 0$. Solving for v_{j+1},

$$v_{j+1} = \frac{1}{b_{j+1}}\Big(u_{j+1} - a_1u_1 - \cdots - a_ju_j - b_{j+2}v_{j+2} - \cdots - b_mv_m\Big)$$

is a linear combination of $u_1, \ldots, u_{j+1}, v_{j+2}, \ldots, v_m$. Every vector of V is a linear combination of $u_1, \ldots, u_j, v_{j+1}, \ldots, v_m$; substituting this expression for v_{j+1} writes it as a linear combination of $u_1, \ldots, u_{j+1}, v_{j+2}, \ldots, v_m$. So these vectors span V, which completes the induction. For $j = k$, we get $k \leq m$. □

Theorem 3.7. *If a vector space V has a basis consisting of n vectors, then every basis of V consists of exactly n vectors.*

Proof. Let $\{v_1, \ldots, v_n\}$ be a basis, and let S be another basis of V. Any k distinct vectors of S are linearly independent, and $v_1, \ldots, v_n$ span V, so $k \leq n$ by Lemma 3.6. Hence S is finite, with $m \leq n$ vectors. Exchanging the roles of the two bases (S spans V, and $v_1, \ldots, v_n$ are linearly independent) gives $n \leq m$. □

This allows us to make the following definition.

Definition 3.8 (Dimension)**.** A vector space V is said to be finite-dimensional if it has a finite basis. In this case, the dimension of V, denoted by $\dim V$, is the number of vectors in any basis of V; by Theorem 3.7, this number does not depend on which basis we choose. If a vector space does not have a finite basis, it is called infinite-dimensional.

By convention, the vector space $\{0\}$ consisting only of the zero vector has dimension 0 (its basis is the empty set).

Definition 3.9 (Standard basis)**.** The standard basis for the vector space $\mathbb{R}^n$ is the set of vectors $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$, where each $\mathbf{e}_i$ is a vector in $\mathbb{R}^n$ with a 1 in the i-th position and 0s in all other positions. Formally, the standard basis vectors are defined as:

$$\mathbf{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0) \qquad \text{(with 1 in the } i\text{-th position)}.$$

Proposition 3.10. *The standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$. In particular, $\mathbb{R}^n$ is finite-dimensional and $\dim \mathbb{R}^n = n$.*

Proof. Every vector $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ can be written as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n,$$

so the standard basis spans $\mathbb{R}^n$. If $a_1\mathbf{e}_1 + \cdots + a_n\mathbf{e}_n = \mathbf{0}$, then $(a_1, \ldots, a_n) = (0, \ldots, 0)$, so $a_1 = \cdots = a_n = 0$. Hence the standard basis is also linearly independent. □

Example 3.11. Consider the vector space $\mathbb{R}^2$. Its standard basis is the set of vectors $\{(1, 0), (0, 1)\}$. These vectors are linearly independent and span the entire space $\mathbb{R}^2$, meaning that any vector $(x, y) \in \mathbb{R}^2$ can be expressed as a linear combination of these basis vectors:

$$(x, y) = x \cdot (1, 0) + y \cdot (0, 1).$$

The dimension of $\mathbb{R}^2$ is 2, as there are two vectors in the basis.

Example 3.12. Consider the vector space $\mathbb{R}^3$. Its standard basis is the set of vectors

$$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

These vectors are linearly independent and span the entire space $\mathbb{R}^3$, meaning that any vector $(x, y, z) \in \mathbb{R}^3$ can be expressed as a linear combination of these basis vectors:

$$(x, y, z) = x \cdot (1, 0, 0) + y \cdot (0, 1, 0) + z \cdot (0, 0, 1).$$

The dimension of $\mathbb{R}^3$ is 3, as there are three vectors in the basis.

Example 3.13. The set of all polynomials with real coefficients,

$$\mathbb{R}[x] = \{a_0 + a_1x + \cdots + a_dx^d : d \geq 0,\ a_0, \ldots, a_d \in \mathbb{R}\},$$

is a vector space over $\mathbb{R}$ with the usual addition of polynomials and multiplication of a polynomial by a real number. This vector space is infinite-dimensional.

You may check for yourself that the set of monomials $\{1, x, x^2, x^3, \ldots\}$ is linearly independent and spans the space of all polynomials, so it is an (infinite) basis of $\mathbb{R}[x]$.

To see that there is no finite basis, take any finitely many polynomials $p_1, \ldots, p_k \in \mathbb{R}[x]$, and choose an integer $d \geq 0$ such that each p_i has degree at most d. Every linear combination of $p_1, \ldots, p_k$ also has degree at most d, so $x^{d+1} \notin \operatorname{span}\{p_1, \ldots, p_k\}$. Hence no finite set of polynomials spans $\mathbb{R}[x]$, and in particular $\mathbb{R}[x]$ has no finite basis.

The exchange lemma has several consequences that we will use all the time.

Corollary 3.14. *Let V be a vector space.*

1. *If $v_1, \dots, v_m$ span V, then some of these vectors form a basis of V. In particular, V is finite-dimensional, and $\dim V \leq m$.*
2. *If $\dim V = n$, then every set of linearly independent vectors in V has at most n elements, and every set of vectors that spans V has at least n elements.*
3. *If $\dim V = n$, then every set of linearly independent vectors $u_1, \dots, u_k$ in V can be extended to a basis $u_1, \dots, u_k, u_{k+1}, \dots, u_n$ of V.*
4. *If $\dim V = n$, then any n linearly independent vectors in V form a basis of V, and so do any n vectors that span V.*
5. *If $\dim V = n$, then every subspace U of V is finite-dimensional, with $\dim U \leq n$, and $\dim U = n$ only if $U = V$.*

Proof. (1) If $v_1, \dots, v_m$ are linearly dependent, then $a_1 v_1 + \cdots + a_m v_m = 0$ with some $a_i \neq 0$, so $v_i = -\frac{1}{a_i} \sum_{l \neq i} a_l v_l$ is a linear combination of the other vectors. Removing v_i leaves a list that still spans V: in every linear combination of $v_1, \dots, v_m$, we can replace v_i by this expression. We repeat this until the remaining vectors are linearly independent, which happens after at most m steps (the empty list, which spans $\{0\}$, is linearly independent). The remaining vectors span V and are linearly independent, so they form a basis.

(2) A basis of V consists of n linearly independent vectors that span V, so both statements follow from Lemma 3.6.

(3) If $u_1, \dots, u_k$ do not span V, choose $w \in V$ with $w \notin \operatorname{span}\{u_1, \dots, u_k\}$. Then $u_1, \dots, u_k, w$ are linearly independent: if $a_1 u_1 + \cdots + a_k u_k + bw = 0$, then $b = 0$ (otherwise $w = -\frac{1}{b}(a_1 u_1 + \cdots + a_k u_k)$ would lie in the span), and then all $a_i = 0$, since the u_i are linearly independent. We repeat this with the longer list. By (2), a linearly independent list in V has at most n vectors, so the process stops, and it stops with a linearly independent list that spans V, i.e., a basis. By Theorem 3.7, this basis has n vectors.

(4) If n linearly independent vectors did not span V, then the argument of (3) would give $n+1$ linearly independent vectors, contradicting (2). If n vectors span V, then by (1) some of them form a basis, which has n vectors, so it consists of all of them.

(5) Linearly independent vectors in U are linearly independent in V, so by (2), there are at most n of them. Choose linearly independent vectors $u_1, \dots, u_k \in U$ with k as large as possible. They span U: otherwise, the argument of (3) would give a vector $w \in U$ such that $u_1, \dots, u_k, w$ are linearly independent. So they form a basis of U, and $\dim U = k \leq n$. If $k = n$, then $u_1, \dots, u_n$ form a basis of V by (4), so $U = V$. □

4 Linear Transformations

Definition 4.1 (Linear transformation)**.** A linear transformation between two vector spaces V and W over the same field $\mathbb{F}$ is a function $T : V \to W$ that satisfies the following two properties for all $u, v \in V$ and all scalars $a \in \mathbb{F}$:

1. Additivity: $T(u + v) = T(u) + T(v)$
2. Homogeneity: $T(a \cdot v) = a \cdot T(v)$

Lemma 4.2. *If $T : V \to W$ is a linear transformation, then $T(0_V) = 0_W$.*

Proof. By additivity, $T(0_V) = T(0_V + 0_V) = T(0_V) + T(0_V)$. Adding $-T(0_V)$ to both sides gives $0_W = T(0_V)$. □

Example 4.3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (2x + 3y, 4x - y)$. We will verify that T is a linear transformation.

First, we check additivity:

$$\begin{aligned} T((x_1, y_1) + (x_2, y_2)) &= T(x_1 + x_2, y_1 + y_2) \\ &= (2(x_1 + x_2) + 3(y_1 + y_2), 4(x_1 + x_2) - (y_1 + y_2)) \\ &= (2x_1 + 3y_1 + 2x_2 + 3y_2, 4x_1 - y_1 + 4x_2 - y_2) \\ &= (2x_1 + 3y_1, 4x_1 - y_1) + (2x_2 + 3y_2, 4x_2 - y_2) \\ &= T(x_1, y_1) + T(x_2, y_2). \end{aligned}$$

Next, we check homogeneity:

$$\begin{aligned} T(a \cdot (x, y)) &= T(ax, ay) \\ &= (2(ax) + 3(ay), 4(ax) - (ay)) \\ &= a(2x + 3y, 4x - y) \\ &= a \cdot T(x, y). \end{aligned}$$

Since both properties hold, T is indeed a linear transformation.

Remark 4.4 (Visualizing linear transformations of the plane). Every vector of $\mathbb{R}^2$ can be written as $(x, y) = x\mathbf{e}_1 + y\mathbf{e}_2$, so for a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$,

$$T(x, y) = x\,T(\mathbf{e}_1) + y\,T(\mathbf{e}_2).$$

Hence T is completely determined by the two vectors $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$. In particular, T maps the unit square $\{x\mathbf{e}_1 + y\mathbf{e}_2 : 0 \le x, y \le 1\}$ to the parallelogram $\{xT(\mathbf{e}_1) + yT(\mathbf{e}_2) : 0 \le x, y \le 1\}$ spanned by $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$. The origin stays where it is (Lemma 4.2), and a grid of equally spaced parallel lines is mapped to a grid of equally spaced parallel lines (or, if $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ lie on a common line, squashed onto that line). Figure 1 shows this for the linear transformation of the previous example, for which $T(\mathbf{e}_1) = (2, 4)$ and $T(\mathbf{e}_2) = (3, -1)$. Figure 2 shows four basic examples; you may check that each of them is linear.

Linear transformations are not only maps between spaces of lists of numbers: the two basic operations of calculus are linear. Let I be an open interval.

Example 4.5 (Differentiation)**.** The map

$$D : C^1(I) \to C(I), \qquad D(f) = f',$$

is well defined, since f' is continuous for $f \in C^1(I)$, and it is a linear transformation: by the sum rule and the constant factor rule,

$$D(f + g) = (f + g)' = f' + g' = D(f) + D(g) \qquad \text{and} \qquad D(af) = (af)' = af' = aD(f)$$

for all $f, g \in C^1(I)$ and $a \in \mathbb{R}$. For example, $D(3x^2 - \sin x) = 6x - \cos x = 3D(x^2) - D(\sin x)$. In the same way, differentiation gives linear transformations $C^{n+1}(I) \to C^n(I)$ and $C^\infty(I) \to C^\infty(I)$, and also $\mathbb{R}[x] \to \mathbb{R}[x]$, since the derivative of a polynomial is a polynomial.

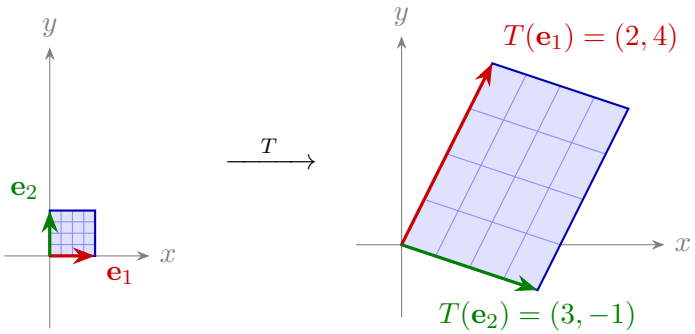


Figure 1: The linear transformation $T(x, y) = (2x + 3y, 4x - y)$ maps the unit square (left) to the parallelogram spanned by $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ (right). The grid lines inside the square are mapped to equally spaced parallel lines inside the parallelogram.

Example 4.6 (Integration)**.** 1. Let $a < b$. The definite integral

$$\varphi : C[a, b] \to \mathbb{R}, \qquad \varphi(f) = \int_a^b f(x)\, dx,$$

is a linear transformation, by the linearity of the integral: $\int_a^b (f + g)(x)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$ and $\int_a^b af(x)\, dx = a \int_a^b f(x)\, dx$. Its values are numbers, so it is a linear functional in the sense of Section 8. For example, the average value $\frac{1}{b-a} \int_a^b f(x)\, dx$ of f on $[a, b]$ depends linearly on f.

2. Let $c \in I$. The integral with a variable upper limit,

$$J : C(I) \to C^1(I), \qquad (Jf)(x) = \int_c^x f(t)\, dt \qquad (x \in I),$$

with the usual convention $\int_c^x = -\int_x^c$ for $x < c$, is linear, again by the linearity of the integral, applied for each fixed x. By the fundamental theorem of calculus, Jf is differentiable with $(Jf)' = f$, which is continuous, so Jf indeed lies in $C^1(I)$: it is the antiderivative of f that vanishes at c.

Differentiation and integration undo each other, but only in one order:

$$D(Jf) = f \quad \text{for all } f \in C(I), \qquad \text{but} \qquad J(Df)(x) = \int_c^x f'(t)\, dt = f(x) - f(c) \quad \text{for } f \in C^1(I).$$

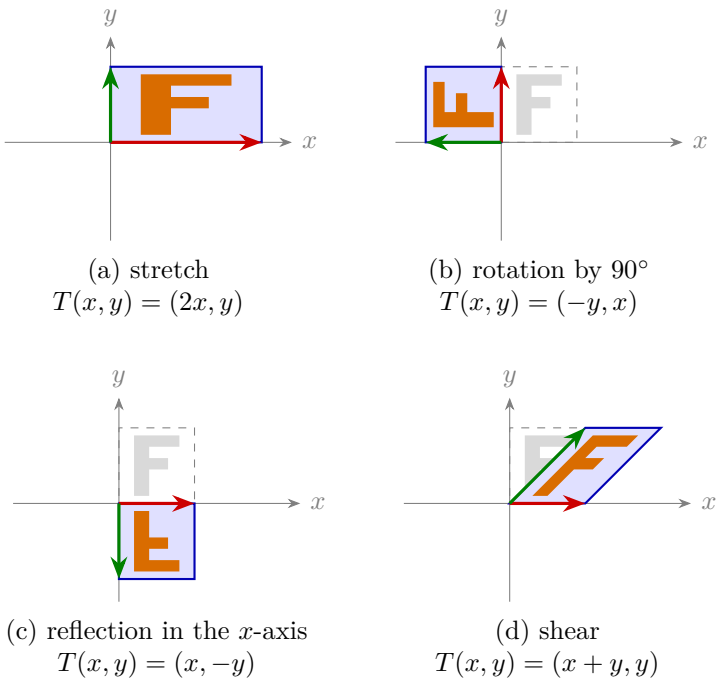
So $D \circ J$ is the identity map of $C(I)$, while $J \circ D$ forgets the value $f(c)$.

Definition 4.7 (Kernel)**.** The kernel of a linear transformation $T : V \to W$ is the set of all vectors in V that are mapped to the zero vector in W. Formally, the kernel is defined as

$$\ker(T) = \{v \in V : T(v) = 0_W\},$$

where 0_W is the zero vector in W.

Example 4.8. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (2x + 3y, 4x - y)$. We will find the kernel of T.



(a) stretch
$T(x, y) = (2x, y)$

(b) rotation by 90°
$T(x, y) = (-y, x)$

(c) reflection in the x-axis
$T(x, y) = (x, -y)$

(d) shear
$T(x, y) = (x + y, y)$

Figure 2: Four linear transformations of the plane. The dashed square and the gray letter F show the unit square before the transformation; the blue parallelogram and the orange letter F show its image. The red and green arrows are $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$. Note that in (c), the image of the letter F is mirrored.

To find the kernel, we need to solve the equation $T(x, y) = (0, 0)$:

$$\begin{aligned} 2x + 3y &= 0 \\ 4x - y &= 0 \end{aligned}$$

From the second equation, we can express y in terms of x: $y = 4x$. Substituting this into the first equation gives:

$$\begin{aligned} 2x + 3(4x) &= 0 \\ 2x + 12x &= 0 \\ 14x &= 0 \\ x &= 0 \end{aligned}$$

Substituting $x = 0$ back into $y = 4x$, we find that $y = 0$. Therefore, the only solution is $(0, 0)$. Thus, the kernel of T is

$$\ker(T) = \{(0, 0)\}.$$

Example 4.9. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$. We will find the kernel of T.

To find the kernel, we need to solve the equation $T(x, y, z) = (0, 0)$:

$$\begin{aligned} x + 2y - z &= 0 \\ 3x - y + 4z &= 0 \end{aligned}$$

From the first equation, we can express z in terms of x and y: $z = x + 2y$. Substituting this into the second equation gives:

$$\begin{aligned} 3x - y + 4(x + 2y) &= 0 \\ 3x - y + 4x + 8y &= 0 \\ 7x + 7y &= 0 \\ x + y &= 0 \end{aligned}$$

From this equation, we can express y in terms of x: $y = -x$. Substituting this back into the expression for z, we get:

$$z = x + 2(-x) = x - 2x = -x.$$

Therefore, any vector in the kernel can be expressed as $(x, -x, -x)$ for some scalar x. Thus, the kernel of T is

$$\ker(T) = \{(x, -x, -x) : x \in \mathbb{R}\}.$$

Definition 4.10 (Image)**.** The image (or range) of a linear transformation $T : V \to W$ is the set of all vectors in W that can be expressed as $T(v)$ for some vector $v \in V$. Formally, the image is defined as

$$\text{im}(T) = \{w \in W : w = T(v) \text{ for some } v \in V\}.$$

Example 4.11. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (2x + 3y, 4x - y)$. We will find the image of T.

To find the image, we need to determine all possible outputs of T. We will show that every vector of $\mathbb{R}^2$ is an output. Let $(u, v) \in \mathbb{R}^2$ be arbitrary. We look for $x, y \in \mathbb{R}$ such that $T(x, y) = (u, v)$, i.e.,

$$\begin{aligned} u &= 2x + 3y \\ v &= 4x - y \end{aligned}$$

We can solve this system of equations for x and y in terms of u and v. From the second equation, we can express y in terms of x and v: $y = 4x - v$. Substituting this into the first equation gives:

$$\begin{aligned} u &= 2x + 3(4x - v) \\ u &= 2x + 12x - 3v \\ u &= 14x - 3v \end{aligned}$$

Solving for x, we get:

$$14x = u + 3v \implies x = \frac{u + 3v}{14}.$$

Substituting this back into the expression for y, we get:

$$y = 4\left(\frac{u + 3v}{14}\right) - v = \frac{4u + 12v - 14v}{14} = \frac{2u - v}{7}.$$

Let us check that these values work:

$$\begin{aligned} 2x + 3y &= \frac{2u + 6v}{14} + \frac{12u - 6v}{14} = u, \\ 4x - y &= \frac{4u + 12v}{14} - \frac{4u - 2v}{14} = v. \end{aligned}$$

Therefore, for any $(u, v) \in \mathbb{R}^2$, we have found $(x, y) \in \mathbb{R}^2$ such that $T(x, y) = (u, v)$. This means that the image of T is all of $\mathbb{R}^2$, i.e.,

$$\text{im}(T) = \mathbb{R}^2.$$

Example 4.12. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$. We will find the image of T.

As before, we will show that every vector of $\mathbb{R}^2$ is an output of T. Let $(u, v) \in \mathbb{R}^2$ be arbitrary. We look for $x, y, z \in \mathbb{R}$ such that $T(x, y, z) = (u, v)$, i.e.,

$$\begin{aligned} u &= x + 2y - z \\ v &= 3x - y + 4z \end{aligned}$$

From the first equation, we can express z in terms of x, y, and u: $z = x + 2y - u$. Substituting this into the second equation gives:

$$\begin{aligned} v &= 3x - y + 4(x + 2y - u) \\ v &= 3x - y + 4x + 8y - 4u \\ v &= 7x + 7y - 4u \end{aligned}$$

Rearranging gives:

$$7x + 7y = 4u + v.$$

For any fixed (u, v), this equation describes a line in the (x, y)-plane, and every point (x, y) on this line, together with $z = x + 2y - u$, is a solution. For example, we can take $y = 0$ and $x = \frac{4u+v}{7}$, and then $z = x - u = \frac{v-3u}{7}$. Let us check:

$$T\left(\frac{4u + v}{7}, 0, \frac{v - 3u}{7}\right) = \left(\frac{(4u + v) - (v - 3u)}{7}, \frac{3(4u + v) + 4(v - 3u)}{7}\right) = (u, v).$$

Therefore, for any $(u, v) \in \mathbb{R}^2$, we have found $(x, y, z) \in \mathbb{R}^3$ such that $T(x, y, z) = (u, v)$. This means that the image of T is all of $\mathbb{R}^2$, i.e.,

$$\operatorname{im}(T) = \mathbb{R}^2.$$

Example 4.13. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x + 2y, 2x + 4y)$. This map is linear, and it can be written as

$$T(x, y) = (x + 2y) \cdot (1, 2).$$

So $T(x, y) = (0, 0)$ if and only if $x + 2y = 0$, and the kernel of T is the line

$$\ker(T) = \{(-2t, t) : t \in \mathbb{R}\} = \operatorname{span}\{(-2, 1)\}.$$

Every output of T is a multiple of $(1, 2)$, and every multiple $s(1, 2) = T(s, 0)$ is an output, so the image of T is the line

$$\operatorname{im}(T) = \{(s, 2s) : s \in \mathbb{R}\} = \operatorname{span}\{(1, 2)\}.$$

More generally, all points of the line $x + 2y = c$, which is parallel to $\ker(T)$, are mapped to the same point $c\,(1, 2)$. So T squashes the whole plane onto the line $\operatorname{im}(T)$, as shown in Figure 3. In particular, T is neither injective nor surjective.

Example 4.14. For differentiation $D : C^1(I) \to C(I)$ (Example 4.5), the kernel consists of the functions with $f' = 0$ on I, which are exactly the constant functions (by the mean value theorem of calculus). Its image is all of $C(I)$: every continuous function g is the derivative of the function Jg of Example 4.6, since $D(Jg) = g$.

For integration $J : C(I) \to C^1(I)$, the kernel is $\{0\}$: if $Jf = 0$, then $f = D(Jf) = D(0) = 0$. Its image consists of the functions $F \in C^1(I)$ with $F(c) = 0$: every Jf vanishes at c, and every $F \in C^1(I)$ with $F(c) = 0$ satisfies $F = F - F(c) = J(DF)$.

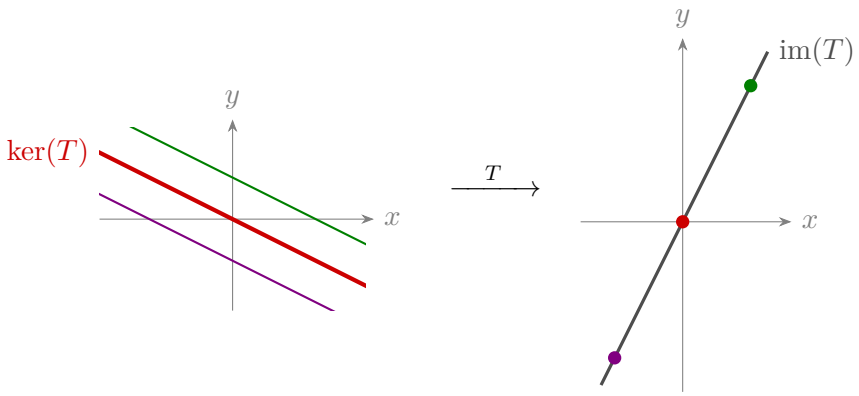


Figure 3: The linear transformation $T(x, y) = (x+2y, 2x+4y)$ of Example 4.13. Each line $x+2y = c$ on the left is mapped to the single point $c\,(1, 2)$ on the right, shown in the same color (here $c = -2$, 0, 2). The kernel (red, $c = 0$) is mapped to the origin, and the whole plane is squashed onto the line $\mathrm{im}(T)$.

4.1 Isomorphisms

Two vector spaces can look very different and still "behave in the same way". For example, a polynomial $a_0 + a_1x + a_2x^2$ is added and multiplied by scalars exactly like the list of its coefficients $(a_0, a_1, a_2) \in \mathbb{R}^3$. Isomorphisms make this idea precise.

Definition 4.15 (Injective, surjective, bijective)**.** A function $f : X \to Y$ between two sets is called

- injective if different inputs give different outputs, i.e., $f(x_1) = f(x_2)$ implies $x_1 = x_2$;
- surjective if every element of Y is an output, i.e., for every $y \in Y$ there exists $x \in X$ with $f(x) = y$;
- bijective if it is both injective and surjective.

A function f is bijective if and only if it has an inverse function $f^{-1} : Y \to X$, i.e., a function with $f^{-1}(f(x)) = x$ for all $x \in X$ and $f(f^{-1}(y)) = y$ for all $y \in Y$. Note that a linear transformation $T : V \to W$ is surjective if and only if $\mathrm{im}(T) = W$.

Definition 4.16 (Isomorphism)**.** Let V and W be vector spaces over the same field $\mathbb{F}$. An isomorphism from V to W is a bijective linear transformation $T : V \to W$. If there exists an isomorphism from V to W, we say that V and W are isomorphic, and we write $V \cong W$.

Remark 4.17. An isomorphism $T : V \to W$ matches every vector of V with exactly one vector of W, in a way that respects addition and scalar multiplication. So any statement about V that only involves addition and scalar multiplication can be translated into a statement about W, and vice versa. In this sense, isomorphic vector spaces are "the same vector space, written in a different way".

To check that a linear transformation is injective, it is enough to look at its kernel.

Lemma 4.18. *A linear transformation $T : V \to W$ is injective if and only if* $\ker(T) = \{0_V\}$. *Hence T is an isomorphism if and only if* $\ker(T) = \{0_V\}$ *and* $\mathrm{im}(T) = W$.

Proof. Suppose that T is injective, and let $v \in \ker(T)$. Then $T(v) = 0_W = T(0_V)$ by Lemma 4.2, so $v = 0_V$.

Conversely, suppose that $\ker(T) = \{0_V\}$, and let $u, v \in V$ with $T(u) = T(v)$. Using linearity and $(-1) \cdot v = -v$ (Exercise 1), we get

$$T(u - v) = T(u + (-1) \cdot v) = T(u) + (-1) \cdot T(v) = T(u) - T(v) = 0_W,$$

so $u - v \in \ker(T) = \{0_V\}$, i.e., $u = v$. □

Example 4.19. The linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (2x + 3y, 4x - y)$, is an isomorphism. Indeed, we found in Example 4.8 that $\ker(T) = \{(0, 0)\}$, so T is injective by Lemma 4.18, and we found in Example 4.11 that $\mathrm{im}(T) = \mathbb{R}^2$, so T is surjective. Its inverse is the function we computed in Example 4.11:

$$T^{-1}(u, v) = \left(\frac{u + 3v}{14}, \frac{2u - v}{7}\right).$$

Example 4.20. The linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$, is surjective by Example 4.12. However, it is not injective, since its kernel is the line $\{(x, -x, -x) : x \in \mathbb{R}\} \neq \{0\}$ (Example 4.9). So T is not an isomorphism.

Similarly, the linear transformation $S : \mathbb{R} \to \mathbb{R}^2$, $S(x) = (x, 0)$, is injective, since $\ker(S) = \{0\}$, but it is not surjective, since $(0, 1)$ is not an output. So S is not an isomorphism either.

Example 4.21. Let $\mathbb{R}[x]_{\leq 2}$ be the set of polynomials of degree at most 2, with the usual addition of polynomials and multiplication by real numbers; one can check that it is a vector space over $\mathbb{R}$. The map

$$T : \mathbb{R}[x]_{\leq 2} \to \mathbb{R}^3, \qquad T(a_0 + a_1 x + a_2 x^2) = (a_0, a_1, a_2),$$

which lists the coefficients of a polynomial, is an isomorphism. It is linear, since adding two polynomials adds their coefficients, and multiplying a polynomial by a scalar multiplies its coefficients by that scalar. It is injective, since its kernel consists only of the zero polynomial, and it is surjective, since every $(a_0, a_1, a_2) \in \mathbb{R}^3$ is the output $T(a_0 + a_1 x + a_2 x^2)$. Hence $\mathbb{R}[x]_{\leq 2} \cong \mathbb{R}^3$: as far as addition and scalar multiplication are concerned, a polynomial of degree at most 2 is the same thing as the list of its three coefficients.

Example 4.22. The complex numbers $\mathbb{C}$ also form a vector space over $\mathbb{R}$, if we only allow multiplication by real scalars. The map $T : \mathbb{C} \to \mathbb{R}^2$, $T(a + bi) = (a, b)$, is linear and bijective (check this!), so it is an isomorphism of vector spaces over $\mathbb{R}$, and $\mathbb{C} \cong \mathbb{R}^2$. This is the familiar picture of complex numbers as points in the plane.

Example 4.23. By Example 4.14 and Lemma 4.18, differentiation $D : C^1(I) \to C(I)$ is surjective but not injective, and integration $J : C(I) \to C^1(I)$ is injective but not surjective. So neither of them is an isomorphism, although $D \circ J$ is the identity. The same happens for differentiation $D : C^\infty(I) \to C^\infty(I)$, which maps a vector space to itself: it is surjective, since J maps smooth functions to smooth functions, but it is not injective. For a linear transformation $T : V \to V$ of a finite-dimensional vector space to itself, this cannot happen: by the rank–nullity theorem (Section 7), T is injective if and only if it is surjective. Infinite-dimensional vector spaces behave differently.

Lemma 4.24. *Let U, V and W be vector spaces over the same field $\mathbb{F}$.*

1. *If $T : U \to V$ and $S : V \to W$ are linear transformations, then their composition $S \circ T : U \to W$, defined by $(S \circ T)(u) = S(T(u))$, is a linear transformation. If S and T are isomorphisms, then so is $S \circ T$.*

2. *If $T : V \to W$ is an isomorphism, then its inverse function $T^{-1} : W \to V$ is also linear, and hence an isomorphism.*

3. *The identity map* $\mathrm{id}_V : V \to V$, $\mathrm{id}_V(v) = v$, *is an isomorphism.*

Proof. (1) For $u_1, u_2, u \in U$ and $a \in \mathbb{F}$, we have

$$(S \circ T)(u_1 + u_2) = S(T(u_1) + T(u_2)) = S(T(u_1)) + S(T(u_2)) = (S \circ T)(u_1) + (S \circ T)(u_2),$$
$$(S \circ T)(au) = S(aT(u)) = aS(T(u)) = a(S \circ T)(u).$$

If S and T are bijective, then $S \circ T$ is bijective, with inverse function $T^{-1} \circ S^{-1}$.

(2) Let $w_1, w_2 \in W$ and $a \in \mathbb{F}$, and let $v_1 = T^{-1}(w_1)$ and $v_2 = T^{-1}(w_2)$. Since $T(v_1 + v_2) = T(v_1) + T(v_2) = w_1 + w_2$, applying T^{-1} gives

$$T^{-1}(w_1 + w_2) = v_1 + v_2 = T^{-1}(w_1) + T^{-1}(w_2).$$

Similarly, $T(av_1) = aT(v_1) = aw_1$ gives $T^{-1}(aw_1) = av_1 = aT^{-1}(w_1)$. Finally, T^{-1} is bijective, with inverse function T.

(3) The identity map is clearly linear and bijective. □

In particular, $V \cong V$; if $V \cong W$, then $W \cong V$; and if $U \cong V$ and $V \cong W$, then $U \cong W$.

Proposition 4.25. *Let $T : V \to W$ be an isomorphism, and let $\{v_1, \dots, v_n\}$ be a basis of V. Then $\{T(v_1), \dots, T(v_n)\}$ is a basis of W. In particular, if V is finite-dimensional and $V \cong W$, then W is also finite-dimensional and* $\dim W = \dim V$.

Proof. Since T is injective, the vectors $T(v_1), \dots, T(v_n)$ are n different vectors.

They span W: let $w \in W$. Since T is surjective, $w = T(v)$ for some $v \in V$, and we can write $v = a_1v_1 + \cdots + a_nv_n$. By linearity, $w = T(v) = a_1T(v_1) + \cdots + a_nT(v_n)$.

They are linearly independent: if $a_1T(v_1) + \cdots + a_nT(v_n) = 0_W$, then by linearity $T(a_1v_1 + \cdots + a_nv_n) = 0_W$, so $a_1v_1 + \cdots + a_nv_n \in \ker(T) = \{0_V\}$ by Lemma 4.18. Since $v_1, \dots, v_n$ are linearly independent, $a_1 = \cdots = a_n = 0$. □

Conversely, every finite-dimensional vector space is isomorphic to some $\mathbb{R}^n$. To see this, we use coordinates. For the rest of this subsection, we work over the field $\mathbb{R}$; everything works in the same way over any field $\mathbb{F}$, with $\mathbb{F}^n$ in place of $\mathbb{R}^n$.

Definition 4.26 (Coordinates)**.** Let V be a vector space over $\mathbb{R}$, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V whose vectors are listed in a fixed order (an ordered basis). By Proposition 3.5, every $v \in V$ can be written in exactly one way as $v = a_1v_1 + \cdots + a_nv_n$. The vector

$$[v]_{\mathcal{B}} = (a_1, \dots, a_n) \in \mathbb{R}^n$$

is called the coordinate vector of v with respect to $\mathcal{B}$, and the map $\varphi_{\mathcal{B}} : V \to \mathbb{R}^n$, $\varphi_{\mathcal{B}}(v) = [v]_{\mathcal{B}}$, is called the coordinate map.

Theorem 4.27. *Let V be a vector space over $\mathbb{R}$ with an ordered basis $\mathcal{B} = (v_1, \dots, v_n)$. Then the coordinate map $\varphi_{\mathcal{B}} : V \to \mathbb{R}^n$ is an isomorphism. In particular, every n-dimensional vector space over $\mathbb{R}$ is isomorphic to $\mathbb{R}^n$.*

Proof. Let $v = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$, and let $c \in \mathbb{R}$. Then

$$v + w = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n \qquad \text{and} \qquad cv = (ca_1)v_1 + \cdots + (ca_n)v_n,$$

and by the uniqueness of these expressions, $[v + w]_{\mathcal{B}} = [v]_{\mathcal{B}} + [w]_{\mathcal{B}}$ and $[cv]_{\mathcal{B}} = c\,[v]_{\mathcal{B}}$. So $\varphi_{\mathcal{B}}$ is linear. It is bijective, because the map $\mathbb{R}^n \to V$, $(a_1, \ldots, a_n) \mapsto a_1v_1 + \cdots + a_nv_n$, is an inverse function of $\varphi_{\mathcal{B}}$, again by the uniqueness of the expressions. □

Example 4.28. With the ordered basis $\mathcal{B} = (1, x, x^2)$ of $\mathbb{R}[x]_{\leq 2}$, the coordinate map $\varphi_{\mathcal{B}}$ is exactly the isomorphism of Example 4.21. With a different ordered basis we get different coordinates: for example, with $\mathcal{B}' = (1, 1 + x, x^2)$, the polynomial $p = 2 + 3x + x^2 = (-1) \cdot 1 + 3 \cdot (1 + x) + 1 \cdot x^2$ has $[p]_{\mathcal{B}'} = (-1, 3, 1)$, while $[p]_{\mathcal{B}} = (2, 3, 1)$.

Corollary 4.29. *Two finite-dimensional vector spaces V and W over $\mathbb{R}$ are isomorphic if and only if $\dim V = \dim W$.*

Proof. If $V \cong W$, then $\dim V = \dim W$ by Proposition 4.25. Conversely, suppose that $\dim V = \dim W = n$, and choose ordered bases $\mathcal{B}$ of V and $\mathcal{C}$ of W. By Theorem 4.27 and Lemma 4.24, the map $\varphi_{\mathcal{C}}^{-1} \circ \varphi_{\mathcal{B}} : V \to W$ is an isomorphism. □

Example 4.30. Since $\dim \mathbb{R}^2 = 2 \neq 3 = \dim \mathbb{R}^3$, the spaces $\mathbb{R}^2$ and $\mathbb{R}^3$ are not isomorphic: there is no bijective linear transformation between them. On the other hand, any two real vector spaces of dimension 3, such as $\mathbb{R}[x]_{\leq 2}$ and $\mathbb{R}^3$, are isomorphic. Finally, the infinite-dimensional space $\mathbb{R}[x]$ is not isomorphic to any $\mathbb{R}^n$, by Proposition 4.25.

5 Subspaces

Definition 5.1 (Subspace)**.** A subspace of a vector space V over $\mathbb{F}$ is a subset $U \subseteq V$ that is itself a vector space under the same operations of vector addition and scalar multiplication defined on V. In practice, a subset $U \subseteq V$ is a subspace if and only if it satisfies the following three conditions:

1. The zero vector of V is in U, i.e., $0_V \in U$.
2. U is closed under vector addition: For any $u, v \in U$, we have $u + v \in U$.
3. U is closed under scalar multiplication: For any $a \in \mathbb{F}$ and any $v \in U$, we have $a \cdot v \in U$.

Remark 5.2. Why are these three conditions enough? Conditions 2 and 3 guarantee that adding vectors of U or multiplying them by scalars never leads outside of U. The axioms of associativity, commutativity and distributivity, and $1 \cdot v = v$, then hold in U automatically, because they already hold in V. Condition 1 provides the additive identity. Finally, the additive inverse of $v \in U$ is $-v = (-1) \cdot v$ (see Exercise 1 in Section 13), which lies in U by condition 3.

Example 5.3. Let $V = \mathbb{R}^3$ and consider the subset $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. We will verify that U is a subspace of V.

First, we check that the zero vector is in U:

$$0_V = (0, 0, 0) \in U \text{ since } 0 + 0 + 0 = 0.$$

Next, we check closure under vector addition. Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in U$. Then we have:

$$x_1 + y_1 + z_1 = 0 \qquad \text{and} \qquad x_2 + y_2 + z_2 = 0.$$

Now consider their sum:

$$(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2).$$

We need to check if this sum is in U:

$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0 + 0 = 0.$$

Thus, $(x_1 + x_2, y_1 + y_2, z_1 + z_2) \in U$.

Finally, we check closure under scalar multiplication. Let $a \in \mathbb{R}$ and $(x, y, z) \in U$. Then we have:

$$x + y + z = 0.$$

Now consider the scalar multiple:

$$a \cdot (x, y, z) = (ax, ay, az).$$

We need to check if this scalar multiple is in U:

$$ax + ay + az = a(x + y + z) = a \cdot 0 = 0.$$

Thus, $(ax, ay, az) \in U$.

Therefore, U is a subspace of $\mathbb{R}^3$.

The proof of Proposition 2.13 was exactly this criterion: $C(I)$, $C^n(I)$ and $C^\infty(I)$ contain the zero function and are closed under addition and scalar multiplication, so they are subspaces of $\mathcal{F}(I)$. Each of them is even a subspace of the next larger one in the chain

$$C^\infty(I) \subseteq \cdots \subseteq C^2(I) \subseteq C^1(I) \subseteq C(I) \subseteq \mathcal{F}(I),$$

since the three conditions do not depend on the larger space. Polynomials fit into this chain at the smooth end.

Proposition 5.4. *Identify every polynomial $p = a_0 + a_1 x + \cdots + a_d x^d \in \mathbb{R}[x]$ with the function $\mathbb{R} \to \mathbb{R}$, $x \mapsto p(x)$. Then $\mathbb{R}[x]$ is a subspace of $C^\infty(\mathbb{R})$.*

Proof. First, every polynomial function is smooth: the derivative of p is the polynomial $p' = a_1 + 2a_2 x + \cdots + d a_d x^{d-1}$, so by induction, all derivatives $p^{(k)}$ exist and are polynomials, which are continuous. Hence $\mathbb{R}[x] \subseteq C^\infty(\mathbb{R})$.

Next, the operations agree: the polynomial $p + q$, which is obtained by adding coefficients, has the values $(p + q)(x) = p(x) + q(x)$, and the polynomial cp has the values $c\,p(x)$. So adding and scaling polynomials is the same as adding and scaling the corresponding functions pointwise. It remains to check the three conditions of Definition 5.1:

1. The zero polynomial is the zero function, so the zero vector of $C^\infty(\mathbb{R})$ lies in $\mathbb{R}[x]$.
2. If p and q are polynomials, then $p + q$ is a polynomial.
3. If p is a polynomial and $c \in \mathbb{R}$, then cp is a polynomial.

Therefore, $\mathbb{R}[x]$ is a subspace of $C^\infty(\mathbb{R})$. □

Identifying polynomials with functions does not lose any information: if two polynomials $p \neq q$ defined the same function, then every real number would be a root of the nonzero polynomial $p - q$, but a nonzero polynomial of degree d has at most d roots. Combining Proposition 5.4 with the inclusions above, we obtain a chain of subspaces

$$\mathbb{R}[x]_{\leq d} \subseteq \mathbb{R}[x] \subseteq C^\infty(\mathbb{R}) \subseteq \cdots \subseteq C^1(\mathbb{R}) \subseteq C(\mathbb{R}) \subseteq \mathcal{F}(\mathbb{R}).$$

All of them from $\mathbb{R}[x]$ on contain the monomials $1, x, x^2, \ldots$, which are infinitely many linearly independent vectors, so these spaces are infinite-dimensional, since a vector space of dimension n contains at most n linearly independent vectors (Corollary 3.14).

Theorem 5.5. *Let $T : V \to W$ be a linear transformation. Then the kernel $\ker(T)$ is a subspace of V, and the image $\operatorname{im}(T)$ is a subspace of W.*

Proof. **1. Image is a subspace of W:**

1. The zero vector of W is in $\operatorname{im}(T)$ since $T(0_V) = 0_W$ by Lemma 4.2.
2. Let $w_1, w_2 \in \operatorname{im}(T)$. Then there exist $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. We have:
$$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2) \in \operatorname{im}(T).$$
3. Let $a \in \mathbb{F}$ and $w \in \operatorname{im}(T)$. Then there exists $v \in V$ such that $T(v) = w$. We have:
$$a \cdot w = a \cdot T(v) = T(a \cdot v) \in \operatorname{im}(T).$$

Therefore, $\operatorname{im}(T)$ is a subspace of W.

2. Kernel is a subspace of V:

1. The zero vector of V is in $\ker(T)$ since $T(0_V) = 0_W$ by Lemma 4.2.
2. Let $v_1, v_2 \in \ker(T)$. Then we have:
$$T(v_1 + v_2) = T(v_1) + T(v_2) = 0_W + 0_W = 0_W,$$
which implies that $v_1 + v_2 \in \ker(T)$.
3. Let $a \in \mathbb{F}$ and $v \in \ker(T)$. Then we have:
$$T(a \cdot v) = a \cdot T(v) = a \cdot 0_W = 0_W,$$
which implies that $a \cdot v \in \ker(T)$.

Therefore, $\ker(T)$ is a subspace of V. □

For example, the constant functions form a subspace of $C^1(I)$, since they are the kernel of differentiation (Example 4.14), and the functions $f \in C^2(\mathbb{R})$ with $f'' + f = 0$, such as sin and cos, form a subspace of $C^2(\mathbb{R})$, since they are the kernel of the linear transformation $f \mapsto f'' + f$.

6 Matrices

Definition 6.1 (Matrix)**.** A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns, and is typically denoted by a capital letter, such as A. The entry in the i-th row and j-th column of the matrix A is denoted by A_{ij}.

Example 6.2. Let A be a 2×3 matrix given by

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

The entry in the first row and second column is $A_{12} = 2$, and the entry in the second row and third column is $A_{23} = 6$.

Definition 6.3 (Transpose)**.** The transpose of an $m \times n$ matrix A is the $n \times m$ matrix $A^\top$ with entries $(A^\top)_{ij} = A_{ji}$. In other words, the rows of A become the columns of $A^\top$.

Example 6.4. For the matrix A from the previous example,

$$A^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

When working with matrices, we write vectors of $\mathbb{R}^n$ as columns, i.e., as $n \times 1$ matrices. To save space, we often write such a column vector as the transpose of a row:

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)^\top = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Definition 6.5 (Matrix–vector product)**.** Let A be an $m \times n$ matrix and $\mathbf{x} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$. The product $A\mathbf{x}$ is the vector in $\mathbb{R}^m$ whose i-th entry is

$$(A\mathbf{x})_i = A_{i1}x_1 + A_{i2}x_2 + \cdots + A_{in}x_n, \qquad i = 1, \ldots, m.$$

Note that $A\mathbf{x}$ is only defined when the number of columns of A equals the number of entries of $\mathbf{x}$.

Remark 6.6. If $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^m$ denote the columns of A, then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n,$$

since the i-th entry of the right-hand side is $x_1A_{i1} + x_2A_{i2} + \cdots + x_nA_{in}$. In words: $A\mathbf{x}$ is the linear combination of the columns of A whose coefficients are the entries of $\mathbf{x}$.

Example 6.7. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and $\mathbf{x} = (1, 0, -1)^\top$. Then

$$A\mathbf{x} = \begin{pmatrix} 1\cdot 1 + 2\cdot 0 + 3\cdot(-1) \\ 4\cdot 1 + 5\cdot 0 + 6\cdot(-1) \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = 1\cdot\begin{pmatrix} 1 \\ 4 \end{pmatrix} + 0\cdot\begin{pmatrix} 2 \\ 5 \end{pmatrix} + (-1)\cdot\begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$

Remark 6.8. For any $m \times n$ matrix A, the map $\mathbf{x} \mapsto A\mathbf{x}$ is a linear transformation $\mathbb{R}^n \to \mathbb{R}^m$. Indeed, for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$(A(\mathbf{x}+\mathbf{y}))_i = \sum_{j=1}^{n} A_{ij}(x_j + y_j) = \sum_{j=1}^{n} A_{ij}x_j + \sum_{j=1}^{n} A_{ij}y_j = (A\mathbf{x})_i + (A\mathbf{y})_i,$$

and similarly $(A(c\mathbf{x}))_i = c\,(A\mathbf{x})_i$. The next theorem shows that, conversely, every linear transformation $\mathbb{R}^n \to \mathbb{R}^m$ is of this form.

Theorem 6.9. *Any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by an $m \times n$ matrix A such that for any vector $\mathbf{x} \in \mathbb{R}^n$, we have*

$$T(\mathbf{x}) = A\mathbf{x}.$$

The i-th column of A is the vector $T(\mathbf{e}_i)$.

Proof. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. We will construct an $m \times n$ matrix A that represents T.

Let $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$, where $\mathbf{e}_i$ is the vector with a 1 in the i-th position and 0s elsewhere. For each basis vector $\mathbf{e}_i$, we compute $T(\mathbf{e}_i)$, which is a vector in $\mathbb{R}^m$. Let

$$T(\mathbf{e}_i) = \begin{pmatrix} T(\mathbf{e}_i)_1 \\ T(\mathbf{e}_i)_2 \\ \vdots \\ T(\mathbf{e}_i)_m \end{pmatrix}.$$

We define the matrix A as the matrix whose i-th column is $T(\mathbf{e}_i)$:

$$A = \begin{pmatrix} T(\mathbf{e}_1)_1 & T(\mathbf{e}_2)_1 & \cdots & T(\mathbf{e}_n)_1 \\ T(\mathbf{e}_1)_2 & T(\mathbf{e}_2)_2 & \cdots & T(\mathbf{e}_n)_2 \\ \vdots & \vdots & \ddots & \vdots \\ T(\mathbf{e}_1)_m & T(\mathbf{e}_2)_m & \cdots & T(\mathbf{e}_n)_m \end{pmatrix}.$$

Now, for any vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)^\top \in \mathbb{R}^n$, we can express $\mathbf{x}$ as a linear combination of the basis vectors:

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n.$$

Applying the linear transformation T to $\mathbf{x}$ and using additivity and homogeneity, we have:

$$T(\mathbf{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n).$$

By Remark 6.6, the right-hand side is exactly the linear combination of the columns of A with coefficients $x_1, \ldots, x_n$, that is,

$$T(\mathbf{x}) = A\mathbf{x}.$$

Therefore, we have shown that any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by an $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. □

Example 6.10. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x - y, 2x + 3y)$. We will find the matrix representation of T.

First, we compute $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$, where $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$ are the standard basis vectors for $\mathbb{R}^2$:

$$T(\mathbf{e}_1) = T(1,0) = (1-0, 2\cdot 1 + 3\cdot 0) = (1,2),$$
$$T(\mathbf{e}_2) = T(0,1) = (0-1, 2\cdot 0 + 3\cdot 1) = (-1,3).$$

Now we can construct the matrix A that represents T:

$$A = \begin{pmatrix} T(\mathbf{e}_1)_1 & T(\mathbf{e}_2)_1 \\ T(\mathbf{e}_1)_2 & T(\mathbf{e}_2)_2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}.$$

Therefore, the matrix representation of the linear transformation T is

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}.$$

Example 6.11. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x,y,z) = (x+2y-z, 3x-y+4z)$. We will find the matrix representation of T.

First, we compute $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, and $T(\mathbf{e}_3)$, where $\mathbf{e}_1 = (1,0,0)$, $\mathbf{e}_2 = (0,1,0)$, and $\mathbf{e}_3 = (0,0,1)$ are the standard basis vectors for $\mathbb{R}^3$:

$$T(\mathbf{e}_1) = T(1,0,0) = (1+2\cdot 0-0, 3\cdot 1 - 0 + 4\cdot 0) = (1,3),$$
$$T(\mathbf{e}_2) = T(0,1,0) = (0+2\cdot 1-0, 3\cdot 0 - 1 + 4\cdot 0) = (2,-1),$$
$$T(\mathbf{e}_3) = T(0,0,1) = (0+2\cdot 0-1, 3\cdot 0 - 0 + 4\cdot 1) = (-1,4).$$

Now we can construct the matrix A that represents T:

$$A = \begin{pmatrix} T(\mathbf{e}_1)_1 & T(\mathbf{e}_2)_1 & T(\mathbf{e}_3)_1 \\ T(\mathbf{e}_1)_2 & T(\mathbf{e}_2)_2 & T(\mathbf{e}_3)_2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}.$$

Therefore, the matrix representation of the linear transformation T is

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}.$$

6.1 Matrix operations

So far, we have used matrices to represent linear transformations. In this subsection, we learn how to add and multiply matrices; inverses of matrices are discussed in Subsection 6.3, after we have developed row reduction. We write $\mathbb{R}^{m\times n}$ for the set of all $m \times n$ matrices with real entries. For $A \in \mathbb{R}^{m\times n}$, we write $T_A : \mathbb{R}^n \to \mathbb{R}^m$ for the linear transformation $T_A(\mathbf{x}) = A\mathbf{x}$.

6.1.1 Matrix addition

Definition 6.12 (Matrix addition and scalar multiplication)**.** Let A and B be $m \times n$ matrices and $c \in \mathbb{R}$. The sum $A + B$ and the scalar multiple cA are the $m \times n$ matrices with entries

$$(A+B)_{ij} = A_{ij} + B_{ij}, \qquad (cA)_{ij} = cA_{ij}.$$

In other words, matrices are added and multiplied by scalars entry by entry. The sum $A + B$ is only defined when A and B have the same size.

Example 6.13. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -1 & 2 \\ 3 & 1 & -2 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 1+0 & 2+(-1) & 3+2 \\ 4+3 & 5+1 & 6+(-2) \end{pmatrix} = \begin{pmatrix} 1 & 1 & 5 \\ 7 & 6 & 4 \end{pmatrix},$$

and

$$2A = \begin{pmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{pmatrix}.$$

On the other hand, $A + A^\top$ is not defined, since A is a 2×3 matrix and $A^\top$ is a 3×2 matrix.

Remark 6.14. With these operations, $\mathbb{R}^{m\times n}$ is a vector space over $\mathbb{R}$. Indeed, an $m \times n$ matrix is just a list of mn real numbers arranged in a rectangle, and the operations are component-wise, exactly as in $\mathbb{R}^{mn}$. The zero vector of this space is the zero matrix 0, all of whose entries are 0, and the additive inverse of A is $-A = (-1)A$. We write $A - B = A + (-B)$. Moreover, $\dim \mathbb{R}^{m\times n} = mn$: a basis consists of the mn matrices that have a single entry equal to 1 and all other entries equal to 0.

Remark 6.15. Matrix addition corresponds to adding linear transformations: for every $\mathbf{x} \in \mathbb{R}^n$,

$$(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}, \qquad (cA)\mathbf{x} = c(A\mathbf{x}).$$

Indeed, $((A + B)\mathbf{x})_i = \sum_j (A_{ij} + B_{ij})x_j = \sum_j A_{ij}x_j + \sum_j B_{ij}x_j = (A\mathbf{x})_i + (B\mathbf{x})_i$, and similarly for cA.

6.1.2 Matrix multiplication

Let A be an $m \times n$ matrix and B an $n \times p$ matrix. Then $T_B : \mathbb{R}^p \to \mathbb{R}^n$ and $T_A : \mathbb{R}^n \to \mathbb{R}^m$, so we can first apply T_B and then T_A. The result is the composition

$$T_A \circ T_B : \mathbb{R}^p \to \mathbb{R}^m, \qquad \mathbf{x} \mapsto A(B\mathbf{x}),$$

which is again a linear transformation by Lemma 4.24. By Theorem 6.9, it is represented by an $m \times p$ matrix, whose j-th column is $A(B\mathbf{e}_j) = A\mathbf{b}_j$, where $\mathbf{b}_j = B\mathbf{e}_j$ is the j-th column of B (Remark 6.6). The i-th entry of $A\mathbf{b}_j$ is $\sum_k A_{ik}B_{kj}$. This motivates the following definition.

Definition 6.16 (Matrix multiplication)**.** Let A be an $m \times n$ matrix and B an $n \times p$ matrix. The product AB is the $m \times p$ matrix with entries

$$(AB)_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j} + \cdots + A_{in}B_{nj} = \sum_{k=1}^{n} A_{ik}B_{kj}.$$

The product AB is only defined when the number of columns of A equals the number of rows of B. An $m \times n$ matrix times an $n \times p$ matrix is an $m \times p$ matrix.

Remark 6.17. There are two useful ways to think about the product AB:

- Entry by entry: $(AB)_{ij}$ is obtained by multiplying the entries of the i-th row of A with the corresponding entries of the j-th column of B, and adding up the results.

- Column by column: the j-th column of AB is $A\mathbf{b}_j$, where $\mathbf{b}_j$ is the j-th column of B:

$$AB = A\begin{pmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p\end{pmatrix} = \begin{pmatrix}A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p\end{pmatrix}.$$

In particular, if $p = 1$, then B is a single column $\mathbf{x}$, and AB is the matrix–vector product $A\mathbf{x}$.

Example 6.18. Let

$$A = \begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}, \qquad B = \begin{pmatrix}1 & 0\\ 0 & 1\\ -1 & 2\end{pmatrix}.$$

Since A is 2×3 and B is 3×2, the product AB is a 2×2 matrix. For example, its $(1,1)$ entry is obtained from the first row of A and the first column of B:

$$(AB)_{11} = 1 \cdot 1 + 2 \cdot 0 + 3 \cdot (-1) = -2.$$

Computing all four entries in the same way, we get

$$AB = \begin{pmatrix}1\cdot 1 + 2\cdot 0 + 3\cdot(-1) & 1\cdot 0 + 2\cdot 1 + 3\cdot 2\\ 4\cdot 1 + 5\cdot 0 + 6\cdot(-1) & 4\cdot 0 + 5\cdot 1 + 6\cdot 2\end{pmatrix} = \begin{pmatrix}-2 & 8\\ -2 & 17\end{pmatrix}.$$

Note that the first column of AB is $A(1,0,-1)^\top = (-2,-2)^\top$, which we computed in Example 6.7.

The product BA is also defined, but it is a 3×3 matrix:

$$BA = \begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix}.$$

In particular, $AB \neq BA$.

Theorem 6.19. *Let A be an $m \times n$ matrix and B an $n \times p$ matrix. Then for every $\mathbf{x} \in \mathbb{R}^p$,*

$$(AB)\mathbf{x} = A(B\mathbf{x}).$$

In other words, AB is the matrix of the composition $T_A \circ T_B$: multiplying matrices corresponds to composing linear transformations.

Proof. This holds by construction, but we can also check it directly. For each $i = 1, \ldots, m$,

$$\begin{aligned}((AB)\mathbf{x})_i &= \sum_{j=1}^{p} (AB)_{ij} x_j = \sum_{j=1}^{p}\sum_{k=1}^{n} A_{ik}B_{kj}x_j\\ &= \sum_{k=1}^{n} A_{ik}\left(\sum_{j=1}^{p} B_{kj}x_j\right) = \sum_{k=1}^{n} A_{ik}(B\mathbf{x})_k = (A(B\mathbf{x}))_i.\end{aligned}$$

□

Example 6.20. Let $S, T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformations $S(x,y) = (2x+3y, 4x-y)$ from Example 4.8 and $T(x,y) = (x-y, 2x+3y)$ from Example 6.10. Their matrices are

$$A_S = \begin{pmatrix}2 & 3\\ 4 & -1\end{pmatrix}, \qquad A_T = \begin{pmatrix}1 & -1\\ 2 & 3\end{pmatrix}.$$

Computing the composition directly, we get

$$\begin{aligned} T(S(x,y)) &= T(2x+3y, 4x-y) \\ &= \big((2x+3y)-(4x-y),\ 2(2x+3y)+3(4x-y)\big) \\ &= (-2x+4y, 16x+3y). \end{aligned}$$

On the other hand,

$$A_T A_S = \begin{pmatrix} 1\cdot 2+(-1)\cdot 4 & 1\cdot 3+(-1)\cdot(-1) \\ 2\cdot 2+3\cdot 4 & 2\cdot 3+3\cdot(-1) \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 16 & 3 \end{pmatrix},$$

which is exactly the matrix of $T \circ S$, as Theorem 6.19 predicts.

Definition 6.21 (Identity matrix)**.** The $n \times n$ identity matrix I_n is the matrix with 1s on the diagonal and 0s everywhere else:

$$(I_n)_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \qquad I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Its columns are the standard basis vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$, so $I_n \mathbf{x} = \mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$, i.e., I_n represents the identity map. When the size is clear, we simply write I.

Proposition 6.22. *Let A, B, C be matrices and $c \in \mathbb{R}$. Whenever the sizes are such that the sums and products below are defined, we have:*

1. *Associativity: $(AB)C = A(BC)$;*
2. *Distributivity: $A(B+C) = AB + AC$ and $(A+B)C = AC + BC$;*
3. *Scalars: $c(AB) = (cA)B = A(cB)$;*
4. *Identity: $I_m A = A = A I_n$ for every $m \times n$ matrix A;*
5. *Transpose: $(AB)^\top = B^\top A^\top$.*

Proof. All five properties are checked by comparing entries. For associativity,

$$\begin{aligned} ((AB)C)_{ij} &= \sum_l (AB)_{il} C_{lj} = \sum_l \sum_k A_{ik} B_{kl} C_{lj} \\ &= \sum_k A_{ik} \left(\sum_l B_{kl} C_{lj} \right) = \sum_k A_{ik} (BC)_{kj} = (A(BC))_{ij}. \end{aligned}$$

For the first distributive law, $(A(B+C))_{ij} = \sum_k A_{ik}(B_{kj} + C_{kj}) = \sum_k A_{ik} B_{kj} + \sum_k A_{ik} C_{kj} = (AB + AC)_{ij}$; the second distributive law and the rule for scalars are checked in the same way. For the identity, $(I_m A)_{ij} = \sum_k (I_m)_{ik} A_{kj} = A_{ij}$, since the only nonzero term is the one with $k = i$; similarly, $A I_n = A$. Finally, for the transpose,

$$((AB)^\top)_{ij} = (AB)_{ji} = \sum_k A_{jk} B_{ki} = \sum_k (B^\top)_{ik} (A^\top)_{kj} = (B^\top A^\top)_{ij}. \qquad \square$$

Remark 6.23. Matrix multiplication is not commutative: even when AB and BA are both defined and have the same size, in general $AB \neq BA$. For example,

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

This matches the fact that composing two linear transformations in a different order usually gives a different result. Another difference from numbers is that a product of two nonzero matrices can be zero:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

As a consequence, we cannot "cancel" matrices: $AB = AC$ does not imply $B = C$ in general.

6.2 Row Reduction

Definition 6.24 (Row echelon form)**.** The leading entry (also called the pivot) of a nonzero row of a matrix is its first nonzero entry from the left. A matrix is said to be in row echelon form (REF) if it satisfies the following conditions:

1. All nonzero rows are above any rows of all zeros.
2. The leading entry of each nonzero row after the first occurs to the right of the leading entry of the previous row.
3. The leading entry of each nonzero row is 1.

Remark 6.25. Conditions 1 and 2 imply that all entries below a leading entry are zero. Some textbooks do not require condition 3, i.e., they allow leading entries to be any nonzero numbers. In these notes, we always scale the leading entries to be 1.

Example 6.26. Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix is in row echelon form because:

1. The nonzero rows are above the row of all zeros.
2. The leading entry of the second row (in column 2) is to the right of the leading entry of the first row (in column 1).
3. Both leading entries are equal to 1.

Note that nonzero entries above a leading 1, such as the 2 in the first row, are allowed.

Example 6.27. Consider the matrix

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

This matrix is also in row echelon form because:

1. There are no rows of all zeros, so the first condition is trivially satisfied.

2. The leading entry of the second row (in column 2) is to the right of the leading entry of the first row (in column 1), and the leading entry of the third row (in column 3) is to the right of the leading entry of the second row.

3. All three leading entries are equal to 1.

Example 6.28. Consider the matrix

$$C = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix is not in row echelon form because the leading entry of the second row is 3, not 1, so the third condition fails. The first two conditions do hold: the row of zeros is at the bottom, and the leading entries are in columns 2 and 3, so the leading entry of the second row is to the right of the leading entry of the first row. (A first column consisting of zeros is allowed.) Multiplying the second row by $1/3$ gives a matrix in row echelon form.

Example 6.29. Consider the matrix

$$D = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 4 \end{pmatrix}.$$

This matrix is not in row echelon form because the nonzero row (the third row) is below a row of all zeros (the second row). For it to be in row echelon form, all nonzero rows must be above any rows of all zeros.

To transform a matrix into row echelon form, we can use elementary row operations, which include:

1. Swapping two rows.

2. Multiplying a row by a nonzero scalar.

3. Adding or subtracting a multiple of one row to another row.

Each elementary row operation can be undone by another elementary row operation: we can swap the same two rows again, multiply the same row by the inverse of the scalar, or subtract (add) the same multiple back. Using these operations, every matrix can be brought into row echelon form; this procedure is called Gaussian elimination.

Example 6.30. Consider the matrix

$$E = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 0 & 0 & 0 \end{pmatrix}.$$

We will transform this matrix into row echelon form using elementary row operations.

First, we swap the first and second rows, so that the first row starts with a 1:

$$\begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 0 & 0 & 0 \end{pmatrix}.$$

Next, we can eliminate the leading entry in the second row by subtracting 2 times the first row from the second row:

$$\begin{pmatrix} 1 & 3 & 5 \\ 0 & -2 & -4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Finally, we can multiply the second row by $-1/2$ to make the leading entry of the second row equal to 1:

$$\begin{pmatrix} 1 & 3 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The resulting matrix is now in row echelon form.

Example 6.31. Consider the matrix

$$F = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

We will transform this matrix into row echelon form using elementary row operations.

First, we can eliminate the leading entry in the second row by subtracting 4 times the first row from the second row:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Next, we can eliminate the leading entry in the third row by subtracting 7 times the first row from the third row:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix}.$$

Now, we can eliminate the leading entry in the third row by subtracting 2 times the second row from the third row:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}.$$

Finally, we can multiply the second row by $-1/3$ to make the leading entry of the second row equal to 1:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The resulting matrix is now in row echelon form.

Sometimes it is useful to further simplify a matrix in row echelon form.

Definition 6.32 (Reduced row echelon form)**.** A matrix is in reduced row echelon form (RREF) if it is in row echelon form and, in addition, each leading 1 is the only nonzero entry in its column.

For example, the matrices A and B in Examples 6.26 and 6.27 are in row echelon form, but not in reduced row echelon form, because they have nonzero entries above some of their leading 1s. Every matrix A can be brought into reduced row echelon form by elementary row operations, and one can show that the result does not depend on which operations we use; we denote it by $\mathrm{RREF}(A)$.

Example 6.33. Consider the matrix

$$G = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix},$$

which is the row echelon form of the matrix F that we found in Example 6.31. We will transform this matrix into reduced row echelon form (RREF).

We can eliminate the entry above the leading 1 in the second row by subtracting 2 times the second row from the first row:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The resulting matrix is now in reduced row echelon form (RREF).

Example 6.34. Consider the matrix

$$H = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}.$$

We will transform this matrix into reduced row echelon form (RREF).

First, we can eliminate the entry above the leading 1 in the third row by subtracting 4 times the third row from the second row:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Next, we can eliminate the entry above the leading 1 in the second row by subtracting 2 times the second row from the first row:

$$\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Finally, we can eliminate the entry above the leading 1 in the third row by subtracting 3 times the third row from the first row:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The resulting matrix is now in reduced row echelon form (RREF).

6.3 The Inverse of a Matrix

6.3.1 Definition and basic properties

For a nonzero number a, dividing by a means multiplying by its inverse a^{-1}, the number with $a \cdot a^{-1} = a^{-1} \cdot a = 1$. For square matrices, the role of the number 1 is played by the identity matrix.

Definition 6.35 (Inverse matrix)**.** An $n \times n$ matrix A is called invertible if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n.$$

Such a matrix B is called an inverse of A. A square matrix that is not invertible is called singular. (We only consider inverses of square matrices.)

Proposition 6.36. *An invertible matrix A has exactly one inverse, which we denote by A^{-1}.*

Proof. Suppose that B and C are both inverses of A. Then, using associativity,

$$B = BI_n = B(AC) = (BA)C = I_nC = C. \qquad \square$$

Example 6.37. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 2 \cdot 1 + 1 \cdot (-1) & 2 \cdot (-1) + 1 \cdot 2 \\ 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and in the same way $BA = I_2$. So A is invertible and $A^{-1} = B$.

Remark 6.38. If A is invertible, then for every $\mathbf{b} \in \mathbb{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution, namely $\mathbf{x} = A^{-1}\mathbf{b}$. Indeed, $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution, since $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I_n\mathbf{b} = \mathbf{b}$; and any solution $\mathbf{x}$ satisfies $\mathbf{x} = I_n\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b}$. In particular, the only solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. In terms of linear transformations, $T_{A^{-1}}$ is the inverse function of T_A: it undoes what T_A does.

Example 6.39. Not every nonzero square matrix is invertible. By Remark 6.38, if $A\mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \neq \mathbf{0}$, then A is not invertible. For example, the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

is singular, since $A(2, -1)^\top = (2 - 2, 4 - 4)^\top = (0, 0)^\top$.

To find the inverse of a matrix A, we can solve the equation $A\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ in terms of $\mathbf{y}$.

Proposition 6.40. *Let A and B be $n \times n$ matrices. Suppose that for every $\mathbf{y} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{y}$ has exactly one solution, and that this solution is $\mathbf{x} = B\mathbf{y}$. Then A is invertible and $A^{-1} = B$.*

Proof. For every $\mathbf{y} \in \mathbb{R}^n$, the vector $B\mathbf{y}$ is a solution of $A\mathbf{x} = \mathbf{y}$, so $(AB)\mathbf{y} = A(B\mathbf{y}) = \mathbf{y}$. Taking $\mathbf{y} = \mathbf{e}_j$ shows that the j-th column of AB is $\mathbf{e}_j$, so $AB = I_n$.

Now let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} = A\mathbf{x}$. Then both $\mathbf{x}$ and $B\mathbf{y}$ are solutions of the equation $Az = \mathbf{y}$. Since this equation has exactly one solution, $\mathbf{x} = B\mathbf{y} = (BA)\mathbf{x}$. Taking $\mathbf{x} = \mathbf{e}_j$ shows that the j-th column of BA is $\mathbf{e}_j$, so $BA = I_n$. $\square$

Example 6.41. Let

$$A = \begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix}$$

be the matrix of the linear transformation $T(x, y) = (2x + 3y, 4x - y)$. In Example 4.11, we solved the equation $T(x, y) = (u, v)$, i.e., $A(x, y)^\top = (u, v)^\top$. Each step of that computation was forced, so the solution is unique, and it is

$$x = \frac{u + 3v}{14}, \qquad y = \frac{2u - v}{7} = \frac{4u - 2v}{14}.$$

In matrix form,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

By Proposition 6.40,

$$A^{-1} = \frac{1}{14} \begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix}.$$

We can double-check this:

$$AA^{-1} = \frac{1}{14} \begin{pmatrix} 2 \cdot 1 + 3 \cdot 4 & 2 \cdot 3 + 3 \cdot (-2) \\ 4 \cdot 1 + (-1) \cdot 4 & 4 \cdot 3 + (-1) \cdot (-2) \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 14 & 0 \\ 0 & 14 \end{pmatrix} = I_2.$$

Proposition 6.42. *Let A and B be invertible $n \times n$ matrices. Then:*

1. *A^{-1} is invertible and $(A^{-1})^{-1} = A$;*
2. *AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$;*
3. *$A^\top$ is invertible and $(A^\top)^{-1} = (A^{-1})^\top$.*

Proof. (1) The equations $AA^{-1} = A^{-1}A = I_n$ say exactly that A is an inverse of A^{-1}.

(2) Using associativity,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n,$$

and similarly $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$.

(3) Using $(AB)^\top = B^\top A^\top$ and $I_n^\top = I_n$, we get

$$A^\top (A^{-1})^\top = (A^{-1}A)^\top = I_n^\top = I_n, \qquad (A^{-1})^\top A^\top = (AA^{-1})^\top = I_n. \qquad \square$$

Note the reversed order in part 2: to undo "first T_B, then T_A", we must first undo T_A and then undo T_B. By induction, every product of invertible matrices is invertible, with $(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}$.

6.3.2 Computing the inverse with row reduction

For larger matrices, solving $A\mathbf{x} = \mathbf{y}$ by hand as in Example 6.41 quickly becomes tedious. We now use row reduction to give a systematic method for deciding whether a square matrix is invertible and for computing its inverse. The key observation is that elementary row operations can be performed by matrix multiplication.

Definition 6.43 (Elementary matrix)**.** An elementary matrix is a matrix obtained from an identity matrix I_m by a single elementary row operation.

Example 6.44. The following are 3×3 elementary matrices. They are obtained from I_3 by swapping rows 1 and 2, by multiplying row 2 by a scalar $c \neq 0$, and by adding c times row 1 to row 3, respectively:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix}.$$

Multiplying a 3×2 matrix from the left by the last of them gives

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix} \begin{pmatrix} p_1 & p_2 \\ q_1 & q_2 \\ r_1 & r_2 \end{pmatrix} = \begin{pmatrix} p_1 & p_2 \\ q_1 & q_2 \\ r_1 + cp_1 & r_2 + cp_2 \end{pmatrix},$$

which is exactly the result of adding c times row 1 to row 3.

Lemma 6.45. *Let E be the elementary matrix obtained from I_m by some elementary row operation. Then for every $m \times n$ matrix A, the product EA is the matrix obtained from A by the same row operation.*

Proof. By the definition of matrix multiplication, $(EA)_{ij} = \sum_k E_{ik} A_{kj}$, so the i-th row of EA is

$$E_{i1} \cdot (\text{row 1 of } A) + E_{i2} \cdot (\text{row 2 of } A) + \cdots + E_{im} \cdot (\text{row } m \text{ of } A).$$

On the other hand, every elementary row operation replaces each row by a linear combination of the old rows, with coefficients that depend only on the operation and not on the matrix. For example, adding c times row 1 to row 3 replaces row 3 by $c \cdot (\text{row } 1) + 1 \cdot (\text{row } 3)$ and keeps all other rows. Say that the operation replaces row i by $c_{i1} \cdot (\text{row } 1) + \cdots + c_{im} \cdot (\text{row } m)$. Applying it to I_m, whose k-th row is $\mathbf{e}_k^\top$, the new i-th row is $c_{i1}\mathbf{e}_1^\top + \cdots + c_{im}\mathbf{e}_m^\top = (c_{i1}, \ldots, c_{im})$. Since this is the i-th row of E, we get $E_{ik} = c_{ik}$ for all i, k. Therefore, the i-th row of EA is $c_{i1} \cdot (\text{row 1 of } A) + \cdots + c_{im} \cdot (\text{row } m \text{ of } A)$, which is the i-th row of the matrix obtained from A by the row operation. □

Lemma 6.46. *Every elementary matrix E is invertible, and E^{-1} is the elementary matrix of the row operation that undoes the operation of E.*

Proof. Let E' be the elementary matrix of the row operation that undoes the operation of E. The matrix E is obtained from I_m by the operation of E, and by Lemma 6.45, $E'E$ is obtained from E by undoing this operation. Hence $E'E = I_m$. In the same way, $EE' = I_m$. □

As a consequence, if a matrix R is obtained from A by row operations with elementary matrices $E_1, E_2, \ldots, E_k$ (in this order), then

$$R = BA, \qquad \text{where } B = E_k \cdots E_2 E_1$$

is invertible as a product of invertible matrices (Proposition 6.42).

Theorem 6.47. *For an $n \times n$ matrix A, the following statements are equivalent:*

1. *A is invertible;*
2. *the only solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$;*

3. $\mathrm{RREF}(A) = I_n$.

Proof. $(1 \Rightarrow 2)$ This was shown in Remark 6.38.

$(2 \Rightarrow 3)$ Suppose that $R = \mathrm{RREF}(A) \neq I_n$; we will find a nonzero solution of $A\mathbf{x} = \mathbf{0}$. If R had n pivots, then each of its n rows would contain a pivot, and the pivot columns $p_1 < p_2 < \cdots < p_n$ would be n different numbers among $1, \ldots, n$, so $p_i = i$ for every i. Since each pivot is a 1 and is the only nonzero entry in its column, this would mean $R = I_n$. So R has fewer than n pivots, and some column j of R contains no pivot. Set $x_j = 1$, and $x_k = 0$ for every other column k without a pivot. Then the equation given by each nonzero row i of R becomes $x_{p_i} + R_{ij} = 0$, so we set $x_{p_i} = -R_{ij}$. This gives a vector $\mathbf{x} \neq \mathbf{0}$ with $R\mathbf{x} = \mathbf{0}$. Writing $R = BA$ with B invertible as above, we get

$$A\mathbf{x} = B^{-1}(BA)\mathbf{x} = B^{-1}(R\mathbf{x}) = B^{-1}\mathbf{0} = \mathbf{0}.$$

$(3 \Rightarrow 1)$ Let $E_1, \ldots, E_k$ be the elementary matrices of the row operations that bring A to its reduced row echelon form $\mathrm{RREF}(A) = I_n$, and let $B = E_k \cdots E_1$. Then $BA = I_n$, and B is invertible. Multiplying from the left by B^{-1} gives $A = B^{-1}$, so A is invertible with $A^{-1} = (B^{-1})^{-1} = B$. □

Corollary 6.48. *Let A and B be $n \times n$ matrices with $BA = I_n$. Then A is invertible and $A^{-1} = B$. In particular, also $AB = I_n$.*

Proof. If $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = I_n\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B\mathbf{0} = \mathbf{0}$. By Theorem 6.47, A is invertible, and

$$B = BI_n = B(AA^{-1}) = (BA)A^{-1} = I_nA^{-1} = A^{-1}.$$ □

So, to check that a square matrix B is the inverse of A, it is enough to check one of the two equations $BA = I_n$ or $AB = I_n$. (In the second case, apply the corollary with the roles of A and B swapped: then B is invertible with $B^{-1} = A$, and hence $A^{-1} = B$.)

Theorem 6.49 (Gauss–Jordan method)**.** *Let A be an $n \times n$ matrix, and row reduce the $n \times 2n$ matrix $(A \mid I_n)$, obtained by writing I_n to the right of A.*

- *If the left block becomes I_n, then A is invertible, and the right block is A^{-1}:*

$$(A \mid I_n) \quad \longrightarrow \quad (I_n \mid A^{-1}).$$

- *If a row of zeros appears in the left block, then A is not invertible.*

Proof. Let $E_1, \ldots, E_k$ be the elementary matrices of the row operations we perform, and let $B = E_k \cdots E_1$. By Lemma 6.45 and the column-by-column description of matrix multiplication (Remark 6.17), the result is

$$B\,(A \mid I_n) = (BA \mid BI_n) = (BA \mid B).$$

In particular, the left block BA is what we get by applying the same row operations to A alone.

If $BA = I_n$, then A is invertible and $A^{-1} = B$ by Corollary 6.48, and B is exactly the right block.

If BA has a row of zeros, then BA is not invertible: for every $n \times n$ matrix M, the product $(BA)M$ also has a row of zeros, so it cannot be equal to I_n. If A were invertible, then BA would be invertible as a product of invertible matrices. Hence A is not invertible. □

One of the two cases always occurs: if $\mathrm{RREF}(A) \neq I_n$, then, as we saw in the proof of Theorem 6.47, $\mathrm{RREF}(A)$ has fewer than n pivots, and hence a row of zeros.

Example 6.50. Let us compute the inverse of the matrix A from Example 6.41 with the Gauss–Jordan method. We start with

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 4 & -1 & 0 & 1 \end{array}\right).$$

Subtracting 2 times the first row from the second row gives

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & -7 & -2 & 1 \end{array}\right).$$

Multiplying the second row by $-1/7$ gives

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & 1 & \frac{2}{7} & -\frac{1}{7} \end{array}\right).$$

Subtracting 3 times the second row from the first row gives

$$\left(\begin{array}{cc|cc} 2 & 0 & \frac{1}{7} & \frac{3}{7} \\ 0 & 1 & \frac{2}{7} & -\frac{1}{7} \end{array}\right).$$

Finally, multiplying the first row by $1/2$ gives

$$\left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{14} & \frac{3}{14} \\ 0 & 1 & \frac{2}{7} & -\frac{1}{7} \end{array}\right).$$

The left block is I_2, so

$$A^{-1} = \begin{pmatrix} \frac{1}{14} & \frac{3}{14} \\ \frac{2}{7} & -\frac{1}{7} \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix},$$

as we found in Example 6.41.

Example 6.51. Let us find the inverse of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

We start with

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

Subtracting the first row from the third row gives

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

The left block is now in row echelon form, and we continue to its reduced row echelon form.

Subtracting the third row from the second row gives

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

Subtracting the second row from the first row gives

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

The left block is I_3, so

$$A^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix}.$$

By Corollary 6.48, to double-check the answer it is enough to verify that $AA^{-1} = I_3$; try it!

Example 6.52. Let us apply the Gauss–Jordan method to the singular matrix of Example 6.39:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right).$$

Subtracting 2 times the first row from the second row gives

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right).$$

A row of zeros has appeared in the left block, so the matrix is not invertible, as we already knew.

Remark 6.53. In practice, if we only want to solve $A\mathbf{x} = \mathbf{b}$ for one particular $\mathbf{b}$, we do not need A^{-1}: it is faster, and numerically more accurate, to row reduce the matrix $(A \mid \mathbf{b})$ directly. If A is invertible, the row operations that turn A into I_n turn $(A \mid \mathbf{b})$ into $(I_n \mid A^{-1}\mathbf{b})$, so the solution can be read off from the last column. This is also what numerical libraries do: for example, in NumPy one uses `np.linalg.solve(A, b)` rather than `np.linalg.inv(A) @ b`.

6.3.3 Invertible matrices and isomorphisms

In Section 4, we defined isomorphisms of vector spaces. We now show that isomorphisms correspond exactly to invertible matrices. We start with the spaces $\mathbb{R}^n$.

Theorem 6.54. *Let A be an $m \times n$ matrix. The linear transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $T_A(\mathbf{x}) = A\mathbf{x}$, is an isomorphism if and only if $m = n$ and A is invertible. In this case, $(T_A)^{-1} = T_{A^{-1}}$.*

Proof. Suppose that $m = n$ and A is invertible. Then $A^{-1}(A\mathbf{x}) = (A^{-1}A)\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, and $A(A^{-1}y) = (AA^{-1})y = y$ for all $y \in \mathbb{R}^n$. So $T_{A^{-1}}$ is an inverse function of T_A. Hence T_A is bijective, i.e., an isomorphism, and $(T_A)^{-1} = T_{A^{-1}}$.

Conversely, suppose that T_A is an isomorphism. Then $\mathbb{R}^n \cong \mathbb{R}^m$, so $n = \dim \mathbb{R}^n = \dim \mathbb{R}^m = m$ by Proposition 4.25. Since T_A is injective, Lemma 4.18 shows that the only solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. By Theorem 6.47, A is invertible. □

To extend this to arbitrary finite-dimensional vector spaces, we represent linear transformations between them by matrices, using coordinates (Definition 4.26). As before, we work over $\mathbb{R}$.

Definition 6.55 (Matrix of a linear transformation)**.** Let V and W be finite-dimensional vector spaces over $\mathbb{R}$, with ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ of V and $\mathcal{C} = (w_1, \dots, w_m)$ of W, and let $T : V \to W$ be a linear transformation. The matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$ is the $m \times n$ matrix $[T]_{\mathcal{C},\mathcal{B}}$ whose j-th column is the coordinate vector $[T(v_j)]_{\mathcal{C}}$, written as a column. We say that this matrix represents T.

Proposition 6.56. *With the notation above, let $A = [T]_{\mathcal{C},\mathcal{B}}$. Then*

$$[T(v)]_{\mathcal{C}} = A\,[v]_{\mathcal{B}} \qquad \text{for every } v \in V.$$

In other words, $\varphi_{\mathcal{C}} \circ T = T_A \circ \varphi_{\mathcal{B}}$.

Proof. Write $v = x_1 v_1 + \cdots + x_n v_n$, so that $[v]_{\mathcal{B}} = (x_1, \dots, x_n)^\top$. Using the linearity of T and of the coordinate map $\varphi_{\mathcal{C}}$ (Theorem 4.27), we get

$$[T(v)]_{\mathcal{C}} = [x_1 T(v_1) + \cdots + x_n T(v_n)]_{\mathcal{C}} = x_1 [T(v_1)]_{\mathcal{C}} + \cdots + x_n [T(v_n)]_{\mathcal{C}}.$$

Since the columns of A are $[T(v_1)]_{\mathcal{C}}, \dots, [T(v_n)]_{\mathcal{C}}$, the right-hand side is $A[v]_{\mathcal{B}}$ by Remark 6.6. □

Remark 6.57. If $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ with their standard bases, then every vector is its own coordinate vector, and $[T]_{\mathcal{C},\mathcal{B}}$ is exactly the matrix of Theorem 6.9. With other bases, the same linear transformation is usually represented by a different matrix.

We can now prove the main result of this subsection.

Theorem 6.58. *Let V and W be finite-dimensional vector spaces over $\mathbb{R}$, with ordered bases $\mathcal{B}$ of V and $\mathcal{C}$ of W.*

1. *A linear transformation $T : V \to W$ is an isomorphism if and only if the matrix $[T]_{\mathcal{C},\mathcal{B}}$ that represents it is invertible. In this case, $[T^{-1}]_{\mathcal{B},\mathcal{C}} = ([T]_{\mathcal{C},\mathcal{B}})^{-1}$.*
2. *Consequently, V and W are isomorphic if and only if there exists a linear transformation $T : V \to W$ that is represented by an invertible matrix.*

Proof. (1) Let $A = [T]_{\mathcal{C},\mathcal{B}}$. By Proposition 6.56, $\varphi_{\mathcal{C}} \circ T = T_A \circ \varphi_{\mathcal{B}}$. Since the coordinate maps are isomorphisms (Theorem 4.27), we can compose with their inverses to get

$$T_A = \varphi_{\mathcal{C}} \circ T \circ \varphi_{\mathcal{B}}^{-1} \qquad \text{and} \qquad T = \varphi_{\mathcal{C}}^{-1} \circ T_A \circ \varphi_{\mathcal{B}}.$$

If T is an isomorphism, then so is T_A, as a composition of isomorphisms (Lemma 4.24), and A is invertible by Theorem 6.54. Conversely, if A is invertible, then T_A is an isomorphism by Theorem 6.54, and hence so is T, again as a composition of isomorphisms.

In this case, $T^{-1} = \varphi_{\mathcal{B}}^{-1} \circ (T_A)^{-1} \circ \varphi_{\mathcal{C}} = \varphi_{\mathcal{B}}^{-1} \circ T_{A^{-1}} \circ \varphi_{\mathcal{C}}$, so $[T^{-1}(w)]_{\mathcal{B}} = A^{-1}[w]_{\mathcal{C}}$ for every $w \in W$. Taking for w the j-th vector of $\mathcal{C}$, whose coordinate vector is e_j, shows that the j-th column of $[T^{-1}]_{\mathcal{B},\mathcal{C}}$ is the j-th column of A^{-1}.

(2) If $V \cong W$, choose an isomorphism $T : V \to W$; by part 1, the matrix $[T]_{\mathcal{C},\mathcal{B}}$ that represents it is invertible. Conversely, if some linear transformation $T : V \to W$ is represented by an invertible matrix, then T is an isomorphism by part 1, so $V \cong W$. □

Since only square matrices can be invertible, Theorem 6.58 gives another explanation of Corollary 4.29: isomorphic spaces must have the same dimension.

Example 6.59. The linear transformation $T(x, y) = (2x + 3y, 4x - y)$ of Example 4.19 is represented, with respect to the standard bases, by the matrix A of Example 6.41. Since A is invertible, T is an isomorphism, and T^{-1} is represented by

$$A^{-1} = \frac{1}{14}\begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix}, \qquad \text{i.e.,} \qquad T^{-1}(u, v) = \left(\frac{u + 3v}{14}, \frac{4u - 2v}{14}\right),$$

in agreement with Example 4.19.

Example 6.60. Let $\mathbb{R}[x]_{\leq 1}$ be the vector space of polynomials of degree at most 1, and consider the linear transformation

$$T : \mathbb{R}[x]_{\leq 1} \to \mathbb{R}^2, \qquad T(p) = (p(0), p(1)),$$

which evaluates a polynomial at 0 and at 1. We use the ordered basis $\mathcal{B} = (1, x)$ of $\mathbb{R}[x]_{\leq 1}$ and the standard basis $\mathcal{C} = (\mathbf{e}_1, \mathbf{e}_2)$ of $\mathbb{R}^2$. Since $T(1) = (1, 1)$ and $T(x) = (0, 1)$,

$$[T]_{\mathcal{C},\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \text{with inverse} \qquad ([T]_{\mathcal{C},\mathcal{B}})^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

(Check that the product of these two matrices is I_2.) By Theorem 6.58, T is an isomorphism: a polynomial of degree at most 1 is uniquely determined by its values at 0 and 1, and every pair of values occurs. The inverse is given by $[T^{-1}(u, v)]_{\mathcal{B}} = (u, v - u)^\top$, i.e.,

$$T^{-1}(u, v) = u + (v - u)x.$$

Indeed, this polynomial takes the value u at 0 and the value v at 1.

6.4 Special Types of Matrices

Some kinds of square matrices appear so often that they have their own names. Recall from Remark 6.14 that the set $\mathbb{R}^{n\times n}$ of all $n \times n$ matrices is a vector space over $\mathbb{R}$. The entries $A_{11}, A_{22}, \ldots, A_{nn}$ of an $n \times n$ matrix A form its (main) diagonal.

6.4.1 Matrices defined by their entries

Definition 6.61 (Special types of square matrices)**.** An $n \times n$ matrix A is called

- diagonal if $A_{ij} = 0$ whenever $i \neq j$, i.e., all entries outside the diagonal are zero. We write $\operatorname{diag}(d_1, \ldots, d_n)$ for the diagonal matrix with diagonal entries $d_1, \ldots, d_n$;
- upper triangular if $A_{ij} = 0$ whenever $i > j$, i.e., all entries below the diagonal are zero;
- lower triangular if $A_{ij} = 0$ whenever $i < j$, i.e., all entries above the diagonal are zero;
- symmetric if $A^\top = A$, i.e., $A_{ij} = A_{ji}$ for all i, j;
- skew-symmetric if $A^\top = -A$, i.e., $A_{ij} = -A_{ji}$ for all i, j.

We denote the sets of diagonal, upper triangular, lower triangular, symmetric and skew-symmetric $n \times n$ matrices by $\mathcal{D}_n$, $\mathcal{U}_n$, $\mathcal{L}_n$, Sym_n and Skew_n, respectively.

Example 6.62. Here are examples of each type for $n = 3$:

$$\underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}}_{\text{diagonal}}, \qquad \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}}_{\text{upper triangular}}, \qquad \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}}_{\text{lower triangular}},$$

$$\underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}}_{\text{symmetric}}, \qquad \underbrace{\begin{pmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{pmatrix}}_{\text{skew-symmetric}}.$$

The diagonal matrix above is $\operatorname{diag}(2, -1, 5)$. The identity matrix I_n and the zero matrix are diagonal, and every diagonal matrix is upper triangular, lower triangular and symmetric at the same time. The diagonal entries of a skew-symmetric matrix are always zero, since $A_{ii} = -A_{ii}$ forces $A_{ii} = 0$. Finally, every square matrix in row echelon form is upper triangular, since the pivot of row i lies in column i or further to the right.

Example 6.63. For any $m \times n$ matrix A, the $n \times n$ matrix $A^\top A$ is symmetric, since by Proposition 6.22,

$$(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A.$$

Matrices of this form appear all the time in machine learning, for example in least squares problems and as covariance matrices.

Remark 6.64. Products of diagonal matrices are easy to compute: the only nonzero term in $(AB)_{ij} = \sum_k A_{ik} B_{kj}$ can occur for $k = i = j$, so

$$\operatorname{diag}(a_1, \ldots, a_n) \operatorname{diag}(b_1, \ldots, b_n) = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n).$$

In particular, if all $d_i \neq 0$, then $\operatorname{diag}(d_1, \ldots, d_n)$ is invertible with inverse $\operatorname{diag}(1/d_1, \ldots, 1/d_n)$. On the other hand, the product of two symmetric matrices need not be symmetric:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The following theorem shows that several of these sets are vector spaces in their own right.

Theorem 6.65. *The sets $\mathcal{D}_n$, $\mathcal{U}_n$ and Sym_n are subspaces of $\mathbb{R}^{n \times n}$. In particular, each of them is a vector space over $\mathbb{R}$. Moreover, $\dim \mathcal{D}_n = n$.*

Proof. We check the three conditions of Definition 5.1. Throughout, let $c \in \mathbb{R}$, and recall that $(A + B)_{ij} = A_{ij} + B_{ij}$ and $(cA)_{ij} = cA_{ij}$.

Diagonal matrices. The zero matrix is diagonal. If $A, B \in \mathcal{D}_n$ and $i \neq j$, then

$$(A + B)_{ij} = 0 + 0 = 0 \qquad \text{and} \qquad (cA)_{ij} = c \cdot 0 = 0,$$

so $A + B \in \mathcal{D}_n$ and $cA \in \mathcal{D}_n$.

Upper triangular matrices. The zero matrix is upper triangular. If $A, B \in \mathcal{U}_n$ and $i > j$, then $(A + B)_{ij} = 0 + 0 = 0$ and $(cA)_{ij} = c \cdot 0 = 0$, so $A + B \in \mathcal{U}_n$ and $cA \in \mathcal{U}_n$.

Symmetric matrices. First, note that the transpose respects addition and scalar multiplication: $((A+B)^\top)_{ij} = (A+B)_{ji} = A_{ji} + B_{ji} = (A^\top + B^\top)_{ij}$, so $(A+B)^\top = A^\top + B^\top$, and similarly $(cA)^\top = cA^\top$. The zero matrix is symmetric. If $A, B \in \mathrm{Sym}_n$, then

$$(A+B)^\top = A^\top + B^\top = A + B \qquad \text{and} \qquad (cA)^\top = cA^\top = cA,$$

so $A + B \in \mathrm{Sym}_n$ and $cA \in \mathrm{Sym}_n$.

Dimension of $\mathcal{D}_n$. For $1 \le i, j \le n$, let e_{ij} denote the $n \times n$ matrix with a 1 in position (i, j) and 0s everywhere else. Every diagonal matrix can be written as

$$\mathrm{diag}(d_1, \ldots, d_n) = d_1 e_{11} + d_2 e_{22} + \cdots + d_n e_{nn},$$

so $e_{11}, \ldots, e_{nn}$ span $\mathcal{D}_n$. If $d_1 e_{11} + \cdots + d_n e_{nn} = 0$, then $\mathrm{diag}(d_1, \ldots, d_n)$ is the zero matrix, so $d_1 = \cdots = d_n = 0$. Hence $e_{11}, \ldots, e_{nn}$ are linearly independent, they form a basis of $\mathcal{D}_n$, and $\dim \mathcal{D}_n = n$. □

Exercise 6.66.

1. Show that $\mathcal{L}_n$ and Skew_n are subspaces of $\mathbb{R}^{n\times n}$.
2. Show that $\dim \mathcal{U}_n = \dim \mathcal{L}_n = \dim \mathrm{Sym}_n = \frac{n(n+1)}{2}$ and that $\dim \mathrm{Skew}_n = \frac{n(n-1)}{2}$. (Hint: use the matrices e_{ij}.)
3. Show that the product of two upper triangular matrices is upper triangular.
4. Show that every $n \times n$ matrix A can be written in exactly one way as $A = S + K$, where S is symmetric and K is skew-symmetric. (Hint: try $S = \frac{1}{2}(A + A^\top)$.)

6.4.2 Matrix groups

The next two sets of matrices are defined using multiplication rather than by conditions on the entries, and they behave very differently.

Definition 6.67 (General linear group)**.** The set of all invertible $n \times n$ matrices is denoted by

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in \mathbb{R}^{n\times n} : A \text{ is invertible}\}.$$

Example 6.68. The identity matrix I_n, all elementary matrices (Lemma 6.46), all diagonal matrices $\mathrm{diag}(d_1, \ldots, d_n)$ with every $d_i \neq 0$, and the matrices of Example 6.41 belong to $\mathrm{GL}(n, \mathbb{R})$ for the appropriate n. The zero matrix and the matrix of Example 6.39 do not.

Proposition 6.69. *The set $\mathrm{GL}(n, \mathbb{R})$ is not a subspace of $\mathbb{R}^{n\times n}$. However, it has the following properties:*

1. *$I_n \in \mathrm{GL}(n, \mathbb{R})$;*
2. *if $A, B \in \mathrm{GL}(n, \mathbb{R})$, then $AB \in \mathrm{GL}(n, \mathbb{R})$;*
3. *if $A \in \mathrm{GL}(n, \mathbb{R})$, then $A^{-1} \in \mathrm{GL}(n, \mathbb{R})$.*

Proof. The zero matrix is not invertible, since $0 \cdot B = 0 \neq I_n$ for every $n \times n$ matrix B. So $0 \notin \mathrm{GL}(n, \mathbb{R})$, and the first condition of Definition 5.1 fails. ($\mathrm{GL}(n, \mathbb{R})$ is not closed under addition either: I_n and $-I_n$ are invertible, but $I_n + (-I_n) = 0$ is not.)

For the three properties: $I_n I_n = I_n$, so I_n is invertible and is its own inverse. Properties 2 and 3 are parts 2 and 1 of Proposition 6.42. □

These three properties say that $\mathrm{GL}(n,\mathbb{R})$ is a group, a structure that appears throughout mathematics.

Definition 6.70 (Group)**.** A group is a set G with an operation that assigns to $a, b \in G$ an element $ab \in G$, such that:

1. (associativity) $(ab)c = a(bc)$ for all $a, b, c \in G$;
2. (identity) there is an element $e \in G$ with $ea = ae = a$ for all $a \in G$;
3. (inverses) for every $a \in G$, there is an element $a^{-1} \in G$ with $aa^{-1} = a^{-1}a = e$.

The group is abelian if moreover $ab = ba$ for all $a, b \in G$. A subgroup of G is a subset that contains e and is closed under the operation and under taking inverses; it is a group itself, with the same operation.

The identity is unique: if e' is another one, then $e = ee' = e'$. So is the inverse: if b and c are inverses of a, then $b = b(ac) = (ba)c = c$. When the operation is an addition, we write $a + b$, 0 and $-a$ instead of ab, e and a^{-1}.

- $\mathbb{Z}$, $\mathbb{R}$ and every vector space are abelian groups under addition (axioms 1 to 3 of Definition 2.1; by commutativity, $v + 0 = v$ and $v + (-v) = 0$ also hold in the other order). The natural numbers $\{0, 1, 2, \ldots\}$ are not a group under addition, since 1 has no inverse.
- The field axioms (Definition 1.4) say that $(\mathbb{F}, +)$ and $(\mathbb{F} \setminus \{0\}, \cdot)$ are abelian groups, connected by the distributive law. (The product of nonzero elements is nonzero: if $ab = 0$ with $a \neq 0$, then $b = a^{-1}ab = a^{-1} \cdot 0 = 0$.) The nonzero integers are not a group under multiplication, since 2 has no inverse.
- By the proposition above, $\mathrm{GL}(n,\mathbb{R})$ is a group under matrix multiplication, with the identity $e = I_n$; it is called the general linear group. For $n \geq 2$, it is not abelian (Remark 6.23).

A subgroup of $\mathrm{GL}(n,\mathbb{R})$, i.e., a set of invertible matrices that contains I_n and is closed under multiplication and under taking inverses, is called a matrix group. Important matrix groups are the orthogonal matrices $\mathrm{O}(n)$ (below), the matrices with determinant 1, which form the special linear group $\mathrm{SL}(n,\mathbb{R})$ (Subsection 6.5), and the rotations, which form the special orthogonal group $\mathrm{SO}(n)$ (Lecture 2). Groups and vector spaces are different kinds of structures: in a vector space we add vectors and multiply them by scalars, while in a matrix group we multiply its elements with each other, and their sums usually leave the group. The matrix groups above are also smooth curved surfaces in $\mathbb{R}^{n\times n}$ on which multiplication and inversion are smooth maps: they are Lie groups, the mathematical description of continuous symmetries such as rotations (see the appendix to Lecture 3).

Definition 6.71 (Orthogonal matrices)**.** An $n \times n$ matrix Q is called orthogonal if $Q^\top Q = I_n$. The set of all orthogonal $n \times n$ matrices is denoted by $\mathrm{O}(n)$.

By Corollary 6.48, every orthogonal matrix Q is invertible with $Q^{-1} = Q^\top$; in particular, also $QQ^\top = I_n$, and $\mathrm{O}(n) \subseteq \mathrm{GL}(n,\mathbb{R})$. So for orthogonal matrices, the inverse comes for free: we only need to transpose.

Example 6.72. For any angle θ, the rotation matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

rotates vectors in the plane counterclockwise by the angle θ. It is orthogonal, since

$$\begin{aligned} R_\theta^\top R_\theta &= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2\theta + \sin^2\theta & -\cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta + \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{pmatrix} = I_2. \end{aligned}$$

Hence $R_\theta^{-1} = R_\theta^\top = R_{-\theta}$: to undo a rotation by the angle θ, we rotate by $-\theta$.

Example 6.73. A permutation matrix is a square matrix in which each row and each column contains exactly one 1, and all other entries are 0. For example,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad P \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_3 \\ x_1 \end{pmatrix},$$

so P permutes the entries of a vector. One checks that $P^\top P = I_3$, so P is orthogonal.

Geometrically, orthogonal matrices are exactly the matrices that preserve lengths and angles, such as rotations, reflections and permutations of the coordinates; this is proved in Lecture 2.

Exercise 6.74. Show that $\mathrm{O}(n)$ is not a subspace of $\mathbb{R}^{n\times n}$, but that it contains I_n and is closed under multiplication and under taking inverses. In other words, $\mathrm{O}(n)$ is a matrix group, called the orthogonal group.

6.5 Trace and Determinant

In this subsection, we introduce two numbers that can be computed from a square matrix: the trace and the determinant. The trace is very easy to compute, and it depends linearly on the matrix. The determinant is harder to compute, but it tells us whether a matrix is invertible, and it measures how the linear transformation T_A changes areas and volumes.

6.5.1 The trace

Definition 6.75 (Trace)**.** The trace of an $n \times n$ matrix A is the sum of its diagonal entries:

$$\mathrm{tr}(A) = A_{11} + A_{22} + \cdots + A_{nn} = \sum_{i=1}^{n} A_{ii}.$$

Example 6.76. For the upper triangular matrix from Subsection 6.4,

$$\mathrm{tr} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} = 1 + 4 + 6 = 11.$$

Moreover, $\mathrm{tr}(I_n) = 1 + \cdots + 1 = n$, and the trace of every skew-symmetric matrix is 0, since its diagonal entries are 0.

Theorem 6.77. *Let A and B be $n \times n$ matrices and $c \in \mathbb{R}$. Then:*

1. $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$ *and* $\mathrm{tr}(cA) = c\,\mathrm{tr}(A)$*; in other words,* $\mathrm{tr} : \mathbb{R}^{n\times n} \to \mathbb{R}$ *is a linear transformation;*

2. $\operatorname{tr}(A^\top) = \operatorname{tr}(A)$;

3. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. *More generally, this holds for every $m \times n$ matrix A and every $n \times m$ matrix B.*

Proof. Parts 1 and 2 follow directly from the definitions, since the diagonal entries of $A + B$, cA and $A^\top$ are $A_{ii} + B_{ii}$, cA_{ii} and A_{ii}, respectively. For part 3, let A be an $m \times n$ matrix and B an $n \times m$ matrix. Then

$$\begin{aligned}\operatorname{tr}(AB) &= \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{k=1}^{n} A_{ik} B_{ki} \\ &= \sum_{k=1}^{n} \sum_{i=1}^{m} B_{ki} A_{ik} = \sum_{k=1}^{n} (BA)_{kk} = \operatorname{tr}(BA).\end{aligned}$$

□

Remark 6.78. Part 3 is surprising, since in general $AB \neq BA$. For example, if P is an invertible $n \times n$ matrix, then

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}\left((P^{-1}A)P\right) = \operatorname{tr}\left(P(P^{-1}A)\right) = \operatorname{tr}(A).$$

However, the trace is not multiplicative: for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

we have $AB = 0$, so $\operatorname{tr}(AB) = 0$, while $\operatorname{tr}(A)\operatorname{tr}(B) = 1 \cdot 1 = 1$.

Example 6.79. For any $m \times n$ matrix A, the diagonal entries of the $n \times n$ matrix $A^\top A$ are

$$(A^\top A)_{jj} = \sum_{i=1}^{m} (A^\top)_{ji} A_{ij} = \sum_{i=1}^{m} A_{ij}^2.$$

Hence

$$\operatorname{tr}(A^\top A) = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2$$

is the sum of the squares of all entries of A. In particular, $\operatorname{tr}(A^\top A) \geq 0$, with equality only for $A = 0$. The number $\|A\|_F = \sqrt{\operatorname{tr}(A^\top A)}$ is called the Frobenius norm of A. It is often used in machine learning to measure the size of a matrix, for example of the weight matrix of a layer in a neural network.

Corollary 6.80. *The set $\{A \in \mathbb{R}^{n \times n} : \operatorname{tr}(A) = 0\}$ of matrices with trace zero is a subspace of $\mathbb{R}^{n \times n}$.*

Proof. This set is the kernel of the linear transformation $\operatorname{tr} : \mathbb{R}^{n \times n} \to \mathbb{R}$, so it is a subspace by Theorem 5.5. □

Exercise 6.81. 1. Show that the subspace of $n \times n$ matrices with trace zero has dimension $n^2 - 1$. (Hint: use the matrices e_{ij} from the proof of Theorem 6.65.)

2. Show that there are no $n \times n$ matrices A and B with $AB - BA = I_n$. (Hint: take the trace of both sides.)

6.5.2 The determinant

The determinant of a square matrix is defined recursively: the determinant of an $n \times n$ matrix is expressed in terms of determinants of $(n-1) \times (n-1)$ matrices.

Definition 6.82 (Determinant)**.** Let A be an $n \times n$ matrix. For $1 \le i, j \le n$, let $A^{(ij)}$ denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting its i-th row and its j-th column. The determinant $\det(A)$ of A is the number defined as follows:

- if $n = 1$, then $\det(A) = A_{11}$;
- if $n \ge 2$, then
$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(A^{(1j)}).$$

Written out, $\det(A) = A_{11}\det(A^{(11)}) - A_{12}\det(A^{(12)}) + A_{13}\det(A^{(13)}) - \cdots$, with alternating signs. This formula is called the cofactor expansion along the first row, and the numbers $C_{ij} = (-1)^{i+j}\det(A^{(ij)})$ are called the cofactors of A. The determinant is also denoted by vertical bars, e.g., $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Example 6.83 (2×2 and 3×3 determinants)**.** For a 2×2 matrix, deleting the first row and the first or the second column leaves the 1×1 matrices (d) and (c), so

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For a 3×3 matrix,

$$\begin{aligned} \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} &= a \det \begin{pmatrix} e & f \\ h & k \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & k \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} \\ &= a(ek - fh) - b(dk - fg) + c(dh - eg). \end{aligned}$$

For example, $\det \begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix} = 2 \cdot (-1) - 3 \cdot 4 = -14$, and

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix} = 1 \cdot (5 \cdot 10 - 6 \cdot 8) - 2 \cdot (4 \cdot 10 - 6 \cdot 7) + 3 \cdot (4 \cdot 8 - 5 \cdot 7) = 2 + 4 - 9 = -3.$$

The following theorem collects the basic properties of the determinant. We prove part 3 and the expansion along rows right after the theorem, directly from the definition. The expansion along columns and part 2 are proved after the product rule for determinants (Theorem 6.92), which is the main tool for them.

Theorem 6.84. *Let A be an $n \times n$ matrix.*

1. *(Expansion along any row or column.) For every row index i and every column index j,*
$$\det(A) = \sum_{l=1}^{n} (-1)^{i+l} A_{il} \det(A^{(il)}) = \sum_{l=1}^{n} (-1)^{l+j} A_{lj} \det(A^{(lj)}).$$

2. *(Transpose.)* $\det(A^\top) = \det(A)$.

3. *(Row operations.)*

   - *Swapping two rows of A multiplies the determinant by -1.*
   - *Multiplying a row of A by a scalar c multiplies the determinant by c.*
   - *Adding a multiple of one row to another row does not change the determinant.*

By part 2, the same rules hold for the corresponding operations on columns. The signs $(-1)^{i+j}$ in part 1 form a checkerboard pattern, starting with + in the top left corner. All proofs below use induction on n and the definition, i.e., the expansion along the first row.

Lemma 6.85. *Let A be an $n \times n$ matrix with rows $\mathbf{r}_1, \dots, \mathbf{r}_n$.*

1. *(Linearity in each row.) Suppose that $\mathbf{r}_i = a\mathbf{u} + b\mathbf{w}$ for row vectors $\mathbf{u}, \mathbf{w} \in \mathbb{R}^{1\times n}$ and scalars a, b. Let A_u and A_w be the matrices obtained from A by replacing its i-th row by $\mathbf{u}$ and by $\mathbf{w}$, respectively. Then $\det(A) = a\det(A_u) + b\det(A_w)$.*

2. *(Swapping neighboring rows.) If $n \geq 2$, swapping two neighboring rows $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$ of A multiplies the determinant by -1.*

Proof. (1) For $n = 1$, $\det(a\mathbf{u} + b\mathbf{w}) = a\mathbf{u} + b\mathbf{w}$. Let $n \geq 2$. If $i = 1$, then the matrices $A^{(1j)}$ do not contain the first row, so they are the same for A, A_u and A_w, and

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j}(au_j + bw_j)\det(A^{(1j)}) = a\det(A_u) + b\det(A_w).$$

If $i \geq 2$, then the entries A_{1j} are the same for the three matrices, and row $i-1$ of $A^{(1j)}$ is a times $\mathbf{u}$ plus b times $\mathbf{w}$, each with its j-th entry deleted. By induction on n, $\det(A^{(1j)}) = a\det(A_u^{(1j)}) + b\det(A_w^{(1j)})$; multiply by $(-1)^{1+j}A_{1j}$ and add over j.

(2) For $n = 2$, swapping the rows of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ gives the determinant $cb - da = -(ad - bc)$. Let $n \geq 3$. If $i \geq 2$, then the swap exchanges two neighboring rows of each matrix $A^{(1j)}$ and does not change the first row of A; by induction, each $\det(A^{(1j)})$ changes its sign, and hence so does $\det(A)$. It remains to swap the first two rows. Let M_{jk} be the matrix obtained from A by deleting the rows 1 and 2 and the columns j and k, so that $M_{jk} = M_{kj}$. In $A^{(1j)}$, the first row consists of the entries A_{2k} with $k \neq j$; the entry A_{2k} stands in column k if $k < j$ and in column $k-1$ if $k > j$, and deleting its row and column leaves M_{jk}. Expanding each $\det(A^{(1j)})$ along its first row therefore gives

$$\det(A) = \sum_{j=1}^{n} \sum_{k\neq j} \varepsilon_{jk} A_{1j} A_{2k} \det(M_{jk}), \qquad \varepsilon_{jk} = \begin{cases} (-1)^{1+j}(-1)^{1+k} = (-1)^{j+k} & \text{if } k < j, \\ (-1)^{1+j}(-1)^{k} = -(-1)^{j+k} & \text{if } k > j. \end{cases}$$

In particular, $\varepsilon_{kj} = -\varepsilon_{jk}$. The matrix B with the first two rows swapped has $B_{1j} = A_{2j}$ and $B_{2k} = A_{1k}$, and the same matrices M_{jk}, so, exchanging the names of the summation indices j and k,

$$\det(B) = \sum_{j=1}^{n} \sum_{k\neq j} \varepsilon_{jk} A_{2j} A_{1k} \det(M_{jk}) = \sum_{k=1}^{n} \sum_{j\neq k} \varepsilon_{kj} A_{1j} A_{2k} \det(M_{jk}) = -\det(A). \qquad \square$$

Proof of Theorem 6.84, part 3, and of the expansion along rows. Swapping the rows $i < k$ can be done by swaps of neighboring rows: move row i down to position k by $k-i$ swaps, and then move the old row k, which is now in position $k-1$, up to position i by $k-i-1$ swaps. The total number $2(k-i)-1$ is odd, so by Lemma 6.85, the determinant changes its sign. Multiplying a row by c multiplies the determinant by c, by Lemma 6.85 with $a = c$ and $b = 0$.

If a matrix has two equal rows, then swapping them does not change the matrix, but changes the sign of its determinant, so the determinant is 0. Now let A' be obtained from A by adding c times row k to row $i \neq k$. By Lemma 6.85, $\det(A') = \det(A) + c\det(A'')$, where A'' is obtained from A by replacing row i by row k. Since A'' has two equal rows, $\det(A'') = 0$, and $\det(A') = \det(A)$.

Finally, let $1 \leq i \leq n$. Moving row i to the top by $i-1$ swaps of neighboring rows gives the matrix B with the rows $\mathbf{r}_i, \mathbf{r}_1, \ldots, \mathbf{r}_{i-1}, \mathbf{r}_{i+1}, \ldots, \mathbf{r}_n$, and $\det(B) = (-1)^{i-1}\det(A)$. The other rows keep their order, so deleting the first row and the l-th column of B gives $A^{(il)}$. By the definition of $\det(B)$,

$$\det(A) = (-1)^{i-1}\sum_{l=1}^{n}(-1)^{1+l}A_{il}\det(A^{(il)}) = \sum_{l=1}^{n}(-1)^{i+l}A_{il}\det(A^{(il)}). \qquad \square$$

Corollary 6.86. *Let A be an $n \times n$ matrix.*

1. *If A has a row or a column of zeros, then $\det(A) = 0$.*
2. *If A has two equal rows, then $\det(A) = 0$.*
3. *If A is upper or lower triangular, then $\det(A) = A_{11}A_{22}\cdots A_{nn}$ is the product of its diagonal entries. In particular, $\det(I_n) = 1$.*
4. *$\det(cA) = c^n\det(A)$ for every $c \in \mathbb{R}$.*

Proof. (1) For a row of zeros, expand along this row: every term of the sum contains a factor 0. For a column of zeros, say column j, we use induction on n; for $n = 1$, $A = (0)$. In the expansion along the first row, $A_{1j} = 0$, and for $l \neq j$, the matrix $A^{(1l)}$ has a column of zeros, so $\det(A^{(1l)}) = 0$ by induction.

(2) Swapping the two equal rows does not change A, but by Theorem 6.84 it multiplies the determinant by -1. So $\det(A) = -\det(A)$, i.e., $\det(A) = 0$.

(3) Let A be lower triangular. The only possibly nonzero entry of its first row is A_{11}, so the definition gives $\det(A) = A_{11}\det(A^{(11)})$. The matrix $A^{(11)}$ is again lower triangular, with diagonal entries $A_{22}, \ldots, A_{nn}$, so repeating this argument (formally, by induction on n) gives $\det(A) = A_{11}A_{22}\cdots A_{nn}$. If A is upper triangular, then the only possibly nonzero entry of its last row is A_{nn}, and the expansion along the last row gives $\det(A) = A_{nn}\det(A^{(nn)})$, where $A^{(nn)}$ is upper triangular with diagonal entries $A_{11}, \ldots, A_{n-1,n-1}$; again, induction gives the claim.

(4) The matrix cA is obtained from A by multiplying each of its n rows by c. $\square$

Theorem 6.84 and Corollary 6.86 give an efficient way to compute determinants: bring the matrix into upper triangular form by row operations, keeping track of how each operation changes the determinant, and then multiply the diagonal entries. For large n, this is much faster than the cofactor expansion, which involves about $n!$ terms.

Example 6.87. Let us compute the determinant of the 3×3 matrix above again, now with row operations. Subtracting 4 times the first row from the second row, and 7 times the first row from

the third row, does not change the determinant:

$$\det\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 10\end{pmatrix} = \det\begin{pmatrix}1 & 2 & 3\\ 0 & -3 & -6\\ 0 & -6 & -11\end{pmatrix}.$$

Subtracting 2 times the second row from the third row does not change it either, and we obtain an upper triangular matrix:

$$\det\begin{pmatrix}1 & 2 & 3\\ 0 & -3 & -6\\ 0 & 0 & 1\end{pmatrix} = 1 \cdot (-3) \cdot 1 = -3,$$

in agreement with the cofactor expansion. The matrix F of Example 6.31 differs from this matrix only in its bottom right entry, and the same row operations turn it into a matrix with a row of zeros (see Example 6.31). Hence $\det(F) = 0$.

Remark 6.88 (Geometric meaning). Let A be a 2×2 matrix. The linear transformation T_A maps the unit square, with sides $\mathbf{e}_1$ and $\mathbf{e}_2$, to the parallelogram with sides $A\mathbf{e}_1$ and $A\mathbf{e}_2$, which are the columns of A. The area of this parallelogram is $|\det(A)|$ (see Figure 4). More generally, T_A multiplies the area of every region by $|\det(A)|$, and for 3×3 matrices, T_A multiplies volumes by $|\det(A)|$. The sign of $\det(A)$ tells us whether T_A preserves the orientation (positive sign) or reverses it, like a mirror (negative sign). We do not prove these facts here.

This picture also makes the rules of Theorem 6.84 for column operations plausible. Adding a multiple of one column to the other shears the parallelogram, but does not change its base or its height, and hence not its area (see Figure 5). Multiplying a column by c stretches the parallelogram by the factor $|c|$ in one direction, which multiplies its area by $|c|$. Swapping the two columns reverses the orientation, which changes the sign of the determinant.

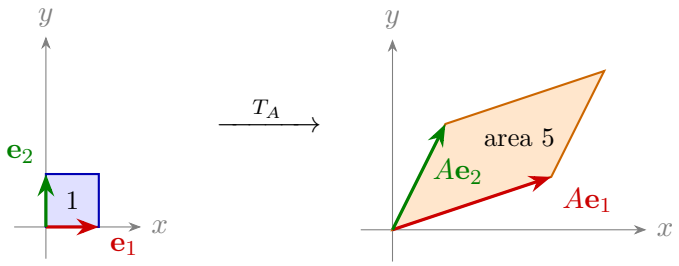


Figure 4: The matrix A with columns $(3, 1)$ and $(1, 2)$ maps the unit square (left, area 1) to the parallelogram spanned by its columns $A\mathbf{e}_1$ and $A\mathbf{e}_2$ (right). The area of this parallelogram is $|\det(A)| = |3 \cdot 2 - 1 \cdot 1| = 5$.

Example 6.89.

- $\det(\mathrm{diag}(2, 3)) = 6$: stretching by a factor 2 in the x-direction and by a factor 3 in the y-direction multiplies areas by 6.
- $\det(R_\theta) = \cos^2\theta + \sin^2\theta = 1$: rotations preserve areas and orientation.
- $\det\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = -1$: the reflection in the x-axis preserves areas, but reverses orientation.

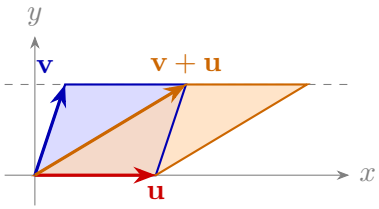


Figure 5: The parallelograms spanned by $\mathbf{u} = (2, 0)$ and $\mathbf{v} = (\frac{1}{2}, \frac{3}{2})$ (blue) and by $\mathbf{u}$ and $\mathbf{v} + \mathbf{u}$ (orange) have the same base $\mathbf{u}$ and the same height (dashed line), hence the same area 3. Correspondingly, adding the first column to the second column does not change the determinant: both matrices with these columns have determinant $2 \cdot \frac{3}{2} = 3$.

- $\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 0$: this singular matrix maps the whole plane onto the line spanned by $(1, 2)$, which has area 0 (see Figure 3).

Compare also with the gallery in Figure 2. The stretch (a) has determinant $\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 2$ and doubles areas. The rotation (b) and the shear (d) have determinants $\det \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = 1$ and $\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1$: they preserve areas, even though the shear changes the shape of the square. The reflection (c) has determinant -1, which is why the image of the letter F is mirrored.

6.5.3 Determinants, invertibility and products

We now connect determinants with the inverse of a matrix, using the elementary matrices of Subsection 6.3.

Lemma 6.90. *Let E be an $n \times n$ elementary matrix. Then $\det(E) = -1$ if E swaps two rows, $\det(E) = c$ if E multiplies a row by $c \neq 0$, and $\det(E) = 1$ if E adds a multiple of one row to another row. In all cases, $\det(E) \neq 0$, and*

$$\det(EA) = \det(E)\det(A) \qquad \text{for every } n \times n \text{ matrix } A.$$

Proof. The matrix E is obtained from I_n by the row operation, and $\det(I_n) = 1$, so the values of $\det(E)$ follow from part 3 of Theorem 6.84. By Lemma 6.45, EA is obtained from A by the same row operation, so $\det(EA)$ is $\det(A)$ multiplied by -1, by c, or by 1, respectively. In each case, this is $\det(E)\det(A)$. □

Theorem 6.91. *An $n \times n$ matrix A is invertible if and only if $\det(A) \neq 0$.*

Proof. Let $E_1, \ldots, E_k$ be the elementary matrices of row operations that bring A to $R = \mathrm{RREF}(A)$, so that $R = E_k \cdots E_1 A$. Applying Lemma 6.90 k times gives

$$\det(R) = \det(E_k) \cdots \det(E_1)\det(A),$$

where all the factors $\det(E_i)$ are nonzero. So $\det(A) \neq 0$ if and only if $\det(R) \neq 0$. If A is invertible, then $R = I_n$ by Theorem 6.47, and $\det(R) = 1 \neq 0$. If A is not invertible, then $R \neq I_n$, so R has a row of zeros (as we saw after Theorem 6.49), and $\det(R) = 0$ by Corollary 6.86. □

Theorem 6.92 (Product rule)**.** *For all $n \times n$ matrices A and B, we have* $\det(AB) = \det(A)\det(B)$.

Proof. Suppose first that A is not invertible, so that $\det(A) = 0$ by Theorem 6.91. Then AB is not invertible either: otherwise, $X = B(AB)^{-1}$ would satisfy $AX = I_n$, and by Corollary 6.48, X would be invertible with $X^{-1} = A$, so A would be invertible. Hence $\det(AB) = 0 = \det(A)\det(B)$.

Now suppose that A is invertible. By the proof of Theorem 6.47, $A = (E_k \cdots E_1)^{-1} = E_1^{-1} \cdots E_k^{-1}$ for some elementary matrices $E_1, \ldots, E_k$, and each $F_i = E_i^{-1}$ is again an elementary matrix by Lemma 6.46. Applying Lemma 6.90 k times, we get

$$\det(AB) = \det(F_1 F_2 \cdots F_k B) = \det(F_1)\det(F_2)\cdots\det(F_k)\det(B).$$

Taking $B = I_n$ gives $\det(A) = \det(F_1)\cdots\det(F_k)$, and hence $\det(AB) = \det(A)\det(B)$. □

Proof of Theorem 6.84, parts 1 and 2. We already proved part 3 and the expansion along rows, and the proofs of Corollary 6.86, Lemma 6.90 and Theorems 6.91 and 6.92 used only these. For part 2, note that A is invertible if and only if $A^\top$ is (Proposition 6.42, applied to A and to $A^\top$). If A is not invertible, then $\det(A^\top) = 0 = \det(A)$ by Theorem 6.91. If A is invertible, then, as in the proof of Theorem 6.92, $A = F_1 \cdots F_k$ with elementary matrices F_i, so $A^\top = F_k^\top \cdots F_1^\top$. The transpose of an elementary matrix is an elementary matrix with the same determinant: the matrices that swap two rows or multiply a row by c are symmetric, and the transpose of the matrix that adds c times row j to row i is the matrix that adds c times row i to row j, with determinant 1. By the product rule,

$$\det(A^\top) = \det(F_k^\top)\cdots\det(F_1^\top) = \det(F_1)\cdots\det(F_k) = \det(A).$$

For the expansion along column j, expand $\det(A^\top)$ along its row j. Since $(A^\top)_{jl} = A_{lj}$, and $(A^\top)^{(jl)} = (A^{(lj)})^\top$ has the same determinant as $A^{(lj)}$ by part 2 (for $(n-1)\times(n-1)$ matrices),

$$\det(A) = \det(A^\top) = \sum_{l=1}^{n} (-1)^{j+l} A_{lj} \det(A^{(lj)}).$$ □

Corollary 6.93. *Let A be an $n \times n$ matrix.*

1. *If A is invertible, then* $\det(A^{-1}) = 1/\det(A)$.
2. *If P is an invertible $n \times n$ matrix, then* $\det(P^{-1}AP) = \det(A)$.
3. *If A is orthogonal, then* $\det(A) = 1$ *or* $\det(A) = -1$.

Proof. (1) $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1$.
(2) $\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A)$, by part 1.
(3) $1 = \det(I_n) = \det(A^\top A) = \det(A^\top)\det(A) = \det(A)^2$. □

Remark 6.94. Unlike the trace, the determinant is not linear. For example, $\det(I_2 + I_2) = \det(2I_2) = 4$, while $\det(I_2) + \det(I_2) = 2$.

Example 6.95 (Inverse of a 2×2 matrix)**.** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det(A) = ad - bc \neq 0$. Then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Indeed,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = (ad-bc)I_2,$$

and by Corollary 6.48, checking one product is enough. For example, for the matrix of Example 6.41, with determinant -14,

$$\begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix}^{-1} = \frac{1}{-14} \begin{pmatrix} -1 & -3 \\ -4 & 2 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix},$$

as we found before.

The matrix $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ is the transpose of the matrix of the cofactors $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$ and $C_{22} = a$. The same holds for all n.

Proposition 6.96 (Cofactor formula for the inverse)**.** *Let A be an $n \times n$ matrix with $n \geq 2$, and let C be the matrix of its cofactors $C_{ij} = (-1)^{i+j} \det(A^{(ij)})$. Then $AC^\top = \det(A)\, I_n$. In particular, if $\det(A) \neq 0$, then*

$$A^{-1} = \frac{1}{\det(A)}\, C^\top.$$

Proof. The (i,k) entry of $AC^\top$ is $\sum_{l=1}^n A_{il} C_{kl} = \sum_{l=1}^n (-1)^{k+l} A_{il} \det(A^{(kl)})$. For $i = k$, this is the expansion of $\det(A)$ along row i (Theorem 6.84). For $i \neq k$, let A' be obtained from A by replacing row k by row i. Deleting row k gives the same matrices for A and A', so $A'^{(kl)} = A^{(kl)}$, and $A'_{kl} = A_{il}$. Hence the sum is the expansion of $\det(A')$ along row k. Since A' has two equal rows, $\det(A') = 0$ by Corollary 6.86. So $AC^\top = \det(A)\, I_n$. If $\det(A) \neq 0$, then $A \cdot \frac{1}{\det(A)} C^\top = I_n$, and by Corollary 6.48, $\frac{1}{\det(A)} C^\top$ is the inverse of A. □

The cofactor formula is useful in theory; for example, it shows that the entries of A^{-1} are rational functions of the entries of A. For computations, the Gauss–Jordan method is much more efficient.

Theorem 6.91 says that $\mathrm{GL}(n,\mathbb{R}) = \{A \in \mathbb{R}^{n\times n} : \det(A) \neq 0\}$. Determinants also give us another matrix group.

Definition 6.97 (Special linear group)**.** The special linear group is the set

$$\mathrm{SL}(n,\mathbb{R}) = \{A \in \mathbb{R}^{n\times n} : \det(A) = 1\}.$$

Proposition 6.98. *The set $\mathrm{SL}(n,\mathbb{R})$ is contained in $\mathrm{GL}(n,\mathbb{R})$ and is not a subspace of $\mathbb{R}^{n\times n}$. However, it contains I_n and is closed under multiplication and under taking inverses, so it is a matrix group.*

Proof. Every matrix with determinant 1 is invertible by Theorem 6.91. The zero matrix has determinant 0, so it is not in $\mathrm{SL}(n,\mathbb{R})$, and $\mathrm{SL}(n,\mathbb{R})$ is not a subspace. We have $\det(I_n) = 1$. If $\det(A) = \det(B) = 1$, then $\det(AB) = 1 \cdot 1 = 1$ by Theorem 6.92, and $\det(A^{-1}) = 1/1 = 1$ by Corollary 6.93. □

Example 6.99. All rotation matrices R_θ belong to $\mathrm{SL}(2,\mathbb{R})$, and so does $\mathrm{diag}(2, \frac{1}{2})$, which stretches the plane by a factor 2 in one direction and shrinks it by a factor 2 in the other. By the geometric meaning of the determinant, the matrices in $\mathrm{SL}(2,\mathbb{R})$ are exactly the 2×2 matrices whose linear transformations preserve areas and orientation.

6.6 Block Matrices, the LU Decomposition and the Woodbury Identity

Large matrices often have a natural structure: the features of a data set fall into groups, a covariance matrix describes several random vectors at once, and a matrix changes by a small update. In this subsection, we compute with matrices in blocks, and we use blocks to describe Gaussian elimination as a factorization of a matrix and to invert low-rank updates.

6.6.1 Block matrices and the Schur complement

A matrix can be split into blocks by horizontal and vertical lines, for example

$$M = \left(\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 4 & 6 \\ \hline 7 & 8 & 9 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad C = \begin{pmatrix} 7 & 8 \end{pmatrix}, \quad D = (9).$$

Block matrices are multiplied like 2×2 matrices, as long as the sizes of the blocks fit.

Proposition 6.100 (Block multiplication)**.** *Let* $M = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ *and* $N = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$*, where* A_{ik} *is an* $m_i \times n_k$ *matrix and* B_{kj} *is an* $n_k \times p_j$ *matrix. Then*

$$MN = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

Proof. Each entry of MN is a sum $\sum_l M_{rl}N_{ls}$ over $l = 1, \ldots, n_1 + n_2$. The terms with $l \le n_1$ form the corresponding entry of $A_{i1}B_{1j}$, and the terms with $l > n_1$ form the entry of $A_{i2}B_{2j}$, where i and j are the block row of r and the block column of s. □

Lemma 6.101. *Let* A *be a* $k \times k$ *matrix and* D *an* $m \times m$ *matrix. Then*

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det(A)\det(D) = \det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$$

for all matrices B *and* C *of the appropriate sizes.*

Proof. Let $M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$. If A is not invertible, then $A\mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \neq \mathbf{0}$, and M maps the nonzero vector $\begin{pmatrix} \mathbf{x} \\ \mathbf{0} \end{pmatrix}$ to $\begin{pmatrix} A\mathbf{x} \\ \mathbf{0} \end{pmatrix} = \mathbf{0}$. So M is not invertible either, and $\det(M) = 0 = \det(A)\det(D)$. If A is invertible, then by Proposition 6.100,

$$M = \begin{pmatrix} A & 0 \\ 0 & I_m \end{pmatrix} \begin{pmatrix} I_k & A^{-1}B \\ 0 & D \end{pmatrix}.$$

Expanding the determinant of the first factor m times along the last row (each time, the only nonzero entry of the last row is a diagonal entry 1) gives $\det(A)$, and expanding the determinant of the second factor k times along the first column gives $\det(D)$. The product rule gives $\det(M) = \det(A)\det(D)$. The second equation follows by taking transposes. □

Definition 6.102 (Schur complement)**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be a square matrix with square blocks A and D, where A is invertible. The Schur complement of A in M is the matrix $S = D - CA^{-1}B$.

Theorem 6.103. *In the situation of Definition 6.102,*

$$M = \begin{pmatrix} I & 0 \\ CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & S \end{pmatrix}, \qquad \text{so} \qquad \det(M) = \det(A)\det(S).$$

Hence M is invertible if and only if S is invertible, and in this case

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}.$$

Proof. By Proposition 6.100, the product of the two factors is $\begin{pmatrix} A & B \\ CA^{-1}A & CA^{-1}B + S \end{pmatrix} = M$. By Lemma 6.101, the first factor has determinant 1 and the second one $\det(A)\det(S)$. Since $\det(A) \neq 0$, M is invertible if and only if $\det(S) \neq 0$. If S is invertible, block multiplication shows that the inverses of the two factors are

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} A^{-1} & -A^{-1}BS^{-1} \\ 0 & S^{-1} \end{pmatrix},$$

and multiplying them in the opposite order, $M^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}BS^{-1} \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$, gives the formula. □

The first factor in Theorem 6.103 is a block version of an elementary matrix: multiplying M by its inverse from the left subtracts CA^{-1} times the first block row from the second one, which eliminates C, just as row reduction eliminates the entries below a pivot. The Schur complement is what remains in the lower right corner.

Example 6.104. For the matrix M above, $A^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$, so $A^{-1}B = \frac{1}{-2}\begin{pmatrix} 8 \\ -9 \end{pmatrix} = \begin{pmatrix} -4 \\ 4.5 \end{pmatrix}$, and the Schur complement is $S = 9 - (7,8)\begin{pmatrix} -4 \\ 4.5 \end{pmatrix} = 9 - 8 = 1$. Hence $\det(M) = \det(A) \cdot 1 = -2$, and the lower right entry of M^{-1} is $S^{-1} = 1$.

In Lecture 4, the Schur complement appears as the covariance matrix of a Gaussian random vector after we condition on some of its entries.

6.6.2 The LU decomposition

Gaussian elimination brings a matrix into upper triangular form by subtracting multiples of rows from the rows below them. We now show that this writes the matrix as a product of a lower and an upper triangular matrix. A lower triangular matrix is called unit lower triangular if all its diagonal entries are 1, and a permutation matrix is a matrix obtained from I_n by reordering its rows; then PA is A with its rows reordered in the same way.

Lemma 6.105. *Products of (unit) lower triangular $n \times n$ matrices are (unit) lower triangular, and so are the inverses of invertible ones. The same holds for upper triangular matrices.*

Proof. Let L and L' be lower triangular. In $(LL')_{ij} = \sum_k L_{ik}L'_{kj}$, only the terms with $j \leq k \leq i$ can be nonzero. So $(LL')_{ij} = 0$ for $i < j$, and $(LL')_{ii} = L_{ii}L'_{ii}$, which is 1 if L and L' are unit lower

triangular. If L is invertible, the j-th column $\mathbf{x}$ of L^{-1} solves $L\mathbf{x} = \mathbf{e}_j$. The first $j-1$ equations $L_{11}x_1 = 0$, $L_{21}x_1 + L_{22}x_2 = 0$, ..., only involve $x_1, \dots, x_{j-1}$, and since the diagonal entries L_{ii} are nonzero ($\det(L) = L_{11} \cdots L_{nn} \neq 0$), they give $x_1 = \cdots = x_{j-1} = 0$; the j-th equation then gives $x_j = \frac{1}{L_{jj}}$. So L^{-1} is lower triangular, with diagonal entries $\frac{1}{L_{jj}}$. Upper triangular matrices are the transposes of lower triangular ones. □

Theorem 6.106 (LU decomposition)**.** *Let A be an $n \times n$ matrix.*

1. *If the upper left $k \times k$ submatrices of A are invertible for $k = 1, \dots, n-1$, then $A = LU$ with a unit lower triangular matrix L and an upper triangular matrix U. If A is invertible, then L and U are unique.*
2. *If A is invertible, then $PA = LU$ for a permutation matrix P, a unit lower triangular matrix L and an upper triangular matrix U.*

Proof. (1) We use induction on n; for $n = 1$, $A = (1)(A_{11})$. Let $n \geq 2$, and write

$$A = \begin{pmatrix} a & \mathbf{r}^\top \\ \mathbf{c} & A' \end{pmatrix}$$

with a number $a = A_{11}$, which is nonzero by assumption, vectors $\mathbf{c}, \mathbf{r} \in \mathbb{R}^{n-1}$ and an $(n-1)\times(n-1)$ matrix A'. By Theorem 6.103 with the 1×1 block A_{11},

$$A = \begin{pmatrix} 1 & 0 \\ \frac{1}{a}\mathbf{c} & I \end{pmatrix} \begin{pmatrix} a & \mathbf{r}^\top \\ 0 & S \end{pmatrix}, \qquad S = A' - \frac{1}{a}\mathbf{c}\mathbf{r}^\top.$$

The upper left $(k+1) \times (k+1)$ submatrix of A has the same form, with $\mathbf{c}$, $\mathbf{r}$ and A' cut down to their first k entries or $k \times k$ submatrix, and its Schur complement is the upper left $k \times k$ submatrix S_k of S. So $a \det(S_k)$ is the determinant of an upper left submatrix of A, which is nonzero for $k \leq n-2$. By induction, $S = L'U'$ with L' unit lower triangular and U' upper triangular, and

$$A = \begin{pmatrix} 1 & 0 \\ \frac{1}{a}\mathbf{c} & L' \end{pmatrix} \begin{pmatrix} a & \mathbf{r}^\top \\ 0 & U' \end{pmatrix},$$

since the product has the lower right block $\frac{1}{a}\mathbf{c}\mathbf{r}^\top + L'U' = \frac{1}{a}\mathbf{c}\mathbf{r}^\top + S = A'$. For uniqueness, let $L_1U_1 = L_2U_2$ with A invertible. Then U_1 and U_2 are invertible (as $\det(U_i) = \det(A) \neq 0$), and $L_2^{-1}L_1 = U_2U_1^{-1}$. By Lemma 6.105, the left side is unit lower triangular and the right side is upper triangular, so both are I_n, and $L_1 = L_2$, $U_1 = U_2$.

(2) Again we use induction on n. Since A is invertible, its first column is not zero. Let P_1 be the permutation matrix that swaps the first row with a row whose first entry is nonzero (or $P_1 = I_n$). Then $P_1A = \begin{pmatrix} a & \mathbf{r}^\top \\ \mathbf{c} & A' \end{pmatrix}$ with $a \neq 0$, and its Schur complement $S = A' - \frac{1}{a}\mathbf{c}\mathbf{r}^\top$ is invertible, since $\pm\det(A) = \det(P_1A) = a\det(S)$. By induction, $P'S = L'U'$, and with the permutation matrix $P = \begin{pmatrix} 1 & 0 \\ 0 & P' \end{pmatrix} P_1$,

$$PA = \begin{pmatrix} a & \mathbf{r}^\top \\ P'\mathbf{c} & P'A' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{a}P'\mathbf{c} & L' \end{pmatrix} \begin{pmatrix} a & \mathbf{r}^\top \\ 0 & U' \end{pmatrix},$$

since $\frac{1}{a}P'cr^\top + L'U' = P'\left(\frac{1}{a}cr^\top + S\right) = P'A'$. □

The proof is Gaussian elimination: the first column of L below the diagonal contains the multipliers $\frac{c_i}{a}$ of the first row that are subtracted from the other rows, the Schur complement S is the rest of the matrix after this step, and the elimination continues with S. So L stores the multipliers, and U is the upper triangular matrix at the end of the elimination. Row swaps are needed when a pivot is 0; in practice, one swaps the row with the largest pivot candidate into place (partial pivoting), which keeps the multipliers at most 1 in absolute value and limits rounding errors.

Example 6.107. Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$. Subtracting 2 times and 4 times the first row from the second and third rows gives the rows $(0,1,1)$ and $(0,3,5)$, and subtracting 3 times the new second row from the third row gives $(0,0,2)$. Hence

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = LU,$$

and $\det(A) = \det(L)\det(U) = 2 \cdot 1 \cdot 2 = 4$. To solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (5,13,31)^\top$, we solve $Ly = \mathbf{b}$ from the top (forward substitution): $y_1 = 5$, $y_2 = 13 - 2 \cdot 5 = 3$, $y_3 = 31 - 4 \cdot 5 - 3 \cdot 3 = 2$; and then $U\mathbf{x} = y$ from the bottom (back substitution): $x_3 = 1$, $x_2 = 3 - 1 = 2$, $x_1 = \frac{1}{2}(5 - 2 - 1) = 1$.

Computing the LU decomposition of an $n \times n$ matrix takes about $\frac{2}{3}n^3$ arithmetic operations (the Schur complement of an $m \times m$ matrix costs about $2m^2$ operations, and $\sum_{m \le n} 2m^2 \approx \frac{2}{3}n^3$), while forward and back substitution take about n^2 operations each. So once $A = LU$ is known, every further system $A\mathbf{x} = \mathbf{b}$ is cheap; this is how `np.linalg.solve` works, and there is no need to compute A^{-1}. The determinant is a by-product: $\det(A) = \pm u_{11} \cdots u_{nn}$, with the sign $\det(P)^{-1} = \pm 1$ of the row permutation.

6.6.3 Low-rank updates and the Woodbury identity

Suppose that we know A^{-1}, and A changes by a matrix of low rank, for example by $\mathbf{u}\mathbf{v}^\top$ when a new data point arrives. Then we do not need to invert the new matrix from scratch.

Theorem 6.108 (Woodbury identity)**.** *Let A be an invertible $n \times n$ matrix, C an invertible $k \times k$ matrix, U an $n \times k$ matrix and V a $k \times n$ matrix, such that $C^{-1} + VA^{-1}U$ is invertible. Then $A + UCV$ is invertible, and*

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U\big(C^{-1} + VA^{-1}U\big)^{-1}VA^{-1}.$$

In particular, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ with $1 + \mathbf{v}^\top A^{-1}\mathbf{u} \neq 0$ (the case $k = 1$, $C = 1$), we have the Sherman–Morrison formula

$$(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1}\mathbf{u}}.$$

Proof. Let X be the right side, and write $K = C^{-1} + VA^{-1}U$. Then

$$\begin{aligned}(A + UCV)X &= I + UCVA^{-1} - UK^{-1}VA^{-1} - UCVA^{-1}UK^{-1}VA^{-1} \\ &= I + UCVA^{-1} - UC\big(C^{-1} + VA^{-1}U\big)K^{-1}VA^{-1} = I + UCVA^{-1} - UCVA^{-1} = I,\end{aligned}$$

and by Corollary 6.48, X is the inverse of $A + UCV$. □

Proposition 6.109 (Determinants of low-rank updates)**.** *For an $n \times k$ matrix U and a $k \times n$ matrix V, $\det(I_n + UV) = \det(I_k + VU)$. In particular, for an invertible matrix A and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,*

$$\det(A + \mathbf{u}\mathbf{v}^\top) = (1 + \mathbf{v}^\top A^{-1}\mathbf{u}) \det(A).$$

Proof. Let $M = \begin{pmatrix} I_n & -U \\ V & I_k \end{pmatrix}$. By Theorem 6.103, with the Schur complement $I_k - VI_n^{-1}(-U) = I_k + VU$ of I_n, $\det(M) = \det(I_k + VU)$. On the other hand, block multiplication gives

$$\begin{pmatrix} I_n & U \\ 0 & I_k \end{pmatrix} M = \begin{pmatrix} I_n + UV & 0 \\ V & I_k \end{pmatrix},$$

and by Lemma 6.101 and the product rule, $\det(M) = \det(I_n + UV)$. For the second statement, $\det(A + \mathbf{u}\mathbf{v}^\top) = \det(A)\det(I_n + A^{-1}\mathbf{u}\mathbf{v}^\top) = \det(A)(1 + \mathbf{v}^\top A^{-1}\mathbf{u})$, by the first statement with $U = A^{-1}\mathbf{u}$ and $V = \mathbf{v}^\top$. □

Example 6.110. Let $A = \operatorname{diag}(2, 4)$ and $\mathbf{u} = \mathbf{v} = (1, 1)^\top$, so that $A + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$. Here $A^{-1}\mathbf{u} = (\frac{1}{2}, \frac{1}{4})^\top$ and $1 + \mathbf{v}^\top A^{-1}\mathbf{u} = \frac{7}{4}$, so $\det(A + \mathbf{u}\mathbf{v}^\top) = \frac{7}{4} \cdot 8 = 14$, and

$$(A + \mathbf{u}\mathbf{v}^\top)^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} - \frac{4}{7}\begin{pmatrix} \frac{1}{4} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{16} \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 5 & -1 \\ -1 & 3 \end{pmatrix},$$

in agreement with the formula for the inverse of a 2×2 matrix.

Remark 6.111 (Low-rank updates in machine learning).

1. Linear regression with the data matrix X, whose rows are the data points $x_i^\top$, uses the matrix $X^\top X = \sum_i x_i x_i^\top$ (Lecture 2). A new data point $\mathbf{x}$ adds the rank-one matrix $\mathbf{x}\mathbf{x}^\top$, and the Sherman–Morrison formula updates $(X^\top X)^{-1}$ with about n^2 operations instead of n^3. This is recursive least squares, which also underlies the Kalman filter and some bandit algorithms.

2. For an $N \times d$ matrix X and $\lambda > 0$, the push-through identity

$$(X^\top X + \lambda I_d)^{-1} X^\top = X^\top (XX^\top + \lambda I_N)^{-1}$$

follows from $X^\top(XX^\top + \lambda I_N) = (X^\top X + \lambda I_d)X^\top$ (both matrices are invertible, since $\mathbf{x}^\top(X^\top X + \lambda I_d)\mathbf{x} = (X\mathbf{x})^\top(X\mathbf{x}) + \lambda\mathbf{x}^\top\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$, and similarly for $XX^\top + \lambda I_N$). In ridge regression with many more features than data points ($d \gg N$), the right side only needs the inverse of an $N \times N$ matrix. The matrix $XX^\top$ contains the inner products of the data points; this is the starting point of kernel methods and Gaussian processes (Lecture 4).

3. Covariance matrices of the form $D + WW^\top$, with a diagonal matrix D and an $n \times k$ matrix W with $k \ll n$ (as in factor analysis), can be inverted with the Woodbury identity and their determinants computed with Proposition 6.109, using only $k \times k$ matrices.

7 Null Space, Column Space and Rank

Row reduction is our main tool for computing the kernel and the image of a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$. To use it, we first introduce the corresponding notions for matrices.

Definition 7.1. Let A be an $m \times n$ matrix.

The null space of A is the set of all vectors $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{0}$. It is denoted as $\text{null}(A)$. If $T(\mathbf{x}) = A\mathbf{x}$, then $\text{null}(A) = \ker(T)$.

The column space of A is the set of all linear combinations of the columns of A. It is denoted as $\text{col}(A)$. By Remark 6.6, the vectors $A\mathbf{x}$ are exactly the linear combinations of the columns of A, so if $T(\mathbf{x}) = A\mathbf{x}$, then $\text{col}(A) = \text{im}(T)$.

The row space of A is the set of all linear combinations of the rows of A. It is denoted as $\text{row}(A)$. Writing the rows as columns, $\text{row}(A) = \text{col}(A^\top)$, which is the image of the linear transformation represented by the transpose $A^\top$.

By Theorem 5.5, $\text{null}(A)$ is a subspace of $\mathbb{R}^n$ and $\text{col}(A)$ is a subspace of $\mathbb{R}^m$. The following theorem explains how row reduction affects these spaces.

Theorem 7.2. *Let A be an $m \times n$ matrix, and let R be obtained from A by a sequence of elementary row operations. Then:*

1. $\text{null}(A) = \text{null}(R)$*, i.e., the equations $A\mathbf{x} = \mathbf{0}$ and $R\mathbf{x} = \mathbf{0}$ have exactly the same solutions;*
2. $\text{row}(A) = \text{row}(R)$*;*
3. *if $R = \text{RREF}(A)$, then the columns of A in the positions of the pivot columns of R (the pivot columns of A) form a basis of* $\text{col}(A)$.

Proof. It is enough to consider a single elementary row operation, since we can then repeat the argument for each operation in the sequence.

(1) Each equation of the new system is a linear combination of the equations of the old system, so every solution of the old system is also a solution of the new one. Since the operation can be undone by another elementary row operation, the converse holds as well.

(2) Similarly, every row of the new matrix is a linear combination of the rows of the old matrix, so the new row space is contained in the old one. Undoing the operation shows the reverse inclusion.

(3) Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be the columns of A and $\mathbf{r}_1, \dots, \mathbf{r}_n$ the columns of R. For any scalars $c_1, \dots, c_n$, part (1) applied to $c = (c_1, \dots, c_n)^\top$ gives

$$c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0} \iff A\mathbf{c} = \mathbf{0} \iff R\mathbf{c} = \mathbf{0} \iff c_1\mathbf{r}_1 + \cdots + c_n\mathbf{r}_n = \mathbf{0}.$$

So the columns of A satisfy exactly the same linear relations as the columns of R. If R has r pivots, then its pivot columns are the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_r$ of $\mathbb{R}^m$, which are linearly independent. Every other column of R has zeros in its last $m - r$ entries, so it is a linear combination of $\mathbf{e}_1, \dots, \mathbf{e}_r$. Transferring these relations back to A, the pivot columns of A are linearly independent, and every other column of A is a linear combination of them. Hence the pivot columns of A span $\text{col}(A)$, and they form a basis of $\text{col}(A)$. □

Remark 7.3. Elementary row operations can change the column space! For example,

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{\text{subtract row 1 from row 2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

but the column space of the first matrix is $\text{span}\{(1,1)^\top\}$, while the column space of the second matrix is $\text{span}\{(1,0)^\top\}$. This is why, in part 3 of Theorem 7.2, we take the pivot columns of the original matrix A, not of R.

Example 7.4. Consider again the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$. The matrix representation of T is

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}.$$

We will transform this matrix into reduced row echelon form to find the null space, column space and row space of A, and hence the kernel and image of T.

First, we can eliminate the leading entry in the second row by subtracting 3 times the first row from the second row:

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & -7 & 7 \end{pmatrix}.$$

Next, we can multiply the second row by $-1/7$ to make the leading entry of the second row equal to 1:

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -1 \end{pmatrix}.$$

Now we can eliminate the entry above the leading 1 in the second row by subtracting 2 times the second row from the first row:

$$\mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

Null space. By Theorem 7.2, the equation $A\mathbf{x} = \mathbf{0}$, where $\mathbf{x} = (x_1, x_2, x_3)^\top$, has the same solutions as $\mathrm{RREF}(A)\,\mathbf{x} = \mathbf{0}$. From the RREF, we have:

$$\begin{aligned} x_1 + x_3 = 0 &\implies x_1 = -x_3, \\ x_2 - x_3 = 0 &\implies x_2 = x_3. \end{aligned}$$

Letting $x_3 = t$ for some scalar t, we have:

$$\mathbf{x} = (-t, t, t)^\top = t(-1, 1, 1)^\top.$$

Therefore, $\ker(T) = \mathrm{null}(A) = \mathrm{span}\{(-1, 1, 1)^\top\}$. This agrees with Example 4.9, where we found $\ker(T) = \{(x, -x, -x) : x \in \mathbb{R}\}$ (take $x = -t$).

Column space. The pivots of $\mathrm{RREF}(A)$ are in columns 1 and 2. By Theorem 7.2, the first two columns of the original matrix A,

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ -1 \end{pmatrix},$$

form a basis of $\mathrm{col}(A)$. These two vectors span all of $\mathbb{R}^2$, since every $(u, v)^\top \in \mathbb{R}^2$ can be written as

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{u + 2v}{7} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \frac{3u - v}{7} \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Hence $\mathrm{im}(T) = \mathrm{col}(A) = \mathbb{R}^2$, which agrees with Example 4.12.

Row space. By Theorem 7.2, $\mathrm{row}(A) = \mathrm{row}(\mathrm{RREF}(A))$, so the row space of A is spanned by the nonzero rows of the RREF:

$$\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & -1 \end{pmatrix}.$$

These two rows are linearly independent (look at their first two entries), so they form a basis of $\mathrm{row}(A)$.

Definition 7.5 (Rank)**.** The rank of a matrix A is the dimension of its column space, $\operatorname{rank}(A) = \dim \operatorname{col}(A)$. If $T(\mathbf{x}) = A\mathbf{x}$, this is the dimension of the image of T.

Theorem 7.6. *Let A be a matrix, and let r be the number of pivots of* $\operatorname{RREF}(A)$*, which is the same as the number of nonzero rows of* $\operatorname{RREF}(A)$*. Then*

$$\operatorname{rank}(A) = r = \dim \operatorname{row}(A).$$

Proof. By part 3 of Theorem 7.2, the r pivot columns of A form a basis of $\operatorname{col}(A)$, so $\operatorname{rank}(A) = r$.

By part 2 of Theorem 7.2, the r nonzero rows of $\operatorname{RREF}(A)$ span $\operatorname{row}(A)$. They are also linearly independent: each of them has a 1 in its pivot column, where all the other rows have a 0. So they form a basis of $\operatorname{row}(A)$, and $\dim \operatorname{row}(A) = r$. □

Remark 7.7. In practice, $\operatorname{rank}(A)$ is the number of nonzero rows in any row echelon form of A: clearing the entries above the leading 1s, as in Examples 6.33 and 6.34, turns a row echelon form into the RREF without changing the number of nonzero rows. Also, Theorem 7.6 says that $\dim \operatorname{col}(A) = \dim \operatorname{row}(A)$: a matrix has as many linearly independent columns as linearly independent rows. This is far from obvious!

Example 7.8. Consider the matrix

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix is already in row echelon form. The number of nonzero rows is 2, so the rank of the matrix M is $\operatorname{rank}(M) = 2$. Similarly, the matrix A in Example 7.4 has rank 2.

Theorem 7.9 (Rank–nullity theorem)**.** *For any $m \times n$ matrix A,*

$$\operatorname{rank}(A) + \dim \operatorname{null}(A) = n.$$

Equivalently, if $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then $\dim \operatorname{im}(T) + \dim \ker(T) = n$.

Idea of proof. Let r be the number of pivots of $\operatorname{RREF}(A)$, so that $\operatorname{rank}(A) = r$ by Theorem 7.6. When we solve $\operatorname{RREF}(A)\,\mathbf{x} = \mathbf{0}$, the r variables corresponding to the pivot columns are expressed in terms of the remaining $n - r$ free variables. Setting one free variable equal to 1 and all others equal to 0 gives $n - r$ solutions. Every solution is a linear combination of these, and they are linearly independent, since each of them has a 1 in a position where all the others have a 0. So they form a basis of $\operatorname{null}(A)$, and $\dim \operatorname{null}(A) = n - r$. □

Example 7.10. For $T(x, y, z) = (x+2y-z, 3x-y+4z)$, we found in Example 7.4 that $\dim \ker(T) = 1$ and $\dim \operatorname{im}(T) = 2$, and indeed $2 + 1 = 3 = n$. For $T(x, y) = (2x + 3y, 4x - y)$, we found in Examples 4.8 and 4.11 that $\ker(T) = \{(0, 0)\}$, of dimension 0, and $\operatorname{im}(T) = \mathbb{R}^2$, of dimension 2; indeed $2 + 0 = 2 = n$.

Combining the rank–nullity theorem with Theorem 6.47, we obtain several ways to recognize an invertible matrix.

Corollary 7.11. *For an $n \times n$ matrix A, the following statements are equivalent:*

1. *A is invertible;*

2. $\operatorname{null}(A) = \{\mathbf{0}\}$;

3. $\operatorname{rank}(A) = n$;

4. *for every* $\mathbf{b} \in \mathbb{R}^n$, *the equation* $A\mathbf{x} = \mathbf{b}$ *has exactly one solution;*

5. $\det(A) \neq 0$.

Proof. Statements 1 and 2 are equivalent by Theorem 6.47. By the rank–nullity theorem, $\operatorname{rank}(A) = n - \dim \operatorname{null}(A)$, so $\operatorname{rank}(A) = n$ if and only if $\dim \operatorname{null}(A) = 0$, i.e., $\operatorname{null}(A) = \{\mathbf{0}\}$; hence statements 2 and 3 are equivalent. Statement 1 implies statement 4 by Remark 6.38, and statement 4 implies statement 2 by taking $\mathbf{b} = \mathbf{0}$. Finally, statements 1 and 5 are equivalent by Theorem 6.91. □

Exercise 7.12. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x - y, 2x + 3y)$. Find the kernel and image of T, and check that your answer agrees with the rank–nullity theorem.

8 Dual Vector Spaces

Many important quantities assign a single number to a vector: a coordinate, the average of the entries, the value of a polynomial at a point, the integral of a function, or the prediction $\mathbf{w}^\top \mathbf{x}$ of a linear model. When such an assignment is linear, it is called a linear functional. The linear functionals on a vector space V form a new vector space, the dual space V^*. Duality explains what the transpose of a matrix really is, gives a convenient way to compute coordinates, and provides "certificates" showing that a system of linear equations has no solution. In machine learning, the weight vector of a linear model and the derivative of a loss function are naturally elements of a dual space.

Throughout this section, V and W are vector spaces over a field $\mathbb{F}$.

8.1 Linear functionals and the dual space

Definition 8.1 (Linear functional, dual space)**.** A linear functional (or linear form) on V is a linear transformation $\varphi : V \to \mathbb{F}$. The set of all linear functionals on V is called the dual space of V and is denoted by V^*. For $\varphi, \psi \in V^*$ and $c \in \mathbb{F}$, we define $\varphi + \psi$ and $c\varphi$ pointwise:

$$(\varphi + \psi)(v) = \varphi(v) + \psi(v), \qquad (c\varphi)(v) = c\,\varphi(v) \qquad (v \in V).$$

Proposition 8.2. *With these operations,* V^* *is a vector space over* $\mathbb{F}$.

Proof. The functions $\varphi+\psi$ and $c\varphi$ are again linear; for example, $(\varphi+\psi)(u+v) = \varphi(u)+\varphi(v)+\psi(u)+\psi(v) = (\varphi+\psi)(u) + (\varphi+\psi)(v)$, and similarly for scalar multiples. The vector space axioms hold because they hold pointwise in $\mathbb{F}$, exactly as for the space $\mathcal{F}(S)$ of all functions (Proposition 2.10). The zero vector of V^* is the zero functional $v \mapsto 0$. □

On $\mathbb{R}^n$, the linear functionals are exactly the row vectors.

Proposition 8.3. *For every row vector* $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^{1\times n}$, *the map* $\varphi_a(\mathbf{x}) = \mathbf{a}\mathbf{x} = a_1x_1 + \cdots + a_nx_n$ *is a linear functional on* $\mathbb{R}^n$. *Conversely, every linear functional* φ *on* $\mathbb{R}^n$ *is of this form, with* $\mathbf{a} = (\varphi(\mathbf{e}_1), \dots, \varphi(\mathbf{e}_n))$. *The map* $\mathbf{a} \mapsto \varphi_a$ *is an isomorphism* $\mathbb{R}^{1\times n} \to (\mathbb{R}^n)^*$.

Proof. The first statement is the case $m = 1$ of matrix–vector multiplication, and the second is the case $m = 1$ of Theorem 6.9: the matrix of $\varphi : \mathbb{R}^n \to \mathbb{R}$ is the $1 \times n$ matrix with columns $\varphi(\mathbf{e}_1), \dots, \varphi(\mathbf{e}_n)$. The map $\mathbf{a} \mapsto \varphi_a$ is linear, since $\varphi_{\mathbf{a}+\mathbf{b}}(\mathbf{x}) = (\mathbf{a}+\mathbf{b})\mathbf{x} = \mathbf{a}\mathbf{x}+\mathbf{b}\mathbf{x}$ and $\varphi_{c\mathbf{a}}(\mathbf{x}) = c\,\mathbf{a}\mathbf{x}$; it is surjective by the second statement, and injective since $\varphi_a = 0$ implies $a_i = \varphi_a(\mathbf{e}_i) = 0$ for all i. □

So vectors of $\mathbb{R}^n$ are columns, and functionals on $\mathbb{R}^n$ are rows; applying a functional to a vector is the product "row times column".

Example 8.4. The following maps are linear functionals.

- The coordinate functional $\mathbf{x} \mapsto x_i$ on $\mathbb{R}^n$, given by the row $\mathbf{e}_i^\top$.
- The mean $\mathbf{x} \mapsto \frac{1}{n}(x_1 + \cdots + x_n)$ on $\mathbb{R}^n$, given by the row $\frac{1}{n}(1, \dots, 1)$.
- The trace $\operatorname{tr} : \mathbb{R}^{n\times n} \to \mathbb{R}$ (Theorem 6.77).
- The evaluation at a point a, $\operatorname{ev}_a : \mathbb{R}[x] \to \mathbb{R}$, $p \mapsto p(a)$, and the maps $p \mapsto p'(0)$ and $p \mapsto \int_0^1 p(x)\,dx$.

On the other hand, $\mathbf{x} \mapsto x_1x_2$, $\mathbf{x} \mapsto x_1 + 1$ and $A \mapsto \det(A)$ (for $n \geq 2$) are not linear, so they are not linear functionals.

Remark 8.5 (Linear models). A linear model in machine learning predicts a number from a feature vector $\mathbf{x} \in \mathbb{R}^n$ by $f(\mathbf{x}) = w_1x_1 + \cdots + w_nx_n = \mathbf{w}^\top\mathbf{x}$. So the model is a linear functional, and learning a linear model means choosing an element of the dual space $(\mathbb{R}^n)^*$. This is one reason why it is natural to write the weights as a row $\mathbf{w}^\top$, and the data as columns.

The kernel of a nonzero linear functional on $\mathbb{R}^n$ is the null space of a nonzero $1 \times n$ matrix a, so by the rank–nullity theorem it is a subspace of dimension $n - 1$, called a hyperplane (through the origin). For example, the kernel of $\mathbf{x} \mapsto x_1 + x_2 + x_3$ is the plane $x_1 + x_2 + x_3 = 0$ in $\mathbb{R}^3$. Using coordinates, the same holds for every n-dimensional vector space.

8.2 The dual basis

For indices i and j, we use the Kronecker delta

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Theorem 8.6 (Dual basis)**.** *Let V be finite-dimensional with an ordered basis $\mathcal{B} = (b_1, \dots, b_n)$. For $i = 1, \dots, n$, let $b^i \in V^*$ be the linear functional that maps a vector to its i-th coordinate with respect to $\mathcal{B}$:*

$$b^i(a_1b_1 + \cdots + a_nb_n) = a_i, \qquad \text{so that} \qquad b^i(b_j) = \delta_{ij}.$$

Then $(b^1, \dots, b^n)$ is an ordered basis of V^*, called the dual basis of $\mathcal{B}$. Moreover, every $\varphi \in V^*$ can be written as

$$\varphi = \varphi(b_1)\,b^1 + \varphi(b_2)\,b^2 + \cdots + \varphi(b_n)\,b^n.$$

In particular, $\dim V^* = \dim V$*, so* $V^* \cong V$.

Here the superscripts in $b^1, \dots, b^n$ are indices, not powers.

Proof. Each b^i is linear, since the coordinate map $v \mapsto [v]_{\mathcal{B}}$ is linear (Theorem 4.27). First, two linear functionals that agree on $b_1, \dots, b_n$ are equal: if $v = \sum_j a_j b_j$, then $\varphi(v) = \sum_j a_j \varphi(b_j)$ is determined by the values $\varphi(b_j)$. Now the functionals φ and $\sum_i \varphi(b_i) b^i$ agree on each b_j, since $\sum_i \varphi(b_i) b^i(b_j) = \sum_i \varphi(b_i)\delta_{ij} = \varphi(b_j)$; hence they are equal, and $b^1, \dots, b^n$ span V^*. If $\sum_i c_i b^i = 0$, then applying both sides to b_j gives $c_j = 0$, so $b^1, \dots, b^n$ are linearly independent. The last statement follows from Corollary 4.29. □

So the coordinates of a functional in the dual basis are simply its values on the basis vectors. For the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ of $\mathbb{R}^n$, the dual basis consists of the coordinate functionals $\mathbf{x} \mapsto x_i$, i.e., of the rows $\mathbf{e}_1^\top, \dots, \mathbf{e}_n^\top$. For other bases of $\mathbb{R}^n$, the dual basis can be computed with an inverse matrix.

Proposition 8.7. *Let $\mathbf{b}_1, \dots, \mathbf{b}_n$ be a basis of $\mathbb{R}^n$, and let P be the (invertible) matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_n$. Then the dual basis functionals are given by the rows of P^{-1}: if $\mathbf{r}_i$ is the i-th row of P^{-1}, then $b^i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$.*

Proof. The (i, j) entry of $P^{-1}P = I_n$ is $\mathbf{r}_i \mathbf{b}_j = \delta_{ij}$. So the functionals $\mathbf{x} \mapsto \mathbf{r}_i \mathbf{x}$ satisfy the defining property of the dual basis. (Equivalently, $P^{-1}\mathbf{x} = [\mathbf{x}]_{\mathcal{B}}$ is the coordinate vector of $\mathbf{x}$.) □

Example 8.8. For the basis $\mathbf{b}_1 = (1, 1)^\top$, $\mathbf{b}_2 = (1, -1)^\top$ of $\mathbb{R}^2$, we have $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, so

$$b^1(\mathbf{x}) = \frac{x_1 + x_2}{2}, \qquad b^2(\mathbf{x}) = \frac{x_1 - x_2}{2}.$$

Indeed, $b^1(\mathbf{b}_1) = 1$, $b^1(\mathbf{b}_2) = 0$, $b^2(\mathbf{b}_1) = 0$ and $b^2(\mathbf{b}_2) = 1$.

Example 8.9 (Lagrange interpolation)**.** Let $V = \mathbb{R}[x]_{\le 2}$. The map $T(p) = (p(-1), p(0), p(1))$ is an isomorphism $V \to \mathbb{R}^3$, since its matrix with respect to the basis $(1, x, x^2)$ is $\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, which has determinant $2 \neq 0$ (Theorem 6.58). Hence the polynomials $L_a = T^{-1}(e_a)$, i.e.,

$$L_{-1}(x) = \frac{x(x-1)}{2}, \qquad L_0(x) = 1 - x^2, \qquad L_1(x) = \frac{x(x+1)}{2},$$

form a basis of V (Proposition 4.25); they satisfy $L_a(c) = \delta_{ac}$ for $a, c \in \{-1, 0, 1\}$. Its dual basis consists of the evaluation functionals $\mathrm{ev}_{-1}, \mathrm{ev}_0, \mathrm{ev}_1$, and Theorem 8.6 gives the Lagrange interpolation formula

$$p = p(-1)\, L_{-1} + p(0)\, L_0 + p(1)\, L_1 \qquad \text{for every } p \in \mathbb{R}[x]_{\le 2}.$$

(Apply T to both sides: both have the values $p(-1)$, $p(0)$, $p(1)$.) For example, the polynomial of degree at most 2 with values $1, 2, 5$ at $-1, 0, 1$ is $L_{-1} + 2L_0 + 5L_1 = 2 + 2x + x^2$.

8.3 The dual map and the transpose

A linear transformation $T : V \to W$ turns functionals on W into functionals on V: first apply T, then the functional.

Definition 8.10 (Dual map)**.** Let $T : V \to W$ be a linear transformation. The dual map (or transpose) of T is

$$T^* : W^* \to V^*, \qquad T^*(\psi) = \psi \circ T.$$

The map $T^*(\psi) = \psi \circ T$ is a linear functional on V by Lemma 4.24, and T^* is linear, since $(\psi_1+\psi_2)\circ T = \psi_1\circ T+\psi_2\circ T$ and $(c\psi)\circ T = c(\psi\circ T)$. Note that T^* goes in the opposite direction: from W^* to V^*. If $S : W \to U$ is another linear transformation, then $(S\circ T)^*(\psi) = \psi\circ S\circ T = T^*(S^*(\psi))$, so

$$(S \circ T)^* = T^* \circ S^*.$$

Theorem 8.11. *Let V and W be finite-dimensional vector spaces over $\mathbb{R}$, with ordered bases $\mathcal{B} = (b_1, \dots, b_n)$ and $\mathcal{C} = (c_1, \dots, c_m)$, and let $\mathcal{B}^* = (b^1, \dots, b^n)$ and $\mathcal{C}^* = (c^1, \dots, c^m)$ be the dual bases. If $T : V \to W$ has the matrix $A = [T]_{\mathcal{C},\mathcal{B}}$, then the dual map has the matrix*

$$[T^*]_{\mathcal{B}^*,\mathcal{C}^*} = A^\top.$$

Proof. By definition of A, the i-th column of A is the coordinate vector of $T(b_i)$, so $c^j(T(b_i)) = A_{ji}$. By Theorem 8.6, the coordinates of the functional $T^*(c^j)$ in the dual basis $\mathcal{B}^*$ are its values on $b_1, \dots, b_n$, which are

$$T^*(c^j)(b_i) = c^j(T(b_i)) = A_{ji} \qquad (i = 1, \dots, n).$$

So the j-th column of $[T^*]_{\mathcal{B}^*,\mathcal{C}^*}$ is $(A_{j1}, \dots, A_{jn})^\top$, which is the j-th column of $A^\top$. □

For $T = T_A : \mathbb{R}^n \to \mathbb{R}^m$ and functionals written as rows (Proposition 8.3), this is very concrete: the functional $\psi(\mathbf{y}) = \mathbf{c}\mathbf{y}$ on $\mathbb{R}^m$, given by a row $\mathbf{c}$, is mapped to

$$T_A^*(\psi)(\mathbf{x}) = \psi(A\mathbf{x}) = \mathbf{c}(A\mathbf{x}) = (\mathbf{c}A)\mathbf{x},$$

i.e., to the row $\mathbf{c}A$. So the dual map multiplies rows from the right by A, while T_A multiplies columns from the left; written as columns, $(\mathbf{c}A)^\top = A^\top\mathbf{c}^\top$. This explains the transpose conceptually, and the rule $(S \circ T)^* = T^* \circ S^*$ explains why $(AB)^\top = B^\top A^\top$ reverses the order.

8.4 Annihilators and solvability of linear systems

Definition 8.12 (Annihilator)**.** Let U be a subspace of V. The annihilator of U is the set of all functionals that vanish on U:

$$U^\circ = \{\varphi \in V^* : \varphi(u) = 0 \text{ for all } u \in U\}.$$

The annihilator is a subspace of V^* (check this). In $\mathbb{R}^n$, it can be computed with a null space.

Theorem 8.13. *Let $U = \operatorname{col}(B)$ be a k-dimensional subspace of $\mathbb{R}^n$, where B is an $n \times k$ matrix with linearly independent columns. A row $\mathbf{a}$ defines a functional in U° if and only if $\mathbf{a}B = \mathbf{0}$, i.e., $\mathbf{a}^\top \in \operatorname{null}(B^\top)$. In particular,*

$$\dim U + \dim U^\circ = n.$$

Proof. The functional $\mathbf{x} \mapsto \mathbf{a}\mathbf{x}$ vanishes on U if and only if it vanishes on the columns $\mathbf{b}_1, \dots, \mathbf{b}_k$ of B (by linearity), i.e., if and only if $\mathbf{a}\mathbf{b}_i = 0$ for all i, which means $\mathbf{a}B = \mathbf{0}$, or equivalently $B^\top\mathbf{a}^\top = \mathbf{0}$. By Proposition 8.3, U° is therefore isomorphic to $\operatorname{null}(B^\top)$. Since $\operatorname{rank}(B^\top) = \operatorname{rank}(B) = k$ (Theorem 7.6), the rank–nullity theorem gives $\dim U^\circ = n - k$. □

Example 8.14. For $U = \operatorname{span}\{(1, 1, 1)^\top\}$ in $\mathbb{R}^3$, the annihilator consists of the rows $\mathbf{a}$ with $a_1 + a_2 + a_3 = 0$, a 2-dimensional space, and indeed $1 + 2 = 3$.

Annihilators give a clean criterion for when a system of linear equations has a solution.

Theorem 8.15 (Fredholm alternative)**.** *Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. The system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{y}\mathbf{b} = 0$ for every row $\mathbf{y} \in \mathbb{R}^{1\times m}$ with $\mathbf{y}A = \mathbf{0}$.*

Proof. If $A\mathbf{x} = \mathbf{b}$ and $\mathbf{y}A = \mathbf{0}$, then $\mathbf{y}\mathbf{b} = \mathbf{y}A\mathbf{x} = 0$. Conversely, suppose that $A\mathbf{x} = \mathbf{b}$ has no solution, i.e., $\mathbf{b} \notin U = \mathrm{col}(A)$. Let $r = \dim U$, choose a basis $u_1, \dots, u_r$ of U, and let $U' = \mathrm{span}\{u_1, \dots, u_r, \mathbf{b}\}$. The vectors $u_1, \dots, u_r, \mathbf{b}$ are linearly independent (a nontrivial relation would have to involve $\mathbf{b}$, and would express $\mathbf{b}$ as an element of U), so $\dim U' = r + 1$. By Theorem 8.13, $\dim U'^\circ = m - r - 1 < m - r = \dim U^\circ$. Since $U \subseteq U'$, we have $U'^\circ \subseteq U^\circ$, and the dimensions show that there is a functional $\mathbf{y} \in U^\circ$ with $\mathbf{y} \notin U'^\circ$. Such a row $\mathbf{y}$ vanishes on the columns of A, i.e., $\mathbf{y}A = \mathbf{0}$, but it does not vanish on all of U', so $\mathbf{y}\mathbf{b} \neq 0$. □

So if a system has no solution, there is always a simple certificate that proves it: a combination $\mathbf{y}$ of the equations whose left-hand sides cancel ($\mathbf{y}A = \mathbf{0}$), but whose right-hand sides do not ($\mathbf{y}\mathbf{b} \neq 0$).

Example 8.16. The system $x_1 + 2x_2 = 1$, $2x_1 + 4x_2 = 3$ has no solution. A certificate is $\mathbf{y} = (2, -1)$: we have $\mathbf{y}A = (2, -1)\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = (0, 0)$, but $\mathbf{y}\mathbf{b} = 2 - 3 = -1 \neq 0$. In words: twice the first equation minus the second one gives $0 = -1$.

Remark 8.17. Certificates of this kind are the starting point of duality theory in optimization (Farkas' lemma, the duality theorem of linear programming, Lagrange multipliers), which is used throughout machine learning, for example to derive the dual problem of support vector machines.

8.5 The double dual

Applying the construction twice gives the double dual $V^{**} = (V^*)^*$. Every vector $v \in V$ gives an element of V^{**}: the functional "evaluate at v".

Theorem 8.18. *For $v \in V$, let $\iota(v) \in V^{**}$ be defined by $\iota(v)(\varphi) = \varphi(v)$ for $\varphi \in V^*$. Then $\iota : V \to V^{**}$ is linear and injective, and if V is finite-dimensional, it is an isomorphism.*

Proof. Each $\iota(v)$ is linear in φ, by the definition of the operations in V^*, and ι is linear in v, since every φ is linear. We only prove the finite-dimensional case. Let $b_1, \dots, b_n$ be a basis of V, with dual basis $b^1, \dots, b^n$. If $\iota(v) = 0$, then $b^i(v) = 0$ for all i, so all coordinates of v are 0, and $v = 0$; hence ι is injective. For surjectivity, let $\Lambda \in V^{**}$, and put $v = \sum_i \Lambda(b^i) b_i$. Then $\iota(v)(b^j) = b^j(v) = \Lambda(b^j)$ for all j, so the linear functionals $\iota(v)$ and Λ on V^* agree on a basis of V^*, and hence $\iota(v) = \Lambda$. □

Remark 8.19. For finite-dimensional V, both V^* and V^{**} are isomorphic to V, since they have the same dimension. But an isomorphism $V \to V^*$ (such as $b_i \mapsto b^i$) depends on the choice of a basis, while $\iota : V \to V^{**}$ is defined without any choices; it is called the natural isomorphism. In Lecture 2, we will see that an inner product also gives a natural way to identify V with V^*: on $\mathbb{R}^n$ with the dot product, the vector $\mathbf{a}^\top$ corresponds to the functional $\mathbf{x} \mapsto \mathbf{a}\mathbf{x}$. (For infinite-dimensional spaces, such as $\mathbb{R}[x]$, the map ι is injective but not surjective; we do not prove this.)

Remark 8.20 (Gradients and backpropagation). For a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, the derivative at a point $\mathbf{x}$ is the linear functional

$$h \mapsto \frac{\partial f}{\partial x_1}(\mathbf{x})\, h_1 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{x})\, h_n,$$

which gives the first-order change of f in the direction h. So the derivative is naturally a row vector; the gradient $\nabla f(\mathbf{x})$ is the corresponding column vector, obtained by transposing. Backpropagation,

the algorithm behind the training of neural networks, computes the derivative of a loss function with respect to all parameters by passing such row vectors backwards through the layers of the network. Each layer applies the dual map (the transpose) of its own derivative, which is the rule $(S \circ T)^* = T^* \circ S^*$ at work.

9 Quotient Spaces and the Isomorphism Theorems

Sometimes the differences between vectors in certain directions do not matter. The antiderivative of a function is only determined up to an additive constant; the outputs of the softmax function in a neural network do not change if the same number is added to all of its inputs; and the information a linear transformation keeps about a vector does not change if we add an element of its kernel. The quotient space V/U formalizes the idea of "the vector space V, where we ignore the directions in a subspace U". The isomorphism theorems then describe how quotients, kernels and images fit together, and they give the rank–nullity theorem for arbitrary linear transformations, as well as a formula for the dimension of the intersection of two subspaces.

Throughout this section, V is a vector space over a field $\mathbb{F}$, and U is a subspace of V.

9.1 Cosets

Definition 9.1 (Coset)**.** For $v \in V$, the coset of U through v is the set

$$v + U = \{v + u : u \in U\}.$$

In $\mathbb{R}^n$, cosets are the shifted copies of the subspace U (affine subspaces, see Lecture 2). For example, if U is a line through the origin in $\mathbb{R}^2$, the cosets of U are the lines parallel to U.

Lemma 9.2. *For all $v, w \in V$:*

1. *$v \in v + U$;*
2. *$v + U = w + U$ if and only if $v - w \in U$;*
3. *two cosets are either equal or disjoint.*

Proof. (1) $v = v + 0$ with $0 \in U$.

(2) If $v+U = w+U$, then $v \in w+U$ by part 1, so $v = w+u$ for some $u \in U$, and $v-w = u \in U$. Conversely, if $v - w = u_0 \in U$, then every element $v + u = w + (u_0 + u)$ of $v + U$ lies in $w + U$, and similarly $w + U \subseteq v + U$, since $w - v = -u_0 \in U$.

(3) If $x \in (v+U) \cap (w+U)$, then $x - v \in U$ and $x - w \in U$, so $v - w = (x - w) - (x - v) \in U$, and $v + U = w + U$ by part 2. □

So the cosets of U cut V into disjoint pieces, and every vector lies in exactly one of them. The vectors in the same coset are those that differ by an element of U; any of them is called a representative of the coset.

Example 9.3. Let $U = \operatorname{span}\{(1,1)^\top\}$ in $\mathbb{R}^2$. Two vectors $\mathbf{v}$ and $\mathbf{w}$ lie in the same coset if and only if $\mathbf{v} - \mathbf{w}$ is a multiple of $(1,1)^\top$, i.e., if and only if $v_1 - v_2 = w_1 - w_2$. So the cosets are the lines $\{x : x_1 - x_2 = c\}$, $c \in \mathbb{R}$, parallel to U. For instance, $(3,1)^\top + U = (2,0)^\top + U$, since $(3,1)^\top - (2,0)^\top = (1,1)^\top \in U$.

Example 9.4. Let U be the subspace of constant functions in the space $\mathcal{F}(\mathbb{R})$ of all functions $\mathbb{R} \to \mathbb{R}$. If F is an antiderivative of f, then the set of all antiderivatives of f is the coset $F + U = \{F + C : C \in \mathbb{R}\}$. The familiar notation $\int f(x)\,dx = F(x) + C$ describes exactly this coset.

9.2 The quotient space

Definition 9.5 (Quotient space)**.** The quotient space of V by U is the set of all cosets of U,

$$V/U = \{v + U : v \in V\},$$

with the operations

$$(v + U) + (w + U) = (v + w) + U, \qquad c\,(v + U) = cv + U \qquad (c \in \mathbb{F}).$$

The map $\pi : V \to V/U$, $\pi(v) = v + U$, is called the quotient map.

Proposition 9.6. *The operations on V/U are well defined, and with them V/U is a vector space over $\mathbb{F}$, whose zero vector is the coset $0 + U = U$. The quotient map π is a surjective linear transformation with $\ker(\pi) = U$.*

Proof. A coset has many representatives, so we must check that the results of the operations do not depend on the choice of representatives. Let $v + U = v' + U$ and $w + U = w' + U$, i.e., $v - v' \in U$ and $w - w' \in U$ (Lemma 9.2). Then $(v + w) - (v' + w') = (v - v') + (w - w') \in U$ and $cv - cv' = c(v - v') \in U$, so $(v + w) + U = (v' + w') + U$ and $cv + U = cv' + U$. The vector space axioms for V/U follow from those for V; for example, $(v + U) + (0 + U) = v + U$, and the additive inverse of $v + U$ is $(-v) + U$. The quotient map is linear by the definition of the operations, and it is clearly surjective. Finally, $\pi(v) = 0 + U$ if and only if $v - 0 \in U$, by Lemma 9.2. □

So in V/U we compute with vectors as usual, except that vectors which differ by an element of U are considered equal; in particular, all vectors of U become 0.

Example 9.7. For $U = \operatorname{span}\{(1, 1)^\top\}$ as in Example 9.3, every coset is determined by the number $x_1 - x_2$, and the map $\mathbb{R}^2/U \to \mathbb{R}$, $\mathbf{x} + U \mapsto x_1 - x_2$, is well defined (it does not depend on the representative $\mathbf{x}$), linear and bijective. So $\mathbb{R}^2/U \cong \mathbb{R}$: collapsing a line in the plane to a point leaves a line.

To compute the dimension of a quotient space, we extend a basis of U to a basis of V.

Theorem 9.8. *If V is finite-dimensional, then $\dim(V/U) = \dim V - \dim U$. More precisely, if $u_1, \dots, u_k, w_1, \dots, w_m$ is a basis of V such that $u_1, \dots, u_k$ is a basis of U, then $w_1 + U, \dots, w_m + U$ is a basis of V/U.*

Proof. By Corollary 3.14, U is finite-dimensional, and every basis of U, which consists of linearly independent vectors of V, can be extended to such a basis of V. The cosets $w_j + U$ span V/U: if $v = \sum_i a_i u_i + \sum_j c_j w_j$, then $v + U = \sum_j c_j (w_j + U)$, since $\sum_i a_i u_i \in U$. They are linearly independent: if $\sum_j c_j (w_j + U) = 0 + U$, then $\sum_j c_j w_j \in U$, so $\sum_j c_j w_j = \sum_i a_i u_i$ for some scalars a_i. Since $u_1, \dots, u_k, w_1, \dots, w_m$ are linearly independent, all c_j (and a_i) are 0. □

Remark 9.9 (Complements). The proof shows that $W = \operatorname{span}\{w_1, \dots, w_m\}$ is a subspace such that every $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in W$; such a W is called a complement of U. Every coset $v + U$ then contains exactly one vector of W, and $w \mapsto w + U$ is an isomorphism $W \to V/U$. For example, the x_1-axis is a complement of $U = \operatorname{span}\{(1, 1)^\top\}$ in $\mathbb{R}^2$: the coset $\{x_1 - x_2 = c\}$ meets the x_1-axis exactly in $(c, 0)$. Complements are not unique, so this description of V/U depends on a choice. In Lecture 2, the orthogonal complement $U^\perp$ will be a canonical choice.

9.3 The isomorphism theorems

Theorem 9.10 (First isomorphism theorem)**.** *Let $T : V \to W$ be a linear transformation, and let $K = \ker(T)$. Then*

$$\bar{T} : V/K \to \operatorname{im}(T), \qquad \bar{T}(v + K) = T(v),$$

is a well-defined isomorphism. In particular, $V/\ker(T) \cong \operatorname{im}(T)$.

Proof. If $v + K = v' + K$, then $v - v' \in K$, so $T(v) - T(v') = T(v - v') = 0$; hence $\bar{T}$ is well defined. It is linear, since T is: $\bar{T}((v + K) + (w + K)) = T(v + w) = T(v) + T(w)$, and similarly for scalar multiples. It is surjective by the definition of $\operatorname{im}(T)$. It is injective: if $\bar{T}(v + K) = T(v) = 0$, then $v \in K$, so $v + K = 0 + K$ is the zero vector of V/K (Lemma 4.18). □

In words: a linear transformation identifies exactly those vectors that differ by an element of its kernel, and after this identification, it becomes an isomorphism onto its image. Combined with Theorem 9.8, the first isomorphism theorem gives the rank–nullity theorem for all linear transformations, not only for matrices.

Corollary 9.11 (Rank–nullity theorem)**.** *Let $T : V \to W$ be a linear transformation, where V is finite-dimensional. Then*

$$\dim \ker(T) + \dim \operatorname{im}(T) = \dim V.$$

Proof. By Theorem 9.10, $\operatorname{im}(T) \cong V/\ker(T)$, which has dimension $\dim V - \dim \ker(T)$ by Theorem 9.8; isomorphic spaces have the same dimension (Proposition 4.25). □

Example 9.12. For $T(x, y) = (x + 2y, 2x + 4y)$ (Example 4.13 and Figure 3), the cosets of $\ker(T) = \operatorname{span}\{(-2, 1)^\top\}$ are the lines $x + 2y = c$, and $\bar{T}$ maps the coset $\{x + 2y = c\}$ to the point $c\,(1, 2)$ of $\operatorname{im}(T)$. Figure 3 is therefore a picture of the isomorphism $\mathbb{R}^2/\ker(T) \cong \operatorname{im}(T)$: each colored line on the left is one element of the quotient space.

Example 9.13. The derivative $D : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 2}$, $p \mapsto p'$, is surjective, and its kernel is the space of constant polynomials. So $\mathbb{R}[x]_{\leq 3}/\{\text{constants}\} \cong \mathbb{R}[x]_{\leq 2}$: a polynomial "up to an additive constant" is the same thing as its derivative. Similarly, the trace $\operatorname{tr} : \mathbb{R}^{n\times n} \to \mathbb{R}$ is surjective, so $\mathbb{R}^{n\times n}$ modulo the matrices with trace zero is isomorphic to $\mathbb{R}$.

To compare two subspaces, we need their sum.

Definition 9.14 (Sum of subspaces)**.** For subspaces U and W of V, their sum is $U + W = \{u + w : u \in U,\ w \in W\}$.

One checks that $U + W$ is a subspace of V; it is the smallest subspace that contains both U and W. Also $U \cap W$ is a subspace (practice problems).

Theorem 9.15 (Second isomorphism theorem)**.** *Let U and W be subspaces of V. Then*

$$(U + W)/W \cong U/(U \cap W).$$

Proof. Consider $T : U \to (U + W)/W$, $T(u) = u + W$, which is linear (it is the restriction of the quotient map). It is surjective: every element of $(U + W)/W$ has the form $(u + w) + W = u + W = T(u)$, since $w \in W$. Its kernel consists of the $u \in U$ with $u \in W$, i.e., $\ker(T) = U \cap W$. Now apply the first isomorphism theorem. □

Corollary 9.16 (Dimension formula)**.** *If U and W are finite-dimensional subspaces of V, then*

$$\dim(U + W) + \dim(U \cap W) = \dim U + \dim W.$$

Proof. By Theorems 9.15 and 9.8, $\dim(U+W) - \dim W = \dim U - \dim(U \cap W)$. □

Example 9.17. Two different planes U and W through the origin in $\mathbb{R}^3$ always meet in a line: since $U+W$ is a subspace that is strictly larger than the 2-dimensional plane U, it is all of $\mathbb{R}^3$, and hence $\dim(U \cap W) = 2+2-3 = 1$. Similarly, two 4-dimensional subspaces of $\mathbb{R}^7$ always have a nonzero intersection, since $\dim(U \cap W) = 8 - \dim(U+W) \geq 8-7 = 1$.

Theorem 9.18 (Third isomorphism theorem)**.** *Let $U \subseteq W \subseteq V$ be subspaces. Then $W/U = \{w+U : w \in W\}$ is a subspace of V/U, and*

$$(V/U)/(W/U) \cong V/W.$$

Proof. W/U is the image of W under the quotient map $V \to V/U$, hence a subspace. Consider $T : V/U \to V/W$, $T(v+U) = v+W$. It is well defined: if $v+U = v'+U$, then $v - v' \in U \subseteq W$, so $v+W = v'+W$. It is linear and surjective, and $T(v+U) = 0+W$ if and only if $v \in W$, so $\ker(T) = W/U$. Now apply the first isomorphism theorem. □

In words: ignoring first the directions in U, and then the remaining directions in W, is the same as ignoring all directions in W at once. For example, for $U = \operatorname{span}\{\mathbf{e}_3\} \subseteq W = \operatorname{span}\{\mathbf{e}_2, \mathbf{e}_3\} \subseteq V = \mathbb{R}^3$, we have $V/U \cong \mathbb{R}^2$ (keep x_1, x_2), $W/U \cong \mathbb{R}$ (keep x_2), and both sides of the theorem are isomorphic to $\mathbb{R}$ (keep only x_1).

9.4 Quotients and duality

Linear functionals on a quotient space are the same as functionals that vanish on the subspace.

Theorem 9.19. *The map $(V/U)^* \to U^\circ$, $\varphi \mapsto \varphi \circ \pi$, is an isomorphism, where $\pi : V \to V/U$ is the quotient map and U° is the annihilator of U (Section 8).*

Proof. For $\varphi \in (V/U)^*$, the functional $\varphi \circ \pi$ vanishes on U, since $\pi(u) = 0$ for $u \in U$; and $\varphi \mapsto \varphi \circ \pi$ is linear. It is injective: if $\varphi \circ \pi = 0$, then $\varphi = 0$, since π is surjective. It is surjective: if $\psi \in U^\circ$, then $\varphi(v+U) = \psi(v)$ is well defined (if $v - v' \in U$, then $\psi(v) - \psi(v') = \psi(v - v') = 0$) and linear, and $\varphi \circ \pi = \psi$. □

For finite-dimensional V, this is consistent with the dimensions: $\dim(V/U)^* = \dim V - \dim U = \dim U^\circ$ (Theorems 9.8, 8.6 and 8.13).

Remark 9.20 (Quotients in machine learning). The softmax function, which turns a vector $\mathbf{z} \in \mathbb{R}^n$ of scores ("logits") into probabilities,

$$\operatorname{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{e^{z_1} + \cdots + e^{z_n}},$$

satisfies $\operatorname{softmax}(\mathbf{z} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{z})$ for every $c \in \mathbb{R}$, where $\mathbf{1} = (1, \ldots, 1)^\top$, since the factor e^c cancels. So the softmax only depends on the coset $\mathbf{z} + \operatorname{span}\{\mathbf{1}\}$, i.e., it is really a function on the quotient space $\mathbb{R}^n / \operatorname{span}\{\mathbf{1}\}$, and logits are only meaningful up to adding a constant. This has a practical consequence: computing e^{z_i} overflows for large scores, so softmax is implemented by first subtracting $\max_i z_i$ from all entries, which picks a different representative of the same coset (Programming Exercise P13). Choosing the representative whose entries sum to 0 is the same as centering the vector; the centered vectors form a complement of $\operatorname{span}\{\mathbf{1}\}$. Quotient spaces also appear in the construction of the function spaces L^2, where two functions are identified if their difference has integral of squares 0.

10 Eigenvalues and Eigenvectors

A linear transformation usually changes the direction of a vector. The vectors whose direction does not change, and which are only stretched or flipped, are called eigenvectors. They reveal the structure of a linear transformation: in a basis of eigenvectors, the transformation simply scales each coordinate. Eigenvalues and eigenvectors appear throughout machine learning, for example in principal component analysis (Lecture 2), in the PageRank algorithm, in the analysis of Markov chains, and in spectral clustering.

Throughout this section, A is a real $n \times n$ matrix.

10.1 Eigenvalues, eigenvectors and eigenspaces

Definition 10.1 (Eigenvalue and eigenvector)**.** A number $\lambda \in \mathbb{R}$ is an eigenvalue of A if there is a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ with

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Such a vector $\mathbf{v}$ is called an eigenvector of A for the eigenvalue λ. Eigenvalues and eigenvectors of a linear transformation $T : V \to V$ are defined in the same way, by $T(v) = \lambda v$ with $v \neq 0$.

The condition $\mathbf{v} \neq \mathbf{0}$ is essential, since $A\mathbf{0} = \lambda\mathbf{0}$ for every λ. On the other hand, $\lambda = 0$ is allowed: the eigenvectors for the eigenvalue 0 are the nonzero vectors of $\mathrm{null}(A)$.

Example 10.2. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Then

$$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

so $(1, 1)^\top$ is an eigenvector for the eigenvalue 3, and $(1, -1)^\top$ is an eigenvector for the eigenvalue 1. On the other hand, $A\mathbf{e}_1 = (2, 1)^\top$ is not a multiple of $\mathbf{e}_1$, so $\mathbf{e}_1$ is not an eigenvector. Figure 6 shows what T_A does to the unit circle: it stretches the direction of $(1, 1)^\top$ by the factor 3, and leaves the direction of $(1, -1)^\top$ unchanged.

Definition 10.3 (Eigenspace)**.** For $\lambda \in \mathbb{R}$, the set

$$E_\lambda = \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \lambda\mathbf{v}\} = \mathrm{null}(A - \lambda I_n)$$

is called the eigenspace of A for λ. It is a subspace of $\mathbb{R}^n$ (Theorem 5.5), and λ is an eigenvalue if and only if $E_\lambda \neq \{0\}$; in that case, E_λ consists of the eigenvectors for λ together with 0.

Proposition 10.4. *A number λ is an eigenvalue of A if and only if $A - \lambda I_n$ is not invertible, if and only if $\det(A - \lambda I_n) = 0$. In particular, 0 is an eigenvalue of A if and only if A is not invertible.*

Proof. $A\mathbf{v} = \lambda\mathbf{v}$ means $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$. So λ is an eigenvalue if and only if $\mathrm{null}(A - \lambda I_n) \neq \{\mathbf{0}\}$, which by Corollary 7.11 means that $A - \lambda I_n$ is not invertible, or equivalently, that $\det(A - \lambda I_n) = 0$. □

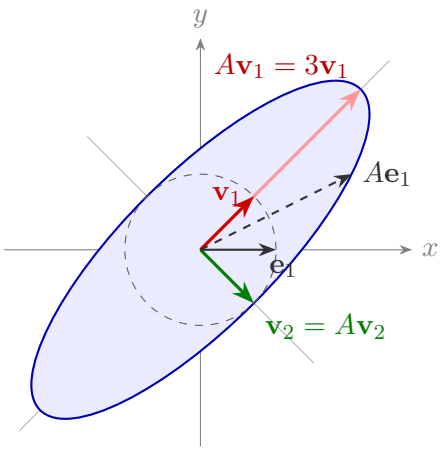


Figure 6: The matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ maps the unit circle (dashed) to an ellipse (blue). The eigenvector $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$ is stretched by its eigenvalue 3, and the eigenvector $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$ is left unchanged, since its eigenvalue is 1. The vector $\mathbf{e}_1$ is not an eigenvector: its image $A\mathbf{e}_1 = (2,1)$ points in a different direction.

10.2 The characteristic polynomial

Definition 10.5 (Characteristic polynomial)**.** The characteristic polynomial of A is

$$p_A(\lambda) = \det(A - \lambda I_n).$$

Expanding the determinant shows that $p_A(\lambda)$ is a polynomial of degree n in λ, with leading term $(-\lambda)^n$, which comes from the product $(A_{11} - \lambda)\cdots(A_{nn} - \lambda)$ of the diagonal entries. By Proposition 10.4, the eigenvalues of A are exactly the real roots of p_A. So, to find the eigenvalues and eigenvectors of A:

1. compute $p_A(\lambda)$ and find its real roots λ;
2. for each root λ, find the eigenspace $E_\lambda = \operatorname{null}(A - \lambda I_n)$ by row reduction.

Proposition 10.6. *For a 2×2 matrix A, $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\,\lambda + \det(A)$.*

Proof. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we get $p_A(\lambda) = (a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + (ad - bc)$. □

Example 10.7. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, we get $p_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$, so the eigenvalues are 1 and 3. The eigenspaces are

$$E_3 = \operatorname{null}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = \operatorname{span}\{(1,1)^\top\}, \qquad E_1 = \operatorname{null}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \operatorname{span}\{(1,-1)^\top\},$$

in agreement with Example 10.2.

Example 10.8. Let $J = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. Expanding along the first row,

$$p_J(\lambda) = \det \begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} = -\lambda(\lambda^2-1) - (-\lambda-1) + (1+\lambda) = -\lambda^3 + 3\lambda + 2 = -(\lambda-2)(\lambda+1)^2.$$

So the eigenvalues are 2 and -1. Row reduction gives

$$E_2 = \operatorname{null}(J - 2I_3) = \operatorname{span}\{(1,1,1)^\top\}, \qquad E_{-1} = \operatorname{null}(J + I_3) = \operatorname{null} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The eigenspace E_{-1} is the plane $x + y + z = 0$, with basis $(1,-1,0)^\top$, $(1,0,-1)^\top$.

Proposition 10.9. *The eigenvalues of a triangular matrix are its diagonal entries.*

Proof. If A is triangular, so is $A - \lambda I_n$, and by Corollary 6.86, $p_A(\lambda) = (A_{11} - \lambda) \cdots (A_{nn} - \lambda)$. □

Example 10.10 (Eigenvectors from geometry)**.** Geometric intuition often tells us the eigenvectors without any computation. The shear and the rotations below also appear in the gallery of Figure 2.

- The projection onto the x-axis, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, has the eigenvalue 1 with eigenspace the x-axis, and the eigenvalue 0 with eigenspace the y-axis.
- The reflection in the line $y = x$, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, fixes this line (eigenvalue 1) and flips the line $y = -x$ (eigenvalue -1).
- The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $p(\lambda) = (1-\lambda)^2$, so its only eigenvalue is 1, and $E_1 = \operatorname{null} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{span}\{\mathbf{e}_1\}$: only the horizontal direction is preserved.
- The rotation R_θ has $p(\lambda) = \lambda^2 - 2\cos\theta\,\lambda + 1$, with discriminant $4\cos^2\theta - 4 < 0$ unless θ is a multiple of π. So a rotation by an angle other than 0 or π has no real eigenvalues: it does not preserve any direction.

Remark 10.11 (Complex eigenvalues)*.* By the fundamental theorem of algebra (which we state without proof), every polynomial of degree n factors into n linear factors over $\mathbb{C}$. Hence

$$p_A(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda)$$

for some $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, which are called the complex eigenvalues of A, counted with multiplicity. For example, the rotation R_θ has the complex eigenvalues $\cos\theta \pm i\sin\theta$, with complex eigenvectors $(1, \mp i)^\top$. In these notes, "eigenvalue" means a real eigenvalue unless we say otherwise.

Theorem 10.12. *Let $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ be the complex eigenvalues of A, counted with multiplicity. Then*

$$\det(A) = \lambda_1\lambda_2 \cdots \lambda_n \qquad \textit{and} \qquad \operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Proof. Setting $\lambda = 0$ in the factorization of p_A gives $\det(A) = p_A(0) = \lambda_1 \cdots \lambda_n$. For the trace, we compare the coefficients of λ^{n-1}. In $(\lambda_1 - \lambda)\cdots(\lambda_n - \lambda)$, this coefficient is $(-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$. In $\det(A - \lambda I_n)$, one can check by induction on n, using the cofactor expansion, that only the product $(A_{11} - \lambda)\cdots(A_{nn} - \lambda)$ of the diagonal entries contributes to λ^{n-1}, since every other term contains at most $n-2$ diagonal entries. So the coefficient is $(-1)^{n-1}(A_{11} + \cdots + A_{nn}) = (-1)^{n-1}\operatorname{tr}(A)$. □

Example 10.13. For the matrix of Example 10.7, $1 + 3 = 4 = \operatorname{tr}(A)$ and $1 \cdot 3 = 3 = \det(A)$. For the matrix J of Example 10.8, $2 + (-1) + (-1) = 0 = \operatorname{tr}(J)$ and $2 \cdot (-1) \cdot (-1) = 2 = \det(J)$. For the rotation R_θ, $(\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta) = 1 = \det(R_\theta)$ and the sum is $2\cos\theta = \operatorname{tr}(R_\theta)$.

Proposition 10.14. *Let $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then:*

1. *$A^k\mathbf{v} = \lambda^k\mathbf{v}$ for every $k \geq 1$;*
2. *if A is invertible, then $\lambda \neq 0$ and $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$;*
3. *$(A + cI_n)\mathbf{v} = (\lambda + c)\mathbf{v}$ for every $c \in \mathbb{R}$.*

Moreover, A and $A^\top$ have the same characteristic polynomial, and hence the same eigenvalues.

Proof. (1) By induction, $A^k\mathbf{v} = A(A^{k-1}\mathbf{v}) = A(\lambda^{k-1}\mathbf{v}) = \lambda^{k-1}A\mathbf{v} = \lambda^k\mathbf{v}$. (2) $\lambda \neq 0$ by Proposition 10.4, and applying A^{-1} to $A\mathbf{v} = \lambda\mathbf{v}$ gives $\mathbf{v} = \lambda A^{-1}\mathbf{v}$. (3) $(A + cI_n)\mathbf{v} = A\mathbf{v} + c\mathbf{v} = (\lambda + c)\mathbf{v}$. Finally, $\det(A^\top - \lambda I_n) = \det\left((A - \lambda I_n)^\top\right) = \det(A - \lambda I_n)$. □

Note that A and $A^\top$ usually have different eigenvectors; for the shear, $E_1 = \operatorname{span}\{\mathbf{e}_1\}$ for A, but $E_1 = \operatorname{span}\{\mathbf{e}_2\}$ for $A^\top$.

10.3 Diagonalization

Theorem 10.15. *Eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ of A for distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ are linearly independent.*

Proof. Suppose not, and let m be the smallest index such that $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are linearly dependent. Then $m \geq 2$ (since $\mathbf{v}_1 \neq \mathbf{0}$), $\mathbf{v}_1, \ldots, \mathbf{v}_{m-1}$ are linearly independent, and $\mathbf{v}_m = c_1\mathbf{v}_1 + \cdots + c_{m-1}\mathbf{v}_{m-1}$ for some scalars c_i. Applying $A - \lambda_m I_n$ to both sides gives

$$0 = c_1(\lambda_1 - \lambda_m)\mathbf{v}_1 + \cdots + c_{m-1}(\lambda_{m-1} - \lambda_m)\mathbf{v}_{m-1}.$$

By independence, $c_i(\lambda_i - \lambda_m) = 0$, and since the eigenvalues are distinct, $c_i = 0$ for all i. But then $\mathbf{v}_m = \mathbf{0}$, a contradiction. □

Definition 10.16 (Diagonalizable matrix)**.** A matrix A is called diagonalizable if there are an invertible matrix P and a diagonal matrix D with

$$A = PDP^{-1}, \qquad \text{or equivalently} \qquad P^{-1}AP = D.$$

More generally, two matrices A and B are called similar if $B = P^{-1}AP$ for some invertible matrix P.

Similar matrices have the same characteristic polynomial, since by the product rule for determinants,

$$\det(P^{-1}AP - \lambda I_n) = \det\left(P^{-1}(A - \lambda I_n)P\right) = \det(P^{-1})\det(A - \lambda I_n)\det(P) = \det(A - \lambda I_n).$$

Hence they also have the same eigenvalues, trace and determinant (compare Subsection 6.5).

Theorem 10.17 (Diagonalization)**.** *An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_n$. In this case, $A = PDP^{-1}$, where P is the matrix with columns $\mathbf{p}_1, \dots, \mathbf{p}_n$, and $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ contains the corresponding eigenvalues, in the same order.*

Proof. Let P have columns $\mathbf{p}_1, \dots, \mathbf{p}_n$, and let $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. By the column-by-column description of matrix multiplication, the i-th column of AP is $A\mathbf{p}_i$, and the i-th column of PD is $\lambda_i \mathbf{p}_i$. So $AP = PD$ if and only if $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$ for all i. Moreover, P is invertible if and only if $\operatorname{null}(P) = \{\mathbf{0}\}$ (Corollary 7.11), i.e., if and only if its columns are linearly independent. Hence: if $\mathbf{p}_1, \dots, \mathbf{p}_n$ are linearly independent eigenvectors, then $AP = PD$ with P invertible, so $A = PDP^{-1}$. Conversely, if $A = PDP^{-1}$, then $AP = PD$, so the columns of P are eigenvectors (they are nonzero, since P is invertible), and they are linearly independent. □

Corollary 10.18. *If an $n \times n$ matrix has n distinct real eigenvalues, then it is diagonalizable.*

Proof. Choose an eigenvector for each eigenvalue. By Theorem 10.15, these n eigenvectors are linearly independent. □

Remark 10.19. If $A = PDP^{-1}$, then the columns $\mathbf{p}_1, \dots, \mathbf{p}_n$ of P form a basis $\mathcal{B}$ of $\mathbb{R}^n$, and $P^{-1}\mathbf{v} = [\mathbf{v}]_{\mathcal{B}}$ is the coordinate vector of $\mathbf{v}$ with respect to this basis. So $P^{-1}AP = D$ says that the matrix of T_A with respect to the basis of eigenvectors is diagonal (Subsection 6.3): in these coordinates, T_A simply multiplies the i-th coordinate by λ_i.

Example 10.20. For the matrix A of Example 10.7,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}}_{D} \underbrace{\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{P^{-1}}.$$

Not every matrix is diagonalizable. To see what can go wrong, we count eigenvectors.

Definition 10.21 (Multiplicities)**.** Let λ be an eigenvalue of A. Its algebraic multiplicity is its multiplicity as a root of p_A, i.e., the number of times the factor $(\lambda - \cdot)$ appears in p_A. Its geometric multiplicity is $\dim E_\lambda = n - \operatorname{rank}(A - \lambda I_n)$ (by the rank–nullity theorem).

Theorem 10.22. *For every eigenvalue, $1 \le$ geometric multiplicity $\le$ algebraic multiplicity. Moreover, A is diagonalizable if and only if p_A has n real roots (counted with multiplicity), and the geometric multiplicity of every eigenvalue equals its algebraic multiplicity.*

Proof. Let λ be an eigenvalue with geometric multiplicity g; then $g \ge 1$, since E_λ contains an eigenvector. Choose a basis $\mathbf{p}_1, \dots, \mathbf{p}_g$ of E_λ, extend it to a basis $\mathbf{p}_1, \dots, \mathbf{p}_n$ of $\mathbb{R}^n$ (Corollary 3.14), and let P be the invertible matrix with these columns, so that $P\mathbf{e}_i = \mathbf{p}_i$. For $i \le g$, the i-th column of $P^{-1}AP$ is $P^{-1}A\mathbf{p}_i = \lambda P^{-1}\mathbf{p}_i = \lambda \mathbf{e}_i$. Hence

$$P^{-1}AP = \begin{pmatrix} \lambda I_g & B \\ 0 & C \end{pmatrix}$$

for some $g \times (n-g)$ matrix B and some $(n-g) \times (n-g)$ matrix C. Similar matrices have the same characteristic polynomial, and expanding $\det(P^{-1}AP - tI_n)$ along the first column g times (each time, the only nonzero entry of the first column is $\lambda - t$) gives $p_A(t) = (\lambda - t)^g \det(C - tI_{n-g})$. So the factor $\lambda - t$ appears at least g times in p_A: the algebraic multiplicity is at least g.

Now suppose that $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then $p_A(t) = p_D(t) = (\lambda_1 - t) \cdots (\lambda_n - t)$ has n real roots. Let λ be an eigenvalue, and let m be the number of indices i with $\lambda_i = \lambda$, which is its algebraic multiplicity. Since $A - \lambda I_n = P(D - \lambda I_n)P^{-1}$, a vector $\mathbf{v}$ lies in E_λ if and only if $P^{-1}\mathbf{v} \in \operatorname{null}(D - \lambda I_n) = \operatorname{span}\{\mathbf{e}_i : \lambda_i = \lambda\}$, i.e., if and only if $\mathbf{v}$ is a linear combination of the m columns $\mathbf{p}_i$ of P with $\lambda_i = \lambda$. These columns are linearly independent, so $\dim E_\lambda = m$.

Conversely, suppose that $p_A(t) = (\mu_1 - t)^{m_1} \cdots (\mu_k - t)^{m_k}$ with distinct real numbers μ_j and $m_1 + \cdots + m_k = n$, and that $\dim E_{\mu_j} = m_j$ for all j. Choose a basis of each eigenspace E_{μ_j}; together, these are n eigenvectors. They are linearly independent: in a linear combination of them that equals 0, collect the terms from each eigenspace, so that $w_1 + \cdots + w_k = 0$ with $w_j \in E_{\mu_j}$. If some w_j were nonzero, then the nonzero ones would be eigenvectors for distinct eigenvalues whose sum is 0, which contradicts Theorem 10.15. So every $w_j = 0$, and since w_j is a linear combination of a basis of E_{μ_j}, all its coefficients are 0. By Theorem 10.17, A is diagonalizable. □

The second part says that we need "enough" eigenvectors: A is diagonalizable exactly when the eigenspaces together contain n linearly independent vectors.

Example 10.23.

- The matrix J of Example 10.8 has the eigenvalue -1 with algebraic multiplicity 2, and geometric multiplicity $3 - \operatorname{rank}(J + I_3) = 3 - 1 = 2$. So J is diagonalizable:

$$J = PDP^{-1} \qquad \text{with} \qquad P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \quad D = \operatorname{diag}(2, -1, -1).$$

- The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the eigenvalue 1 with algebraic multiplicity 2, but geometric multiplicity 1. So it is not diagonalizable.
- A rotation R_θ with θ not a multiple of π has no real eigenvalues, so it is not diagonalizable with real matrices P and D. (It is diagonalizable with complex matrices.)

Remark 10.24. Symmetric matrices are always diagonalizable, and their eigenvectors can even be chosen to be orthonormal. This is the spectral theorem, which we prove in Lecture 2 once we have inner products.

10.4 The Cayley–Hamilton theorem and the minimal polynomial

Matrices can be substituted into polynomials: for $q(t) = a_0 + a_1 t + \cdots + a_m t^m$ and an $n \times n$ matrix A, let $q(A) = a_0 I_n + a_1 A + \cdots + a_m A^m$. Since powers of A commute with each other, $(qr)(A) = q(A)r(A)$ for all polynomials q and r. If $A\mathbf{v} = \lambda\mathbf{v}$, then $q(A)\mathbf{v} = q(\lambda)\mathbf{v}$, by Proposition 10.14.

Theorem 10.25 (Cayley–Hamilton)**.** *Every square matrix A satisfies $p_A(A) = 0$.*

Substituting A for λ in $\det(A - \lambda I_n)$ would give $\det(A - A) = 0$, a number rather than the zero matrix, so this is not a proof; we use the cofactor formula instead.

Proof. For $n = 1$, $p_A(t) = A_{11} - t$, and $p_A(A) = 0$. Let $n \geq 2$, and let $B(t)$ be the transposed cofactor matrix of $A - tI_n$, so that $(A - tI_n)B(t) = p_A(t)\, I_n$ for every $t \in \mathbb{R}$, by Proposition 6.96. Each entry of $B(t)$ is $\pm$ the determinant of an $(n-1) \times (n-1)$ matrix whose entries are polynomials of degree at most 1 in t, so it is a polynomial of degree at most $n-1$. Collecting powers of t, we can write $B(t) = B_0 + B_1 t + \cdots + B_{n-1} t^{n-1}$ with constant matrices B_j. Let $p_A(t) = c_0 + c_1 t + \cdots + c_n t^n$.

Comparing the coefficients of t^j on both sides of $(A - tI_n)B(t) = p_A(t)\, I_n$ (entry by entry, two polynomials that agree for all t have the same coefficients) gives

$$AB_0 = c_0 I_n, \qquad AB_j - B_{j-1} = c_j I_n \quad (1 \le j \le n-1), \qquad -B_{n-1} = c_n I_n.$$

We multiply the equation for t^j by A^j from the left, and add all of them:

$$p_A(A) = AB_0 + \sum_{j=1}^{n-1} (A^{j+1}B_j - A^j B_{j-1}) - A^n B_{n-1} = 0,$$

since the sum telescopes: each term $A^{j+1}B_j$ with $0 \le j \le n-1$ appears once with a plus sign and once with a minus sign. □

So A^n is a linear combination of $I_n, A, \ldots, A^{n-1}$, and by induction, so is every higher power of A. If A is invertible, then $c_0 = p_A(0) = \det(A) \neq 0$, and multiplying $p_A(A) = 0$ by A^{-1} gives

$$A^{-1} = -\frac{1}{c_0}(c_1 I_n + c_2 A + \cdots + c_n A^{n-1}) :$$

the inverse is a polynomial in A. This is the idea behind the Krylov subspace methods for large linear systems, such as the conjugate gradient method, which approximate $A^{-1}\mathbf{b}$ by vectors in $\operatorname{span}\{\mathbf{b}, A\mathbf{b}, \ldots, A^{k-1}\mathbf{b}\}$ and only need matrix-vector products.

Example 10.26. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $p_A(t) = t^2 - 4t + 3$, so $A^2 = 4A - 3I_2$. Hence $A^3 = 4A^2 - 3A = 13A - 12I_2 = \begin{pmatrix} 14 & 13 \\ 13 & 14 \end{pmatrix}$, and $A(4I_2 - A) = 3I_2$ gives $A^{-1} = \frac{1}{3}(4I_2 - A) = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

The characteristic polynomial is not always the polynomial of smallest degree that A satisfies: I_n satisfies $I_n - I_n = 0$, a polynomial equation of degree 1, while $p_{I_n}(t) = (1-t)^n$.

Definition 10.27 (Minimal polynomial)**.** The minimal polynomial m_A of a square matrix A is a polynomial of smallest degree with leading coefficient 1 such that $m_A(A) = 0$.

Such polynomials exist, since $(-1)^n p_A$ has leading coefficient 1 and annihilates A by the Cayley–Hamilton theorem.

Proposition 10.28. *Let A be a square matrix.*

1. *If $q(A) = 0$ for a polynomial q, then m_A divides q. In particular, m_A divides p_A, and m_A is unique.*
2. *The real roots of m_A are exactly the eigenvalues of A.*

Proof. (1) Division with remainder gives $q = sm_A + r$ with polynomials s and r, where $r = 0$ or $\deg r < \deg m_A$. Then $r(A) = q(A) - s(A)m_A(A) = 0$, so $r = 0$ by the minimality of $\deg m_A$ (otherwise, r divided by its leading coefficient would be a smaller polynomial annihilating A). If m and m' are two minimal polynomials, each divides the other, and since both have leading coefficient 1, $m = m'$.

(2) If $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{0} = m_A(A)\mathbf{v} = m_A(\lambda)\mathbf{v}$, so $m_A(\lambda) = 0$. Conversely, let $m_A(\lambda) = 0$, and write $m_A(t) = (t - \lambda)s(t)$. Since $\deg s < \deg m_A$, $s(A) \neq 0$, so there is a vector $\mathbf{w}$ with $\mathbf{v} = s(A)\mathbf{w} \neq \mathbf{0}$. Then $(A - \lambda I_n)\mathbf{v} = m_A(A)\mathbf{w} = \mathbf{0}$, so $\mathbf{v}$ is an eigenvector for λ. □

Theorem 10.29. *A square matrix A is diagonalizable if and only if $m_A(t) = (t-\mu_1)\cdots(t-\mu_k)$ for distinct real numbers $\mu_1, \ldots, \mu_k$.*

Proof. Let $A = PDP^{-1}$, and let $\mu_1, \ldots, \mu_k$ be the distinct diagonal entries of D, the eigenvalues of A. For $q(t) = (t-\mu_1)\cdots(t-\mu_k)$, the matrix $q(D)$ is diagonal with the entries $q(\lambda_i) = 0$, so $q(A) = Pq(D)P^{-1} = 0$. By Proposition 10.28, m_A divides q, so m_A is a product of some of the distinct factors $t - \mu_j$; since every eigenvalue μ_j is a root of m_A, all of them appear, and $m_A = q$.

Conversely, let $m_A(t) = (t-\mu_1)\cdots(t-\mu_k)$ with distinct μ_j, and consider the Lagrange polynomials

$$\ell_j(t) = \prod_{i \neq j} \frac{t-\mu_i}{\mu_j - \mu_i}, \qquad \text{with } \ell_j(\mu_i) = \delta_{ij}.$$

The polynomial $\ell_1 + \cdots + \ell_k - 1$ has degree at most $k-1$ and the k roots $\mu_1, \ldots, \mu_k$, so it is 0 (a nonzero polynomial of degree at most $k-1$ has at most $k-1$ roots). Hence $I_n = \ell_1(A) + \cdots + \ell_k(A)$, and every $\mathbf{v} \in \mathbb{R}^n$ is the sum of the vectors $\ell_j(A)\mathbf{v}$. Each of them lies in the eigenspace E_{μ_j}, since $(t-\mu_j)\ell_j(t)$ is a constant multiple of $m_A(t)$, so that $(A - \mu_j I_n)\ell_j(A)\mathbf{v} = \mathbf{0}$. So the eigenspaces together span $\mathbb{R}^n$. Bases of the eigenspaces together are linearly independent, as in the proof of Theorem 10.22, so they form a basis of $\mathbb{R}^n$ consisting of eigenvectors, and A is diagonalizable by Theorem 10.17. □

Example 10.30.

- For the shear $N = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, the only eigenvalue is 1, so m_N is $t-1$ or $(t-1)^2$. Since $N - I_2 \neq 0$ but $(N-I_2)^2 = 0$, $m_N = (t-1)^2$ has a repeated root, and N is not diagonalizable.
- For the matrix J of Example 10.8, $(J - 2I_3)(J + I_3) = 0$, and both 2 and -1 must be roots of m_J, so $m_J = (t-2)(t+1)$, and J is diagonalizable, without computing any eigenvectors.
- If $P^2 = P$, then m_P divides $t^2 - t = t(t-1)$, so its roots lie in $\{0, 1\}$ and are not repeated, and P is diagonalizable. Similarly, every matrix with $A^2 = I_n$, such as a reflection, is diagonalizable, with eigenvalues in $\{1, -1\}$.

10.5 Powers of matrices and applications

Proposition 10.31. *If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then for every $k \geq 1$,*

$$A^k = PD^kP^{-1}, \qquad \textit{where} \quad D^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k).$$

Proof. $A^k = (PDP^{-1})(PDP^{-1})\cdots(PDP^{-1}) = PD(P^{-1}P)D(P^{-1}P)\cdots DP^{-1} = PD^kP^{-1}$, and D^k is computed entry by entry, since products of diagonal matrices are computed on the diagonal (Subsection 6.4). □

Example 10.32. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$,

$$A^k = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

Computing A^{100} by multiplying A with itself 99 times would be much more work.

Example 10.33 (Fibonacci numbers)**.** The Fibonacci numbers $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$ satisfy

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} = \cdots = M^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \text{where} \quad M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial of M is $\lambda^2 - \lambda - 1$, with the roots

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618 \quad \text{(the golden ratio)} \qquad \text{and} \qquad \psi = \frac{1-\sqrt{5}}{2} \approx -0.618.$$

One checks that $(\varphi, 1)^\top$ and $(\psi, 1)^\top$ are eigenvectors, and that $(1,0)^\top = \frac{1}{\sqrt{5}}\big((\varphi,1)^\top - (\psi,1)^\top\big)$, since $\varphi - \psi = \sqrt{5}$. By Proposition 10.14,

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \frac{1}{\sqrt{5}} \left(\varphi^n \begin{pmatrix} \varphi \\ 1 \end{pmatrix} - \psi^n \begin{pmatrix} \psi \\ 1 \end{pmatrix} \right), \qquad \text{so} \qquad F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

This is Binet's formula. Since $|\psi| < 1$, the term ψ^n quickly becomes tiny, and F_n is the integer closest to $\varphi^n/\sqrt{5}$. In particular, $F_{n+1}/F_n \to \varphi$.

Example 10.34 (Markov chains)**.** Suppose that the weather is either sunny or rainy, and that a sunny day is followed by a sunny day with probability 0.9, while a rainy day is followed by a sunny day with probability 0.5. If $\mathbf{x}_k = (s_k, r_k)^\top$ contains the probabilities that day k is sunny or rainy, then

$$\mathbf{x}_{k+1} = A\mathbf{x}_k, \qquad \text{where} \quad A = \begin{pmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{pmatrix}.$$

The columns of A are probability vectors (nonnegative entries summing to 1). By Proposition 10.6, $p_A(\lambda) = \lambda^2 - 1.4\lambda + 0.4 = (\lambda - 1)(\lambda - 0.4)$, with eigenvectors $(5,1)^\top$ for $\lambda = 1$ and $(1,-1)^\top$ for $\lambda = 0.4$. If today is rainy, then $\mathbf{x}_0 = (0,1)^\top = \big(\frac{5}{6}, \frac{1}{6}\big)^\top - \frac{5}{6}(1,-1)^\top$, so

$$\mathbf{x}_k = A^k \mathbf{x}_0 = \begin{pmatrix} 5/6 \\ 1/6 \end{pmatrix} - \frac{5}{6}\,(0.4)^k \begin{pmatrix} 1 \\ -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 5/6 \\ 1/6 \end{pmatrix} \qquad (k \to \infty).$$

In the long run, $\frac{5}{6}$ of the days are sunny, whatever the weather today. The limit $\big(\frac{5}{6}, \frac{1}{6}\big)^\top$ is the eigenvector for the eigenvalue 1, scaled to be a probability vector; it is called the stationary distribution of the Markov chain.

Remark 10.35 (PageRank and power iteration). The PageRank algorithm, originally used by Google to rank web pages, models a random surfer who follows a random link on each page. The importance of a page is its probability in the stationary distribution of this Markov chain, i.e., an entry of the eigenvector for the eigenvalue 1 of a huge matrix. Such eigenvectors are computed by power iteration: if A has an eigenvalue λ_1 with $|\lambda_1| > |\lambda_i|$ for all other eigenvalues, and A is diagonalizable with eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, then writing $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ gives

$$A^k\mathbf{x} = \lambda_1^k \Big(c_1\mathbf{v}_1 + \sum_{i \geq 2} c_i \Big(\frac{\lambda_i}{\lambda_1}\Big)^k \mathbf{v}_i \Big),$$

so after rescaling, $A^k\mathbf{x}$ points more and more in the direction of $\mathbf{v}_1$ (if $c_1 \neq 0$). You will implement this in Programming Exercise P10.

10.6 Convolutions and circulant matrices

Convolutional neural networks process images and signals with filters that are applied in the same way at every position. We study the simplest version of this: signals $\mathbf{x} \in \mathbb{R}^n$ whose indices $0, 1, \ldots, n-1$ are taken modulo n, so that, for example, $x_{-1} = x_{n-1}$, as if the signal were periodic. In this subsection, the entries of vectors and matrices are numbered from 0.

Definition 10.36 (Cyclic convolution and circulant matrices)**.** The cyclic convolution of $\mathbf{c}, \mathbf{x} \in \mathbb{R}^n$ is the vector $\mathbf{c} * \mathbf{x} \in \mathbb{R}^n$ with the entries

$$(\mathbf{c} * \mathbf{x})_j = \sum_{k=0}^{n-1} c_k x_{j-k} \qquad (j = 0, \ldots, n-1),$$

where indices are taken modulo n. For fixed $\mathbf{c}$, it is linear in $\mathbf{x}$, with the matrix $\mathrm{circ}(\mathbf{c})$ whose entries are $\mathrm{circ}(\mathbf{c})_{jk} = c_{j-k}$. Such matrices are called circulant: the first column is $\mathbf{c}$, and each further column is the previous one, shifted down cyclically by one position.

For example, for $n = 4$ and $\mathbf{c} = (2, 1, 0, 1)^\top$, we get $(\mathbf{c} * \mathbf{x})_j = 2x_j + x_{j-1} + x_{j+1}$, which smooths $\mathbf{x}$ by averaging each entry with its neighbors (up to the factor 4), and

$$\mathrm{circ}(\mathbf{c}) = \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \end{pmatrix}, \qquad \mathbf{c} * \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \\ 12 \\ 12 \end{pmatrix}.$$

A convolutional layer learns a filter $\mathbf{c}$ with only a few nonzero entries; for images, the filters are small two-dimensional arrays, such as 3×3 filters, and the convolution is defined with two indices in the same way. (Deep learning libraries compute $\sum_k c_k x_{j+k}$ and still call it a convolution; this is the same operation with $\mathbf{c}$ reversed.)

Let $S = \mathrm{circ}(\mathbf{e}_1)$, where $\mathbf{e}_1 = (0, 1, 0, \ldots, 0)^\top$ in the numbering from 0, be the cyclic shift, with $(S\mathbf{x})_j = x_{j-1}$, which moves every entry one position down and the last entry to the top.

Proposition 10.37. *1.* $\mathrm{circ}(\mathbf{c}) = c_0 I_n + c_1 S + c_2 S^2 + \cdots + c_{n-1} S^{n-1}$.

2. An $n \times n$ matrix A commutes with S, i.e., $AS = SA$, if and only if A is circulant.

Proof. (1) S^k shifts by k positions, $(S^k \mathbf{x})_j = x_{j-k}$, so $\left(\sum_k c_k S^k \mathbf{x}\right)_j = \sum_k c_k x_{j-k} = (\mathbf{c} * \mathbf{x})_j$.

(2) By (1), a circulant matrix is a polynomial in S, so it commutes with S. Conversely, the k-th column of AS is A applied to the k-th column $\mathbf{e}_{k+1}$ of S, so $(AS)_{jk} = A_{j,k+1}$, and SA is A with its rows shifted down, so $(SA)_{jk} = A_{j-1,k}$. Hence $AS = SA$ means that $A_{j+1,k+1} = A_{jk}$ for all j and k: the entries of A are constant along the cyclic diagonals, i.e., A_{jk} only depends on $j - k$ modulo n, and $A = \mathrm{circ}(\mathbf{c})$ with $c_m = A_{m0}$. □

Part 2 says that the linear maps that commute with shifts, i.e., that treat every position in the same way (shifting the input shifts the output), are exactly the convolutions. This is why convolutional networks use convolutions: their layers are translation equivariant, and a pattern is detected wherever it appears in an image. The eigenvectors of circulant matrices are complex. Let $\omega = e^{2\pi i/n} = \cos\frac{2\pi}{n} + i \sin\frac{2\pi}{n}$, so that $\omega^n = 1$, and let

$$\mathbf{f}_m = (1, \omega^m, \omega^{2m}, \ldots, \omega^{(n-1)m})^\top \in \mathbb{C}^n \qquad (m = 0, \ldots, n-1),$$

the discrete Fourier basis, which consists of the sampled oscillations with the frequencies m. Let F be the $n \times n$ matrix with the columns $\mathbf{f}_0, \dots, \mathbf{f}_{n-1}$, i.e., $F_{jm} = \omega^{jm}$, and let $\overline{F}$ be its complex conjugate. The discrete Fourier transform (DFT) of $\mathbf{x} \in \mathbb{R}^n$ is $\hat{\mathbf{x}} = \overline{F}\mathbf{x}$, with $\hat{x}_m = \sum_j x_j \omega^{-jm}$. We use complex numbers as in the remark on complex eigenvalues; the rules of matrix algebra hold over $\mathbb{C}$ as well.

Theorem 10.38 (The DFT diagonalizes convolutions)**.** *1. $\overline{F}F = nI_n$, so F is invertible, with $F^{-1} = \frac{1}{n}\overline{F}$.*

2. $\operatorname{circ}(\mathbf{c})\mathbf{f}_m = \hat{c}_m \mathbf{f}_m$ for every m, so $\operatorname{circ}(\mathbf{c}) = F \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1}) F^{-1}$: the eigenvalues of a convolution are the DFT of its filter.

3. (Convolution theorem.) $\widehat{\mathbf{c} * \mathbf{x}}_m = \hat{c}_m \hat{x}_m$ for every m.

Proof. (1) The (l, m) entry of $\overline{F}F$ is $\sum_j \omega^{-jl}\omega^{jm} = \sum_{j=0}^{n-1} r^j$ with $r = \omega^{m-l}$. For $l = m$, this is n. Otherwise $0 < |m - l| < n$, so $r \neq 1$, and the geometric sum is $\frac{r^n - 1}{r - 1} = 0$, since $r^n = (\omega^n)^{m-l} = 1$.

(2) Since $\omega^n = 1$, powers of ω only depend on exponents modulo n, and

$$\big(\operatorname{circ}(\mathbf{c})\mathbf{f}_m\big)_j = \sum_k c_k \omega^{(j-k)m} = \Big(\sum_k c_k \omega^{-km}\Big)\omega^{jm} = \hat{c}_m (\mathbf{f}_m)_j.$$

So $\operatorname{circ}(\mathbf{c})F = F \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1})$, and we multiply by F^{-1} from the right.

(3) By (1) and (2), $\overline{F}(\mathbf{c} * \mathbf{x}) = \overline{F}F \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1}) \frac{1}{n}\overline{F}\mathbf{x} = \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1})\hat{\mathbf{x}}$. □

All circulant matrices have the same eigenvectors $\mathbf{f}_m$, so they commute with each other. For the smoothing filter $\mathbf{c} = (2, 1, 0, 1)^\top$ above, $\omega = i$ and $\hat{\mathbf{c}} = (4, 2, 0, 2)$: the constant signal $\mathbf{f}_0$ is multiplied by 4, and the fastest oscillation $\mathbf{f}_2 = (1, -1, 1, -1)^\top$ is mapped to $\mathbf{0}$. Smoothing damps high frequencies, and since 0 is an eigenvalue, it cannot be undone. Theorem 10.38 is also how convolutions are computed in practice: the fast Fourier transform (FFT) computes $\hat{\mathbf{x}}$ with $O(n \log n)$ operations instead of the $O(n^2)$ operations of a matrix-vector product, so $\mathbf{c} * \mathbf{x}$ is computed as the inverse DFT of $\hat{c}_m \hat{x}_m$.

11 Tensors

Data in machine learning often have more than two indices. A color image of $H \times W$ pixels has three color channels, so it is described by $3HW$ numbers x_{cij}, indexed by a channel c, a row i and a column j; a batch of N such images has four indices; and a batch of N sentences of L tokens, each represented by an embedding of dimension d, has three. Deep learning libraries such as PyTorch and TensorFlow call such arrays tensors. In mathematics, a tensor is an element of a tensor product of vector spaces, a vector space in which vectors can be multiplied with each other, bilinearly. Arrays are the coordinates of tensors with respect to bases, just as a list of numbers is the coordinate vector of a vector (Theorem 4.27). In this section, we construct tensor products with the quotient spaces of Section 9, and we derive the properties of tensors, including their description by arrays, from this construction. We work over $\mathbb{R}$, and all vector spaces are finite-dimensional, except for the auxiliary spaces $\mathcal{F}_0(S)$ below.

11.1 Multilinear maps

Definition 11.1 (Multilinear map)**.** Let $V_1, \dots, V_k$ and W be vector spaces. A map $\tau : V_1 \times \cdots \times V_k \to W$ is multilinear (or k-linear) if it is linear in each argument when the other arguments are

fixed: for each j, all $v_j, v_j' \in V_j$ and all $a \in \mathbb{R}$,

$$\tau(\dots, v_j + v_j', \dots) = \tau(\dots, v_j, \dots) + \tau(\dots, v_j', \dots),$$
$$\tau(\dots, av_j, \dots) = a\,\tau(\dots, v_j, \dots),$$

where the dots stand for the same vectors in all terms. A multilinear map with $W = \mathbb{R}$ is called a multilinear form; for $k = 2$, we speak of bilinear maps and bilinear forms.

A multilinear map is *not* a linear map on the product space $V_1 \times \cdots \times V_k$: for a bilinear map, $\tau(2v_1, 2v_2) = 4\,\tau(v_1, v_2)$, and not $2\,\tau(v_1, v_2)$.

Example 11.2.

1. The dot product $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^\top \mathbf{y} = x_1 y_1 + \cdots + x_n y_n$ is a bilinear form on $\mathbb{R}^n \times \mathbb{R}^n$.

2. For every $m \times n$ matrix A, the map

$$\beta_A(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y} = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} x_i y_j$$

   is a bilinear form on $\mathbb{R}^m \times \mathbb{R}^n$: by Proposition 6.22, $\mathbf{x}^\top A\mathbf{y}$ is linear in $\mathbf{y}$ for fixed $\mathbf{x}$, and linear in $\mathbf{x}$ for fixed $\mathbf{y}$.

3. Matrix multiplication $(A, B) \mapsto AB$ is a bilinear map $\mathbb{R}^{m\times n} \times \mathbb{R}^{n\times p} \to \mathbb{R}^{m\times p}$, by the distributive laws.

4. The determinant of an $n \times n$ matrix is an n-linear form of its columns: by the expansion along the j-th column (Theorem 6.84), $\det(A) = \sum_l (-1)^{l+j} A_{lj} \det(A^{(lj)})$, where the matrices $A^{(lj)}$ do not involve the j-th column, so $\det(A)$ is a linear function of the j-th column when the other columns are fixed. But the determinant is not a linear function of the matrix (Example 8.4).

Bilinear and multilinear maps appear everywhere, but they are not linear, so the theory of the previous sections does not apply to them directly. The tensor product is the vector space on which they become linear.

11.2 The tensor product of two vector spaces

We want to multiply a vector $v \in V$ and a vector $w \in W$ to a product $v \otimes w$ that depends bilinearly on v and w, and that satisfies no other rules. To construct it, we first form all formal linear combinations of pairs (v, w), and then we impose the rules of bilinearity by a quotient.

Definition 11.3 (Free vector space)**.** Let S be a set. The free vector space on S is the set $\mathcal{F}_0(S)$ of all functions $f : S \to \mathbb{R}$ with $f(s) \neq 0$ for only finitely many $s \in S$, with the pointwise operations. For $s \in S$, let $\delta_s \in \mathcal{F}_0(S)$ be the function with $\delta_s(s) = 1$ and $\delta_s(t) = 0$ for $t \neq s$.

$\mathcal{F}_0(S)$ is a subspace of $\mathcal{F}(S)$ (Proposition 2.10): it contains the zero function, and sums and multiples of functions that vanish outside finite sets again vanish outside finite sets. Every $f \in \mathcal{F}_0(S)$ can be written in exactly one way as a finite linear combination $f = \sum_s f(s)\,\delta_s$ of the functions δ_s, since two functions are equal if they agree at every point. So $\{\delta_s : s \in S\}$ is a basis of $\mathcal{F}_0(S)$. We think of δ_s as the element s itself, and of $\mathcal{F}_0(S)$ as the space of formal linear combinations $a_1 s_1 + \cdots + a_r s_r$ of elements of S. For $S = \{1, \dots, n\}$, it is $\mathbb{R}^n$, with $\delta_i = \mathbf{e}_i$. For $S = V \times W$, however, S is an infinite set, and $\mathcal{F}_0(V \times W)$ is a huge space, in which, for example, $\delta_{(2v,w)}$ and $2\,\delta_{(v,w)}$ are different vectors (for $v \neq 0$).

Definition 11.4 (Tensor product)**.** Let V and W be vector spaces, and let $R \subseteq \mathcal{F}_0(V \times W)$ be the subspace spanned by all the vectors

$$\delta_{(v+v',w)} - \delta_{(v,w)} - \delta_{(v',w)}, \qquad \delta_{(v,w+w')} - \delta_{(v,w)} - \delta_{(v,w')},$$
$$\delta_{(av,w)} - a\,\delta_{(v,w)}, \qquad \delta_{(v,aw)} - a\,\delta_{(v,w)},$$

for all $v, v' \in V$, $w, w' \in W$ and $a \in \mathbb{R}$. The tensor product of V and W is the quotient space

$$V \otimes W = \mathcal{F}_0(V \times W)/R,$$

and for $v \in V$ and $w \in W$, we write $v \otimes w = \delta_{(v,w)} + R$. The elements of $V \otimes W$ are called tensors (of order 2), and the tensors of the form $v \otimes w$ are called pure tensors.

Proposition 11.5. *1. The map $V \times W \to V \otimes W$, $(v, w) \mapsto v \otimes w$, is bilinear: for all $v, v' \in V$, $w, w' \in W$ and $a \in \mathbb{R}$,*

$$(v + v') \otimes w = v \otimes w + v' \otimes w, \qquad v \otimes (w + w') = v \otimes w + v \otimes w',$$
$$(av) \otimes w = a(v \otimes w) = v \otimes (aw).$$

2. Every tensor in $V \otimes W$ is a finite sum of pure tensors.

Proof. (1) Two cosets $f + R$ and $g + R$ are equal if and only if $f - g \in R$ (Section 9). For example, $\delta_{(v+v',w)} - (\delta_{(v,w)} + \delta_{(v',w)}) \in R$, so $(v + v') \otimes w = v \otimes w + v' \otimes w$. The other rules follow in the same way from the other generators of R.

(2) Every $f \in \mathcal{F}_0(V \times W)$ is a finite linear combination $f = \sum_l a_l\,\delta_{(v_l,w_l)}$, so $f + R = \sum_l a_l\,(v_l \otimes w_l) = \sum_l (a_l v_l) \otimes w_l$. □

For example, $0 \otimes w = (0 \cdot 0) \otimes w = 0\,(0 \otimes w) = 0$ for every $w \in W$. Not every tensor is pure; we will see below that $e_1 \otimes e_1 + e_2 \otimes e_2 \in \mathbb{R}^2 \otimes \mathbb{R}^2$ cannot be written as $v \otimes w$. The construction is designed so that the following theorem holds.

Theorem 11.6 (Universal property)**.** *Let $\beta : V \times W \to U$ be a bilinear map into a vector space U. Then there is exactly one linear transformation $\tilde{\beta} : V \otimes W \to U$ with*

$$\tilde{\beta}(v \otimes w) = \beta(v, w) \qquad \text{for all } v \in V \text{ and } w \in W.$$

Proof. Define $L : \mathcal{F}_0(V \times W) \to U$ by $L(f) = \sum_{(v,w)} f(v, w)\,\beta(v, w)$, where the sum runs over the finitely many pairs (v, w) with $f(v, w) \neq 0$. This map is linear, and $L(\delta_{(v,w)}) = \beta(v, w)$. It vanishes on the generators of R, since β is bilinear; for example,

$$L(\delta_{(v+v',w)} - \delta_{(v,w)} - \delta_{(v',w)}) = \beta(v + v', w) - \beta(v, w) - \beta(v', w) = 0.$$

Since $\ker(L)$ is a subspace (Theorem 5.5), it contains the whole span R of these vectors. Now define $\tilde{\beta}(f + R) = L(f)$. This does not depend on the choice of the representative f of the coset: if $f + R = g + R$, then $f - g \in R \subseteq \ker(L)$, so $L(f) = L(g)$. The map $\tilde{\beta}$ is linear, since L is, and $\tilde{\beta}(v \otimes w) = L(\delta_{(v,w)}) = \beta(v, w)$. It is the only such map, since a linear map on $V \otimes W$ is determined by its values on the pure tensors, which span $V \otimes W$ by Proposition 11.5. □

So bilinear maps $V \times W \to U$ are the same thing as linear maps $V \otimes W \to U$: the tensor product turns bilinear problems into linear ones. For example, a bilinear model $f(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$, which combines two inputs $\mathbf{x}$ and $\mathbf{y}$, is a linear model in the features $\mathbf{x} \otimes \mathbf{y}$; this is called bilinear pooling.

Theorem 11.7. *Let $b_1, \dots, b_m$ be a basis of V, and let $c_1, \dots, c_n$ be a basis of W. Then the mn pure tensors $b_i \otimes c_j$ form a basis of $V \otimes W$. In particular,*

$$\dim(V \otimes W) = \dim V \cdot \dim W,$$

and every tensor $t \in V \otimes W$ can be written in exactly one way as

$$t = \sum_{i=1}^{m} \sum_{j=1}^{n} T_{ij}\, b_i \otimes c_j.$$

For $v = \sum_i x_i b_i$ and $w = \sum_j y_j c_j$, the coefficients of $v \otimes w$ are $T_{ij} = x_i y_j$.

Proof. By bilinearity, $v \otimes w = \sum_{i,j} x_i y_j\, b_i \otimes c_j$. Since the pure tensors span $V \otimes W$, so do the tensors $b_i \otimes c_j$. To show that they are linearly independent, let $b^1, \dots, b^m$ and $c^1, \dots, c^n$ be the dual bases (Theorem 8.6). For fixed k and l, the map $(v, w) \mapsto b^k(v)\, c^l(w)$ is bilinear, so by Theorem 11.6, there is a linear functional λ_{kl} on $V \otimes W$ with $\lambda_{kl}(v \otimes w) = b^k(v)\, c^l(w)$; in particular, $\lambda_{kl}(b_i \otimes c_j) = \delta_{ki}\delta_{lj}$. If $\sum_{i,j} T_{ij}\, b_i \otimes c_j = 0$, then applying λ_{kl} gives $T_{kl} = 0$, for all k and l. □

Corollary 11.8. *The coordinate map with respect to the standard bases,*

$$\mathbb{R}^m \otimes \mathbb{R}^n \to \mathbb{R}^{m \times n}, \qquad \sum_{i,j} T_{ij}\, \mathbf{e}_i \otimes \mathbf{e}_j \mapsto (T_{ij})_{i,j},$$

is an isomorphism, and it maps the pure tensor $\mathbf{u} \otimes \mathbf{v}$ to the matrix $\mathbf{u}\mathbf{v}^\top$.

Proof. By Theorem 11.7, the tensors $\mathbf{e}_i \otimes \mathbf{e}_j$ form a basis, so the coordinate map is an isomorphism (Theorem 4.27, with the mn coordinates arranged as a matrix). The coordinates of $\mathbf{u} \otimes \mathbf{v}$ are $u_i v_j = (\mathbf{u}\mathbf{v}^\top)_{ij}$. □

From now on, we identify $\mathbb{R}^m \otimes \mathbb{R}^n$ with $\mathbb{R}^{m \times n}$, and the pure tensor $\mathbf{u} \otimes \mathbf{v}$ with the matrix $\mathbf{u}\mathbf{v}^\top$, which is called the outer product of $\mathbf{u}$ and $\mathbf{v}$. For example,

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \end{pmatrix}.$$

All columns of $\mathbf{u}\mathbf{v}^\top$ are multiples of $\mathbf{u}$, so a pure tensor in $\mathbb{R}^m \otimes \mathbb{R}^n$ is a matrix of rank at most 1. Hence $\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2 = I_2$, which has rank 2, is not a pure tensor.

11.3 Tensors of higher order and arrays

The same construction works for more factors. For vector spaces $V_1, \dots, V_k$, the tensor product $V_1 \otimes \cdots \otimes V_k$ is the quotient of $\mathcal{F}_0(V_1 \times \cdots \times V_k)$ by the subspace spanned by the rules of additivity and homogeneity in each of the k arguments, and $v_1 \otimes \cdots \otimes v_k$ is the coset of $\delta_{(v_1,\dots,v_k)}$. The proofs of Proposition 11.5 and Theorems 11.6 and 11.7 carry over word by word:

- $(v_1, \dots, v_k) \mapsto v_1 \otimes \cdots \otimes v_k$ is multilinear, and the pure tensors $v_1 \otimes \cdots \otimes v_k$ span $V_1 \otimes \cdots \otimes V_k$;
- for every multilinear map $\tau : V_1 \times \cdots \times V_k \to U$, there is exactly one linear map $\tilde{\tau} : V_1 \otimes \cdots \otimes V_k \to U$ with $\tilde{\tau}(v_1 \otimes \cdots \otimes v_k) = \tau(v_1, \dots, v_k)$;
- if $b_1^{(j)}, \dots, b_{n_j}^{(j)}$ is a basis of V_j for $j = 1, \dots, k$, then the tensors $b_{i_1}^{(1)} \otimes \cdots \otimes b_{i_k}^{(k)}$ form a basis of $V_1 \otimes \cdots \otimes V_k$, and $\dim(V_1 \otimes \cdots \otimes V_k) = n_1 n_2 \cdots n_k$.

One can also show, with the universal property, that grouping the factors does not matter: $u \otimes v \otimes w \mapsto (u \otimes v) \otimes w$ defines an isomorphism $U \otimes V \otimes W \to (U \otimes V) \otimes W$, and similarly for $U \otimes (V \otimes W)$.

Definition 11.9 (Tensor)**.** A tensor of order k is an element t of a tensor product $V_1 \otimes \cdots \otimes V_k$. With respect to bases of $V_1, \ldots, V_k$ as above, its coordinates are the numbers $T_{i_1 \ldots i_k}$ with

$$t = \sum_{i_1, \ldots, i_k} T_{i_1 \ldots i_k} \, b^{(1)}_{i_1} \otimes \cdots \otimes b^{(k)}_{i_k},$$

which form an array with k indices.

For $V_j = \mathbb{R}^{n_j}$ with the standard bases, we identify a tensor with the array of its coordinates, as we did for matrices. The arrays of shape $n_1 \times \cdots \times n_k$ form a vector space $\mathbb{R}^{n_1 \times \cdots \times n_k}$ with the entrywise operations (it is $\mathcal{F}(S)$ for the set S of all index tuples, Proposition 2.10), and the coordinate map is an isomorphism

$$\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_k} \cong \mathbb{R}^{n_1 \times \cdots \times n_k}, \qquad \text{with} \qquad (u^{(1)} \otimes \cdots \otimes u^{(k)})_{i_1 \ldots i_k} = u^{(1)}_{i_1} u^{(2)}_{i_2} \cdots u^{(k)}_{i_k}.$$

These arrays are the tensors of deep learning libraries. A tensor of order 1 is a vector, a tensor of order 2 is a matrix, and a number is sometimes called a tensor of order 0. A color image of $H \times W$ pixels is a tensor in $\mathbb{R}^3 \otimes \mathbb{R}^H \otimes \mathbb{R}^W$, and a batch of 64 color images of 224×224 pixels is a tensor in $\mathbb{R}^{64} \otimes \mathbb{R}^3 \otimes \mathbb{R}^{224} \otimes \mathbb{R}^{224}$, a space of dimension almost 10 million.

Example 11.10 (Reshaping)**.** Listing the coordinates of a tensor in a fixed order gives an isomorphism $\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_k} \to \mathbb{R}^{n_1 \cdots n_k}$ (Theorem 4.27). The usual order is row by row, with the last index changing fastest; for example,

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \mapsto (a, b, c, d, e, f).$$

This is what `reshape` does in NumPy and PyTorch. For vectors $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$, it maps $\mathbf{u} \otimes \mathbf{v}$ to the vector $(u_1 v_1, \ldots, u_1 v_n, u_2 v_1, \ldots, u_m v_n)$, which NumPy computes as `np.kron(u, v)`. Grouping only some of the factors gives matrices: by the isomorphism $\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2} \otimes \mathbb{R}^{n_3} \cong \mathbb{R}^{n_1} \otimes (\mathbb{R}^{n_2} \otimes \mathbb{R}^{n_3}) \cong \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2 n_3}$, a tensor of shape $n_1 \times n_2 \times n_3$ becomes an $n_1 \times (n_2 n_3)$ matrix, its unfolding, whose i-th row lists the coordinates T_{ijk} with the first index i. Reshaping changes only the arrangement of the numbers, so it costs no computation at all.

11.4 Multilinear forms and linear maps as tensors

By the universal property, a multilinear form on $V_1 \times \cdots \times V_k$ is the same thing as a linear functional on $V_1 \otimes \cdots \otimes V_k$. The dual space of a tensor product is again a tensor product.

Theorem 11.11. *1. There is an isomorphism $V_1^* \otimes \cdots \otimes V_k^* \to (V_1 \otimes \cdots \otimes V_k)^*$ that maps $\varphi_1 \otimes \cdots \otimes \varphi_k$ to the functional with $v_1 \otimes \cdots \otimes v_k \mapsto \varphi_1(v_1)\, \varphi_2(v_2) \cdots \varphi_k(v_k)$. So the multilinear forms on $V_1 \times \cdots \times V_k$ correspond to the tensors in $V_1^* \otimes \cdots \otimes V_k^*$.*

2. Every multilinear form τ on $\mathbb{R}^{n_1} \times \cdots \times \mathbb{R}^{n_k}$ is given by an array $T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$ as

$$\tau(\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(k)}) = \sum_{i_1, \ldots, i_k} T_{i_1 \ldots i_k} \, \mathbf{x}^{(1)}_{i_1} \cdots \mathbf{x}^{(k)}_{i_k}, \qquad \textit{with} \quad T_{i_1 \ldots i_k} = \tau(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_k}).$$

Proof. (1) For fixed $\varphi_1, \dots, \varphi_k$, the map $(v_1, \dots, v_k) \mapsto \varphi_1(v_1) \cdots \varphi_k(v_k)$ is multilinear, so by the universal property, it defines a linear functional $\Lambda(\varphi_1, \dots, \varphi_k)$ on $V_1 \otimes \cdots \otimes V_k$. This functional depends multilinearly on $(\varphi_1, \dots, \varphi_k)$ (it suffices to check this on pure tensors, which span), so by the universal property again, we get a linear map $\tilde{\Lambda} : V_1^* \otimes \cdots \otimes V_k^* \to (V_1 \otimes \cdots \otimes V_k)^*$. Choose bases of all V_j and the dual bases. Then $\tilde{\Lambda}$ maps the basis tensor $b^{(1)i_1} \otimes \cdots \otimes b^{(k)i_k}$, made of dual basis vectors, to the functional that is 1 on $b^{(1)}_{i_1} \otimes \cdots \otimes b^{(k)}_{i_k}$ and 0 on all other basis tensors. So $\tilde{\Lambda}$ maps a basis to the dual basis of the basis of $V_1 \otimes \cdots \otimes V_k$, and it is an isomorphism.

(2) Writing each argument in the standard basis, $\mathbf{x}^{(j)} = \sum_i \mathbf{x}^{(j)}_i \mathbf{e}_i$, and using the linearity in each argument, we get

$$\tau(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}) = \sum_{i_1, \dots, i_k} \mathbf{x}^{(1)}_{i_1} \cdots \mathbf{x}^{(k)}_{i_k} \, \tau(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k}). \qquad \square$$

For $k = 1$, part (2) is Proposition 8.3. For $k = 2$, it says that every bilinear form on $\mathbb{R}^m \times \mathbb{R}^n$ is $\beta_A(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$ for exactly one $m \times n$ matrix A, namely the one with $A_{ij} = \beta(\mathbf{e}_i, \mathbf{e}_j)$; this matrix is the coordinate array of a tensor in $(\mathbb{R}^m)^* \otimes (\mathbb{R}^n)^*$. Matrices also describe linear maps, and linear maps are tensors, too.

Proposition 11.12. *Let* $\mathrm{Hom}(V, W)$ *be the vector space of linear transformations* $V \to W$*, with the pointwise operations. The linear map* $\Psi : W \otimes V^* \to \mathrm{Hom}(V, W)$ *with*

$$\Psi(w \otimes \varphi)(v) = \varphi(v)\, w$$

is an isomorphism. If $b_1, \dots, b_n$ *and* $c_1, \dots, c_m$ *are bases of* V *and* W*, and* $T : V \to W$ *has the matrix* $A = [T]_{\mathcal{C},\mathcal{B}}$*, then* $T = \Psi\big(\sum_{i,j} A_{ij}\, c_i \otimes b^j\big)$*: the matrix of* T *is the coordinate array of the tensor* $\Psi^{-1}(T)$*.*

Proof. The map $(w, \varphi) \mapsto \big(v \mapsto \varphi(v) w\big)$ is bilinear, so Ψ exists by Theorem 11.6. For a tensor $t = \sum_{i,j} A_{ij}\, c_i \otimes b^j$, we get

$$\Psi(t)(b_l) = \sum_{i,j} A_{ij}\, b^j(b_l)\, c_i = \sum_i A_{il}\, c_i.$$

If A is the matrix of T, this is $T(b_l)$, since the l-th column of A is the coordinate vector of $T(b_l)$. Two linear maps that agree on a basis are equal, so $\Psi(t) = T$, and Ψ is surjective. If $\Psi(t) = 0$, then the same computation gives $\sum_i A_{il}\, c_i = 0$, so $A_{il} = 0$ for all i and l, and $t = 0$; so Ψ is injective. $\square$

For $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, with a functional φ given by a row $\mathbf{a}$, the map $\Psi(\mathbf{u} \otimes \varphi)$ is $\mathbf{x} \mapsto (\mathbf{a}\mathbf{x})\, \mathbf{u} = (\mathbf{u}\mathbf{a})\, \mathbf{x}$, multiplication by the rank-one matrix $\mathbf{u}\mathbf{a}$. So one and the same $m \times n$ array can be the coordinate array of a tensor in $\mathbb{R}^m \otimes \mathbb{R}^n$, of a linear map in $\mathbb{R}^m \otimes (\mathbb{R}^n)^*$, or of a bilinear form in $(\mathbb{R}^m)^* \otimes (\mathbb{R}^n)^*$. The difference becomes visible when we change the basis (Subsection 11.7).

11.5 Contractions and Einstein notation

Definition 11.13 (Contraction)**.** The contraction on $V \otimes V^*$ is the linear functional $C : V \otimes V^* \to \mathbb{R}$ with $C(v \otimes \varphi) = \varphi(v)$. It exists by Theorem 11.6, since $(v, \varphi) \mapsto \varphi(v)$ is bilinear.

Proposition 11.14. *If a linear transformation* $T : V \to V$ *has the matrix* A *with respect to a basis of* V*, then* $\mathrm{tr}(A) = C(\Psi^{-1}(T))$*. In particular, the trace of the matrix of* T *is the same for all bases.*

Proof. By Proposition 11.12, $\Psi^{-1}(T) = \sum_{i,j} A_{ij}\, b_i \otimes b^j$, so $C(\Psi^{-1}(T)) = \sum_{i,j} A_{ij}\, b^j(b_i) = \sum_i A_{ii} = \mathrm{tr}(A)$. The second statement holds since C and Ψ are defined without choosing a basis. $\square$

Since similar matrices A and $P^{-1}AP$ are the matrices of the same linear transformation in two bases, this explains why $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(A)$, which also follows from $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (Theorem 6.77). Contractions are also defined for tensors with more factors, by pairing a factor V with a factor V^*. For example, the composition of linear maps is a contraction: for $S = \Psi(u \otimes \varphi) : V \to W$ and $T = \Psi(w \otimes \psi) : W \to U$, we have $(T \circ S)(v) = \varphi(v)\,\psi(u)\,w$, so

$$T \circ S = \Psi\big(\psi(u)\,w \otimes \varphi\big) :$$

the middle pair $u \in W$ and $\psi \in W^*$ is contracted to the number $\psi(u)$. In coordinates, this is the product of matrices, $(AB)_{ik} = \sum_j A_{ij}B_{jk}$, which sums over the pair of indices that belong to the middle factor. In general, a contraction sums over an index that appears twice. Since such sums appear everywhere, it is convenient to leave out the summation signs.

Definition 11.15 (Einstein summation convention)**.** In an expression with indices, every index that appears twice is summed over its whole range. The other indices are free, and the result is a tensor indexed by the free indices.

In this notation, the product of two matrices is $A_{ij}B_{jk}$, with the free indices i and k, the trace is A_{ii}, with no free index (a number), and the bilinear form of a matrix is $x_iA_{ij}y_j$. The NumPy function `np.einsum` implements this convention: the indices of the arguments are given as letters, and the letters after the arrow are the free indices, in the desired order (Table 1).

operation	Einstein notation	NumPy
matrix product AB	$A_{ij}B_{jk}$	`np.einsum("ij,jk->ik", A, B)`
trace $\operatorname{tr}(A)$	A_{ii}	`np.einsum("ii->", A)`
dot product $\mathbf{x}^\top\mathbf{y}$	x_iy_i	`np.einsum("i,i->", x, y)`
outer product $\mathbf{x} \otimes \mathbf{y}$	x_iy_j	`np.einsum("i,j->ij", x, y)`
transpose $A^\top$	A_{ji}	`np.einsum("ij->ji", A)`
bilinear form $\mathbf{x}^\top A\mathbf{y}$	$x_iA_{ij}y_j$	`np.einsum("i,ij,j->", x, A, y)`
batch of products A_bB_b	$A_{bij}B_{bjk}$	`np.einsum("bij,bjk->bik", A, B)`

Table 1: Operations on vectors, matrices and tensors in Einstein notation, and the corresponding calls of `np.einsum`. In the last row, A and B are tensors of order 3, which store a batch of matrices A_b and B_b.

Example 11.16 (Attention scores)**.** In a transformer, a sentence of L tokens is represented by a query matrix $Q \in \mathbb{R}^{L\times d}$ and a key matrix $K \in \mathbb{R}^{L\times d}$, and the attention scores $S_{ij} = \sum_c Q_{ic}K_{jc}$ measure how much token i attends to token j, i.e., $S = QK^\top$ (before a scaling by $\frac{1}{\sqrt{d}}$ and a softmax). For a batch of N sentences, Q and K are tensors of shape $N \times L \times d$, and the scores are $S_{bij} = Q_{bic}K_{bjc}$ in Einstein notation, or `np.einsum("bic,bjc->bij", Q, K)`. This contraction needs NL^2d multiplications, one for each choice of b, i, j and c.

In general, a contraction needs one multiplication for each combination of values of all the indices involved, free and summed. This is why the order of contractions matters: for $n \times n$ matrices A and B and a vector $\mathbf{x} \in \mathbb{R}^n$, both $(AB)\mathbf{x}$ and $A(B\mathbf{x})$ compute the contraction $A_{ij}B_{jk}x_k$, but the first needs $n^3 + n^2$ multiplications, and the second only $2n^2$.

Remark 11.17 (The Kronecker product). For an $m \times n$ matrix A and a $p \times q$ matrix B, the tensor $A \otimes B$ of order 4 has the coordinates $A_{ij}B_{kl}$. Reshaping it into an $mp \times nq$ matrix, with the index pairs (i,k) as rows and (j,l) as columns, gives the Kronecker product of A and B (`np.kron` in NumPy), which is also written $A \otimes B$. It appears whenever a linear map acts on matrices: for $X \in \mathbb{R}^{n\times q}$, we have $(AXB^\top)_{ik} = A_{ij}X_{jl}B_{kl}$, so after flattening both X and $AXB^\top$ row by row, the linear map $X \mapsto AXB^\top$ is multiplication by the Kronecker product of A and B.

11.6 Tensor rank

Definition 11.18 (Tensor rank)**.** The rank of a nonzero tensor $t \in V_1 \otimes \cdots \otimes V_k$ is the smallest number r such that t is a sum of r pure tensors. The zero tensor has rank 0. A tensor of rank one is a nonzero pure tensor.

Proposition 11.19. *For tensors in $\mathbb{R}^m \otimes \mathbb{R}^n = \mathbb{R}^{m\times n}$, the tensor rank is the rank of Section 7: an $m \times n$ matrix A has rank r if and only if it is a sum of r, but not fewer, outer products $\mathbf{u}\otimes\mathbf{v} = \mathbf{u}\mathbf{v}^\top$.*

Proof. Let $\operatorname{rank}(A) = r$, and let $\mathbf{c}_1, \dots, \mathbf{c}_r$ be a basis of $\operatorname{col}(A)$. Each column $\mathbf{a}_j$ of A is a linear combination $\mathbf{a}_j = \sum_{l=1}^{r} R_{lj}\,\mathbf{c}_l$, so $A = CR$, where C is the $m \times r$ matrix with the columns $\mathbf{c}_l$, and R is the $r \times n$ matrix of the coefficients. If $\boldsymbol{\rho}_l$ denotes the l-th row of R, then

$$A = CR = \sum_{l=1}^{r} \mathbf{c}_l \boldsymbol{\rho}_l = \sum_{l=1}^{r} \mathbf{c}_l \otimes \boldsymbol{\rho}_l^\top,$$

a sum of r outer products. Conversely, if $A = \sum_{l=1}^{s} u_l \otimes v_l = \sum_{l=1}^{s} u_l v_l^\top$, then every column of A, $\mathbf{a}_j = \sum_l (v_l)_j\, u_l$, lies in $\operatorname{span}\{u_1, \dots, u_s\}$, so $\operatorname{rank}(A) = \dim \operatorname{col}(A) \le s$, since a subspace spanned by s vectors has dimension at most s (Corollary 3.14). □

For tensors of order 3 or more, the rank is no longer the rank of a matrix, but matrices still help to compute it.

Example 11.20. Let $t = \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2 \in \mathbb{R}^2 \otimes \mathbb{R}^2 \otimes \mathbb{R}^2$, with the coordinates $T_{111} = T_{222} = 1$ and all other coordinates 0. So t has rank at most 2. It does not have rank one: the coordinates of a pure tensor $\mathbf{u} \otimes \mathbf{v} \otimes w$ are $u_i v_j w_k$, so its two slices $(T_{ij1})_{i,j}$ and $(T_{ij2})_{i,j}$ are the matrices $w_1\,\mathbf{u}\mathbf{v}^\top$ and $w_2\,\mathbf{u}\mathbf{v}^\top$, which are multiples of the same matrix. For our t, the slices are $\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)$, which are not multiples of each other. Hence t has rank 2.

A pure tensor of order k in $\mathbb{R}^n \otimes \cdots \otimes \mathbb{R}^n$ is described by the kn coordinates of its factors instead of n^k numbers, so tensors of low rank can be stored and processed much more cheaply.

Remark 11.21 (Low-rank tensors in machine learning). For matrices, the rank is computed by row reduction, and the best approximation of a matrix by matrices of rank r is given by the singular value decomposition (Lecture 2). For tensors of order 3 and more, the rank is much harder to handle: computing it is NP-hard in general, and it can even depend on the field, since there are real $2 \times 2 \times 2$ tensors of rank 3 that have rank 2 as complex tensors. Nevertheless, approximating a tensor by a sum of a few pure tensors (the CP decomposition) is a standard tool to compress the weight tensors of neural networks, to find structure in data with several indices (such as users × products × time), and to learn mixture models.

11.7 Change of basis

Let $b_1, \dots, b_n$ and $b'_1, \dots, b'_n$ be two bases of V, related by $b'_j = \sum_i P_{ij}\, b_i$ with an invertible matrix P. (For $V = \mathbb{R}^n$ and $b_i = e_i$, P is the matrix with the columns b'_j.) Then $b_i = \sum_k (P^{-1})_{ki}\, b'_k$, and the dual bases are related by

$$b^i = \sum_l P_{il}\, b'^l,$$

since both sides take the value P_{ij} at b'_j. Substituting these formulas into the expansion of a tensor in the old basis gives its coordinates in the new basis:

- a vector $v = \sum_i x_i b_i \in V$ has the new coordinates $x' = P^{-1}x$;
- a linear transformation, $t = \sum_{i,j} A_{ij}\, b_i \otimes b^j \in V \otimes V^*$, has the new coordinates $A' = P^{-1}AP$, since $A'_{kl} = \sum_{i,j} (P^{-1})_{ki}\, A_{ij}\, P_{jl}$;
- a bilinear form, $t = \sum_{i,j} A_{ij}\, b^i \otimes b^j \in V^* \otimes V^*$, has the new coordinates $A' = P^\top AP$, since $A'_{kl} = \sum_{i,j} P_{ik}\, A_{ij}\, P_{jl}$;
- a tensor $t = \sum_{i,j} A_{ij}\, b_i \otimes b_j \in V \otimes V$ has the new coordinates $A' = P^{-1}A(P^{-1})^\top$.

In general, every index that belongs to a factor V transforms with P^{-1}, and every index that belongs to a factor V^* transforms with P. So the same array A transforms in different ways, depending on the tensor it describes. For example, the identity matrix describes both the identity map of V, i.e., the tensor $\sum_i b_i \otimes b^i \in V \otimes V^*$, and the dot product on $\mathbb{R}^n$, i.e., the tensor $\sum_i e^i \otimes e^i \in (\mathbb{R}^n)^* \otimes (\mathbb{R}^n)^*$. In every basis, the identity map is still described by $P^{-1}I_nP = I_n$, but the dot product is described by $P^\top P$, which is I_n only if P is orthogonal. Similarly, the trace is a property of linear transformations (Proposition 11.14), while the trace of the matrix of a bilinear form depends on the basis.

Remark 11.22 (Tensors in mathematics, physics and machine learning). A tensor of type (p, q) on V is an element of $V^{\otimes p} \otimes (V^*)^{\otimes q}$, the tensor product of p copies of V and q copies of V^*. Vectors are tensors of type $(1, 0)$, linear functionals of type $(0, 1)$, linear maps $V \to V$ of type $(1, 1)$, and bilinear forms of type $(0, 2)$. By Theorem 11.11 and the isomorphism $V \cong V^{**}$ (Theorem 8.18), a tensor of type (p, q) is the same thing as a multilinear form on $(V^*)^p \times V^q$, and in physics, tensors are often introduced as arrays of numbers that transform as above under a change of basis. In machine learning, the word "tensor" usually just means an array of coordinates in the standard bases, which are never changed, and the meaning of the indices is kept track of separately. The distinction still matters: the derivative of a loss with respect to the parameters is a linear functional (Section 8), so under a linear change of the parameters, it transforms like an index of type V^*, and not like the parameters themselves.

12 Key Takeaways

- A field is a set of "numbers" in which we can add, subtract, multiply, and divide by nonzero elements. $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are fields; $\mathbb{Z}$ is not.
- A vector space over a field $\mathbb{F}$ is a set of vectors that can be added and multiplied by scalars from $\mathbb{F}$, subject to the vector space axioms. Examples: $\mathbb{F}$ itself, products $V \times W$, $\mathbb{R}^n$, the polynomials $\mathbb{R}[x]$, and spaces of functions with the pointwise operations: all functions $\mathcal{F}(S)$ on a set S, the continuous functions $C(I)$, the n times continuously differentiable functions $C^n(I)$, and the smooth functions $C^\infty(I)$.

- A basis is a linearly independent set that spans the space; every vector is a unique linear combination of the basis vectors. All bases of a finite-dimensional space have the same number of vectors, the dimension: $\dim \mathbb{R}^n = n$, while $\mathbb{R}[x]$ is infinite-dimensional.

- A linear transformation $T : V \to W$ preserves addition and scalar multiplication, and $T(0_V) = 0_W$. Its kernel is a subspace of V, and its image is a subspace of W. Differentiation $D : C^1(I) \to C(I)$ and integration $Jf(x) = \int_c^x f(t)\,dt$ are linear, with $D \circ J = \mathrm{id}$ but $J(Df) = f - f(c)$; on $C^\infty(I)$, D is surjective but not injective, which is impossible for a map of a finite-dimensional space to itself.

- A subspace is a subset that contains 0 and is closed under addition and scalar multiplication. The polynomials and the function spaces form a chain of subspaces $\mathbb{R}[x] \subseteq C^\infty(\mathbb{R}) \subseteq C^n(\mathbb{R}) \subseteq C(\mathbb{R}) \subseteq \mathcal{F}(\mathbb{R})$, all of them infinite-dimensional.

- An isomorphism is a bijective linear transformation, i.e., one with $\ker(T) = \{0\}$ and $\mathrm{im}(T) = W$. Via coordinates, every n-dimensional real vector space is isomorphic to $\mathbb{R}^n$, and $V \cong W$ if and only if $\dim V = \dim W$, if and only if some linear transformation $V \to W$ is represented by an invertible matrix.

- Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is multiplication by an $m \times n$ matrix A, whose i-th column is $T(\mathbf{e}_i)$.

- Matrices of the same size are added entry by entry. The product of an $m \times n$ and an $n \times p$ matrix is the $m \times p$ matrix with $(AB)_{ij} = \sum_k A_{ik}B_{kj}$; it represents the composition $T_A \circ T_B$. Matrix multiplication is associative but not commutative, and $(AB)^\top = B^\top A^\top$.

- A square matrix A is invertible if $AB = BA = I_n$ for some matrix $B = A^{-1}$; then $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$, and $(AB)^{-1} = B^{-1}A^{-1}$. The inverse is computed without determinants by row reducing $(A \mid I_n)$ to $(I_n \mid A^{-1})$.

- Diagonal, upper and lower triangular, symmetric ($A^\top = A$) and skew-symmetric ($A^\top = -A$) matrices form subspaces of $\mathbb{R}^{n\times n}$. The invertible matrices $\mathrm{GL}(n, \mathbb{R})$ and the orthogonal matrices $\mathrm{O}(n)$ (with $Q^\top Q = I_n$, so $Q^{-1} = Q^\top$) do not form subspaces, but they contain I_n and are closed under multiplication and inverses: they are matrix groups, i.e., subgroups of the group $\mathrm{GL}(n, \mathbb{R})$. A group is a set with an associative operation, an identity element and inverses; examples are $(\mathbb{R}, +)$, every vector space under addition, and $(\mathbb{F} \setminus \{0\}, \cdot)$ for a field $\mathbb{F}$.

- The trace $\mathrm{tr}(A) = \sum_i A_{ii}$ is linear and satisfies $\mathrm{tr}(AB) = \mathrm{tr}(BA)$. The determinant is defined by cofactor expansion and is best computed by row reduction to triangular form; it satisfies $\det(AB) = \det(A)\det(B)$ and $\det(A^\top) = \det(A)$, and $|\det(A)|$ is the factor by which T_A scales areas and volumes.

- Block matrices multiply like 2×2 matrices. With the Schur complement $S = D - CA^{-1}B$, $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A)\det(S)$. Gaussian elimination computes the LU decomposition $A = LU$ (or $PA = LU$ with row swaps), which solves $A\mathbf{x} = \mathbf{b}$ by forward and back substitution and gives $\det(A)$ as a product of pivots. Low-rank updates are inverted with the Woodbury identity, for example $(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1+\mathbf{v}^\top A^{-1}\mathbf{u}}$, and $\det(A + \mathbf{u}\mathbf{v}^\top) = (1 + \mathbf{v}^\top A^{-1}\mathbf{u})\det(A)$.

- Elementary row operations bring a matrix into (reduced) row echelon form. They do not change the null space or the row space, but they can change the column space; the pivot columns of the original matrix form a basis of its column space.

- $\operatorname{rank}(A) = \dim \operatorname{col}(A) = \dim \operatorname{row}(A)$ is the number of pivots, and $\operatorname{rank}(A) + \dim \operatorname{null}(A) = n$ for every $m \times n$ matrix A.

- For an $n \times n$ matrix A: A is invertible $\iff$ $\operatorname{null}(A) = \{\mathbf{0}\}$ $\iff$ $\operatorname{rank}(A) = n$ $\iff$ $\operatorname{RREF}(A) = I_n$ $\iff$ $\det(A) \neq 0$.

- The dual space V^* consists of the linear functionals $V \to \mathbb{F}$; on $\mathbb{R}^n$, these are the row vectors. A basis $b_1, \ldots, b_n$ has a dual basis $b^1, \ldots, b^n$ with $b^i(b_j) = \delta_{ij}$ (in $\mathbb{R}^n$: the rows of P^{-1}), and $\dim V^* = \dim V$. The dual map $T^*(\psi) = \psi \circ T$ is represented by the transpose $A^\top$. A system $Ax = b$ is solvable if and only if $yb = 0$ for all rows y with $yA = 0$.

- The quotient space V/U consists of the cosets $v + U$, where $v + U = w + U$ if and only if $v - w \in U$; it has dimension $\dim V - \dim U$. First isomorphism theorem: $V/\ker(T) \cong \operatorname{im}(T)$, which gives rank–nullity for every linear map. Second: $(U + W)/W \cong U/(U \cap W)$, so $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$. Third: $(V/U)/(W/U) \cong V/W$.

- An eigenvector $\mathbf{v} \neq \mathbf{0}$ of A satisfies $A\mathbf{v} = \lambda\mathbf{v}$; the eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I_n)$, and the eigenspace $E_\lambda = \operatorname{null}(A - \lambda I_n)$ is found by row reduction. The (complex) eigenvalues add up to $\operatorname{tr}(A)$ and multiply to $\det(A)$.

- A is diagonalizable, $A = PDP^{-1}$, exactly when it has n linearly independent eigenvectors (the columns of P); this holds, for example, when A has n distinct eigenvalues. Then $A^k = PD^kP^{-1}$, which explains Fibonacci numbers, the long-run behavior of Markov chains, and power iteration (PageRank).

- Cayley–Hamilton: $p_A(A) = 0$, so every power of A, and A^{-1} if it exists, is a polynomial in A of degree less than n. The minimal polynomial m_A divides every polynomial that annihilates A, its real roots are the eigenvalues, and A is diagonalizable if and only if m_A is a product of distinct real linear factors.

- A cyclic convolution $\mathbf{c} * \mathbf{x}$ is multiplication by the circulant matrix $\operatorname{circ}(\mathbf{c})$; the matrices that commute with the cyclic shift are exactly the circulant ones. The Fourier vectors $\mathbf{f}_m$ are eigenvectors of every circulant matrix, with the eigenvalues $\hat{c}_m$, so $\widehat{\mathbf{c} * \mathbf{x}}_m = \hat{c}_m \hat{x}_m$, and the FFT computes convolutions in $O(n \log n)$ operations.

- The tensor product $V \otimes W$ is the quotient of the free vector space on $V \times W$ by the rules of bilinearity; $(v, w) \mapsto v \otimes w$ is bilinear, and bilinear maps $V \times W \to U$ are the same as linear maps $V \otimes W \to U$ (universal property). If b_i and c_j are bases, the pure tensors $b_i \otimes c_j$ form a basis, so $\dim(V \otimes W) = \dim V \cdot \dim W$. A tensor is an element of $V_1 \otimes \cdots \otimes V_k$; its coordinates form an array, the "tensor" of deep learning libraries, and $\mathbb{R}^m \otimes \mathbb{R}^n = \mathbb{R}^{m \times n}$ with $u \otimes v = uv^\top$.

- Multilinear forms are tensors in $V_1^* \otimes \cdots \otimes V_k^*$, and linear maps $V \to W$ are tensors in $W \otimes V^*$; the contraction $v \otimes \varphi \mapsto \varphi(v)$ is the trace. Under a change of basis, indices of type V transform with P^{-1} and indices of type V^* with P: linear maps change to $P^{-1}AP$, bilinear forms to $P^\top AP$.

- The rank of a tensor is the smallest number of pure tensors with sum t; for matrices, it is the usual rank. Contractions such as $A_{ij}B_{jk}$ in Einstein notation, computed with `np.einsum`, describe matrix products, traces, batched products and attention scores.

13 Exercises

1. **Homework.** Using only the vector space axioms, prove that in every vector space V over a field $\mathbb{F}$:

   (a) $0 \cdot v = 0_V$ for every $v \in V$. (Hint: $0 \cdot v = (0+0) \cdot v$.)

   (b) $(-1) \cdot v = -v$ for every $v \in V$. (Hint: compute $v + (-1) \cdot v$.)

   (c) The axiom $u + v = v + u$ is redundant: it follows from the other axioms. (Hint: first show that $(-v) + v = 0$ and $0 + v = v$, using only associativity and axioms 2 and 3; for the first, add $w = (-v) + v$ to itself. Then expand $(1+1)(u+v)$ in two ways, and cancel u on the left and v on the right.)

2. Which of the following subsets of $\mathbb{R}^2$ are subspaces? Justify your answers.

   (a) $\{(x, y) \in \mathbb{R}^2 : y = 2x\}$;

   (b) $\{(x, y) \in \mathbb{R}^2 : y = 2x + 1\}$;

   (c) $\{(x, y) \in \mathbb{R}^2 : xy = 0\}$.

3. **Homework.** Show that $\{(1, 2), (3, 4)\}$ is a basis of $\mathbb{R}^2$, and write $(5, 6)$ as a linear combination of these two vectors.

4. Let $\mathbb{R}[x]_{\leq 2}$ be the set of polynomials with real coefficients of degree at most 2. Show that it is a subspace of $\mathbb{R}[x]$, and find its dimension.

5. **Homework.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y, y + z, x + 2y + z)$. Find the matrix A of T and $\mathrm{RREF}(A)$. Then find a basis of $\mathrm{null}(A)$, a basis of $\mathrm{col}(A)$, and $\mathrm{rank}(A)$, and check that your answers agree with the rank–nullity theorem.

6. **Homework.** Write down explicit isomorphisms $\mathbb{R}^{2\times 2} \to \mathbb{R}^4$ and $\mathrm{Sym}_2 \to \mathbb{R}^3$, and prove that they are isomorphisms. Are $\mathbb{R}^{2\times 2}$ and Sym_2 isomorphic?

7. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Compute AB and BA. Describe in words what multiplying a 2×2 matrix by B from the left does, and what multiplying it by B from the right does.

8. **Homework.** Use the Gauss–Jordan method to find the inverse of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix},$$

   and check your answer.

9. Show that the matrix F from Example 6.31 is not invertible, and find a nonzero vector $\mathbf{x}$ with $F\mathbf{x} = \mathbf{0}$.

10. **Homework.** Let A be an $n \times n$ matrix with $A^2 = A$ (where $A^2 = AA$) and $A \neq I_n$. Show that A is not invertible.

11. Show that an upper triangular matrix is invertible if and only if all of its diagonal entries are nonzero, in two ways: using Theorem 6.47, and using determinants.

12. **Homework.** Compute the determinant of

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

in two ways: by cofactor expansion, and by row reduction. Is A invertible?

13. **Homework.** Let A be a skew-symmetric $n \times n$ matrix, where n is odd. Show that $\det(A) = 0$. (Hint: compute $\det(A^\top)$ in two ways.)

14. Show that $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$ for all $n \times n$ matrices A, B, C, but find 2×2 matrices with $\operatorname{tr}(ABC) \neq \operatorname{tr}(ACB)$.

15. **Homework.** Let $A = \begin{pmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix}$.

    (a) Find the LU decomposition of A, and use it to compute $\det(A)$ and to solve $A\mathbf{x} = (2, 8, 10)^\top$.

    (b) Let D be the diagonal matrix with the diagonal entries of U. Check that $U = DL^\top$, so that $A = LDL^\top$. Show that every invertible symmetric matrix that has an LU decomposition can be written as $A = LDL^\top$ in this way. (Hint: uniqueness.)

16. Show that $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ has no LU decomposition, and find a permutation matrix P such that PA has one.

17. Let A be an invertible $k \times k$ matrix and D an invertible $m \times m$ matrix. Show that $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{pmatrix}$, and that $\det \begin{pmatrix} I_k & B \\ B^\top & I_m \end{pmatrix} = \det(I_m - B^\top B)$ for every $k \times m$ matrix B.

18. **Homework.** Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Show that $I_n + \mathbf{u}\mathbf{v}^\top$ is invertible if and only if $\mathbf{v}^\top\mathbf{u} \neq -1$, and find its inverse. Show that every nonzero vector $\mathbf{x}$ with $\mathbf{v}^\top\mathbf{x} = 0$ is an eigenvector with the eigenvalue 1, and that $\mathbf{u}$ (if $\mathbf{u} \neq \mathbf{0}$) is an eigenvector with the eigenvalue $1 + \mathbf{v}^\top\mathbf{u}$.

19. **Homework.** Find the eigenvalues and eigenspaces of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. Diagonalize A, and use the diagonalization to find a formula for A^k.

20. Show that $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ is not diagonalizable. Find all of its eigenvectors.

21. Find the complex eigenvalues of the rotation by 90°, $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and check Theorem 10.12 for this matrix.

22. Let A be a 3×3 matrix with eigenvalues 1, 2 and 3. Compute $\operatorname{tr}(A^2)$, $\det(A)$ and $\det(A^{-1})$. (Hint: A is diagonalizable.)

23. Let A be an $n \times n$ matrix and λ an eigenvalue of A.

(a) Show that if $A^2 = A$, then $\lambda = 0$ or $\lambda = 1$.

(b) Show that if $A^k = 0$ for some $k \geq 1$, then $\lambda = 0$.

(c) Show that if A is orthogonal, then $\lambda = 1$ or $\lambda = -1$.

24. A Markov chain has the transition matrix $A = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$. Find its eigenvalues and its stationary distribution, and describe what happens to $A^k \mathbf{x}_0$ as $k \to \infty$ for a probability vector $\mathbf{x}_0$.

25. **Homework.** Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Use the Cayley–Hamilton theorem to write A^{-1} and A^5 in the form $aA + bI_2$, and check your answers by multiplying out.

26. Find the minimal polynomials of $\operatorname{diag}(2,2,3)$ and of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, and decide which of the two matrices is diagonalizable. Show that every matrix A with $A^3 = A$ is diagonalizable, and that its eigenvalues lie in $\{-1, 0, 1\}$.

27. (Circulant matrices.) Let $\mathbf{c}, \mathbf{d} \in \mathbb{R}^n$.

(a) Show that $\operatorname{circ}(\mathbf{c}) \operatorname{circ}(\mathbf{d}) = \operatorname{circ}(\mathbf{c} * \mathbf{d})$, and conclude that $\mathbf{c} * \mathbf{d} = \mathbf{d} * \mathbf{c}$. (Hint: Proposition 10.37 and $S^n = I_n$.)

(b) Find the eigenvalues of the difference filter, $(\mathbf{c} * \mathbf{x})_j = x_j - x_{j-1}$. Which signals does it map to $\mathbf{0}$?

(c) Show that $\operatorname{circ}(\mathbf{c})$ is invertible if and only if $\hat{c}_m \neq 0$ for all m, and that its inverse is again circulant.

28. **Homework.** Find the dual basis of the basis $(1,0,0)^\top$, $(1,1,0)^\top$, $(1,1,1)^\top$ of $\mathbb{R}^3$, and check that $b^i(\mathbf{b}_j) = \delta_{ij}$.

29. On $V = \mathbb{R}[x]_{\leq 2}$, consider the functionals $\varphi_1(p) = p(0)$, $\varphi_2(p) = p'(0)$ and $\varphi_3(p) = \int_{-1}^{1} p(x)\,dx$. Show that $\varphi_1, \varphi_2, \varphi_3$ is a basis of V^*, and find the basis q_1, q_2, q_3 of V whose dual basis it is, i.e., with $\varphi_i(q_j) = \delta_{ij}$.

30. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x,y) = (x + 2y, 3y)$. Compute $T^*(\psi)$ for the coordinate functionals $\psi(x,y) = x$ and $\psi(x,y) = y$, and check that the matrix of T^* with respect to the dual of the standard basis is the transpose of the matrix of T.

31. **Homework.** Let φ and ψ be nonzero linear functionals on a finite-dimensional vector space with $\ker(\varphi) = \ker(\psi)$. Show that $\psi = c\varphi$ for some $c \neq 0$. (Hint: choose v with $\varphi(v) = 1$, and show that every vector is of the form $w + tv$ with $w \in \ker(\varphi)$.)

32. **Homework.** Let $U = \operatorname{span}\{(1,0,1)^\top, (0,1,1)^\top\} \subseteq \mathbb{R}^3$. Show that the cosets of U are the planes $x_1 + x_2 - x_3 = c$, and that $\mathbf{x} + U \mapsto x_1 + x_2 - x_3$ is an isomorphism $\mathbb{R}^3/U \to \mathbb{R}$. Check Theorem 9.8.

33. Use the first isomorphism theorem and the map $A \mapsto \frac{1}{2}(A + A^\top)$ to show that $\mathbb{R}^{n\times n}/\mathrm{Skew}_n \cong \mathrm{Sym}_n$, and check that the dimensions agree.

34. **Homework.** Let U and W be 3-dimensional subspaces of $\mathbb{R}^5$. Show that $U \cap W \neq \{0\}$. Which values can $\dim(U \cap W)$ take?

35. Show that $V/U = \{0\}$ if and only if $U = V$, and that $V/\{0\} \cong V$.

36. For $n \geq 0$, let $f_n(x) = x^n|x|$.

    (a) Show that for $n \geq 1$, f_n is differentiable on $\mathbb{R}$, with $f_n' = (n+1)f_{n-1}$. (Hint: $f_n(x) = x^{n+1}$ for $x \geq 0$, and $f_n(x) = -x^{n+1}$ for $x < 0$; at $x = 0$, use the definition of the derivative.)

    (b) Conclude that $f_n \in C^n(\mathbb{R})$, but $f_n \notin C^{n+1}(\mathbb{R})$, so that $C^{n+1}(\mathbb{R}) \neq C^n(\mathbb{R})$.

37. **Homework.** Let $D : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $Df = f'$, and $M : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $(Mf)(x) = xf(x)$.

    (a) Show that M is linear, and that $D \circ M - M \circ D = \mathrm{id}$.

    (b) Show that there are no $n \times n$ matrices A and B with $AB - BA = I_n$. (Hint: take the trace.) So the identity in (a) has no analogue for matrices.

38. **Homework.** Let $D : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 2}$, $Dp = p'$, and $J : \mathbb{R}[x]_{\leq 2} \to \mathbb{R}[x]_{\leq 3}$, $(Jp)(x) = \int_0^x p(t)\,dt$. Find the matrices of D and J with respect to the bases of monomials, and compute the products of these two matrices in both orders. Which product is the identity matrix? Why can the other product not be the identity?

39. Which of the following maps $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ are bilinear? For the bilinear ones, find the matrix A with $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$. (a) $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$; (b) $\beta(\mathbf{x}, \mathbf{y}) = x_1 + y_1$; (c) $\beta(\mathbf{x}, \mathbf{y}) = (x_1 + x_2)(y_1 - 3y_2)$; (d) $\beta(\mathbf{x}, \mathbf{y}) = x_1 x_2 y_1$.

40. Let $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$ be a bilinear form on $\mathbb{R}^n$.

    (a) Show that $\beta(\mathbf{x}, \mathbf{y}) = \beta(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$ if and only if A is symmetric.

    (b) Let P be invertible. Show that the matrix $P^\top A P$ of β in the basis of the columns of P is symmetric if A is, and that $\det(P^\top A P) = \det(P)^2 \det(A)$. So the sign of the determinant of the matrix of a bilinear form does not depend on the basis, but its value does.

41. **Homework.**

    (a) Let $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$. Show that $(\mathbf{u} \otimes \mathbf{v})\,\mathbf{y} = (\mathbf{v}^\top \mathbf{y})\,\mathbf{u}$ for every $\mathbf{y} \in \mathbb{R}^n$, and that $\mathrm{im}(T_{\mathbf{u} \otimes \mathbf{v}}) = \mathrm{span}\{\mathbf{u}\}$ if $\mathbf{v} \neq \mathbf{0}$.

    (b) Write the $2 \times 2 \times 2$ tensor with the entries $T_{ijk} = i + j + k$ as a sum of three rank-one tensors. (Hint: $i + j + k = i \cdot 1 \cdot 1 + 1 \cdot j \cdot 1 + 1 \cdot 1 \cdot k$.) Show that T does not have rank one.

42. **Homework.** Write the following in Einstein notation and as a call of `np.einsum`, and count the multiplications: (a) the vector $A\mathbf{x}$ for $A \in \mathbb{R}^{m \times n}$; (b) $\mathrm{tr}(A^\top B)$ for $A, B \in \mathbb{R}^{m \times n}$; (c) the matrix $XX^\top$ of all dot products of the rows of $X \in \mathbb{R}^{N \times d}$; (d) the number $\mathbf{x}^\top AB\mathbf{y}$ for $n \times n$ matrices A and B, computed as $(\mathbf{x}^\top A)(B\mathbf{y})$ and as $\mathbf{x}^\top (AB)\mathbf{y}$.

43. Let V and W be vector spaces.

    (a) Show that $(2v) \otimes w = v \otimes (2w)$ and $v \otimes 0 = 0$ in $V \otimes W$.

    (b) Use the universal property to construct linear maps $V \otimes W \to W \otimes V$ with $v \otimes w \mapsto w \otimes v$, and $W \otimes V \to V \otimes W$ with $w \otimes v \mapsto v \otimes w$, and show that they are inverse to each other. So $V \otimes W \cong W \otimes V$.

(c) Show that $\mathbb{R} \otimes V \cong V$, via $a \otimes v \mapsto av$.

44. Let $b_1, \dots, b_n$ be a basis of V, with the dual basis $b^1, \dots, b^n$. Show that the tensor $\sum_{i=1}^{n} b_i \otimes b^i \in V \otimes V^*$ corresponds to the identity map of V under the isomorphism Ψ of Proposition 11.12. Conclude that this tensor does not depend on the choice of the basis, and that its contraction is $\dim V$.

14 Programming Exercises

In these exercises, you implement the main algorithms of this lecture in Python. You need Python 3 with NumPy, and Matplotlib for Exercises P6 and P12. Implement the algorithms yourself, using only loops, indexing and arithmetic on NumPy arrays; use NumPy's built-in linear algebra routines (`np.linalg`) only to check your results. Start each file with

```
import numpy as np
rng = np.random.default_rng(0)  # random number generator for test matrices
```

A word on floating-point numbers. Computers store real numbers with only about 16 significant digits, so almost every computation has tiny rounding errors; for example, `0.1 + 0.2 == 0.3` is `False` in Python. Therefore, never test whether a computed number is exactly zero: test `abs(x) < tol` for a small tolerance such as `tol = 1e-10` instead, and compare arrays with `np.allclose(X, Y)`. Also remember that Python counts from 0: the entry A_{ij} of these notes is `A[i-1, j-1]` in NumPy.

P1. Matrices of linear transformations. Write a function that takes a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, given as a Python function that maps NumPy vectors of length n to vectors of length m, and returns its $m \times n$ matrix, whose i-th column is $T(\mathbf{e}_i)$ (Theorem 6.9).

```
def matrix_of(T, n):
    """Return the matrix of the linear transformation T: R^n -> R^m."""
    ...

T = lambda v: np.array([2*v[0] + 3*v[1], 4*v[0] - v[1]])
print(matrix_of(T, 2))  # should print [[ 2.  3.] [ 4. -1.]]
```

- Also test your function on $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$, and compare with the matrix computed in the examples after Theorem 6.9.
- Write a function `looks_linear(T, n, trials=100)` that checks additivity and homogeneity for random vectors `u = rng.normal(size=n)`, `v = rng.normal(size=n)` and random scalars `a = rng.normal()`. Test it on the T above and on the maps $(x, y) \mapsto (x + 1, y)$ and $(x, y) \mapsto (xy, y)$.
- Why can such a random test prove that a map is *not* linear, but never prove that a map *is* linear?

P2. Matrix multiplication from scratch. Implement `matmul(A, B)` with three nested loops, directly from the definition $(AB)_{ij} = \sum_k A_{ik} B_{kj}$. Raise a `ValueError` if the number of columns of `A` is not equal to the number of rows of `B`.

- Compare `matmul(A, B)` with NumPy's `A @ B` for random matrices of several shapes.

- Use your function to check numerically that $(AB)C = A(BC)$ and $(AB)^\top = B^\top A^\top$ (Proposition 6.22), and find random 2×2 matrices with $AB \neq BA$.
- Measure the running times (with `time.perf_counter()`) of `matmul` and of `@` for random $n \times n$ matrices with $n = 50, 100, 200$. By roughly what factor does the running time of `matmul` grow when n doubles? Explain this by counting the multiplications that are performed. (NumPy is much faster because it runs highly optimized compiled code.)

P3. Homework. Row reduction. Implement a function that computes the reduced row echelon form of a matrix and its pivot columns.

```
def rref(A, tol=1e-10):
    """Return (R, pivots): the RREF of A and the list of pivot columns."""
    R = np.array(A, dtype=float)  # work on a copy of A
    ...
```

Go through the columns from left to right, keeping track of the row r in which the next pivot will be placed. For each column c, find the entry with the largest absolute value among the rows $r, r+1, \ldots$ of that column. If it is smaller than `tol`, the column has no pivot. Otherwise, swap its row with row r, divide row r by the pivot, and subtract multiples of row r from all the other rows so that all other entries of column c become zero. (Choosing the largest available entry as the pivot is called partial pivoting; it reduces rounding errors.)

- Test your function: the matrix $\begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}$ should give $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$ with pivots `[0, 1]`, and the matrix F of Example 6.31 should give $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.
- Using `rref`, write the functions `rank(A)` and `null_space_basis(A)`, which returns one basis vector for each free variable as in the proof of the rank–nullity theorem, and `col_space_basis(A)`, which returns the pivot columns of the *original* matrix (Theorem 7.2).
- For 20 random 5×7 matrices A, check that `A @ v` is close to zero for every vector `v` in `null_space_basis(A)`, that `rank(A) + len(null_space_basis(A)) == 7` (rank–nullity), and that `rank(A)` agrees with `np.linalg.matrix_rank(A)`.
- Random matrices almost always have the largest possible rank. Construct a 6×8 matrix of rank 4 as the product of a random 6×4 matrix and a random 4×8 matrix, and check that your functions find rank 4 and a 4-dimensional null space.

P4. The inverse with the Gauss–Jordan method. Implement `inverse(A)`: row reduce the $n \times 2n$ matrix $(A \mid I_n)$ with your `rref` (use `np.hstack` and `np.eye`) and return the right block. If the left block is not I_n, raise `np.linalg.LinAlgError("matrix is singular")` (Theorem 6.49).

- Check your function on the examples of Subsection 6.3: the inverse of $\begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix}$ is $\frac{1}{14}\begin{pmatrix} 1 & 3 \\ 4 & -2 \end{pmatrix}$, the inverse of $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ is $\begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is singular.

- For a random 10×10 matrix A, check that `A @ inverse(A)` and `inverse(A) @ A` are both close to I_{10}, and that `inverse(A)` is close to `np.linalg.inv(A)`.

P5. Determinants in two ways. Implement `det_cofactor(A)`, which computes the determinant recursively by cofactor expansion along the first row (Subsection 6.5), and `det_rowreduce(A)`, which brings A into upper triangular form by row operations, keeps track of the sign changes caused by row swaps, and multiplies the diagonal entries (Theorem 6.84 and Corollary 6.86). Hint: `np.delete(A[1:], j, axis=1)` is the matrix obtained from `A` by deleting its first row and the column with index `j`.

- Both functions should return -3 for $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix}$, and 0 (up to rounding) for the matrix F of Example 6.31. Compare both functions with `np.linalg.det` on random matrices.
- Check the product rule $\det(AB) = \det(A)\det(B)$ and the rule $\det(A^\top) = \det(A)$ on random 5×5 matrices.
- Measure the running time of `det_cofactor` for random $n \times n$ matrices with $n = 2, 3, \ldots, 8$. By roughly what factor does it grow from $n-1$ to n, and why? Estimate how long `det_cofactor` would need for $n = 20$, and compare with the running time of `det_rowreduce` for $n = 200$.

P6. Visualizing linear transformations. Using Matplotlib, write a function that draws the unit square, a few grid lines inside it and the letter F, before and after applying a 2×2 matrix A, as in Figure 2.

```
def plot_transformation(A):
    """Draw the unit square, a grid and the letter F, before and after A."""
    ...
```

Store a shape as an array `P` with one point per row; then its image under A is `P @ A.T`. The letter F in the figures has the corners

$$(0.2, 0.1),\ (0.4, 0.1),\ (0.4, 0.45),\ (0.65, 0.45),\ (0.65, 0.6),$$
$$(0.4, 0.6),\ (0.4, 0.75),\ (0.8, 0.75),\ (0.8, 0.9),\ (0.2, 0.9).$$

- Reproduce the four panels of Figure 2, and also plot the singular matrix $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.
- The signed area of a polygon with corners $(x_1, y_1), \ldots, (x_k, y_k)$ is given by the shoelace formula
$$\frac{1}{2}\sum_{i=1}^{k}(x_i y_{i+1} - x_{i+1} y_i), \qquad \text{where } (x_{k+1}, y_{k+1}) = (x_1, y_1).$$
It is positive if the corners are listed counterclockwise, and negative if they are listed clockwise. Implement it, apply it to the image of the unit square with corners listed in the order $(0,0), (1,0), (1,1), (0,1)$, and check that the result equals $\det(A)$, including the sign, for all the matrices above and for the matrix $\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ of Figure 4.

P7. Special matrices.

- Write functions `is_symmetric(A, tol=1e-10)` and `is_orthogonal(Q, tol=1e-10)`.
- For a random 3×5 matrix A and a random 5×3 matrix B, check that `A.T @ A` is symmetric, that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (even though AB and BA have different sizes!), and that $\operatorname{tr}(A^\top A)$ equals `np.linalg.norm(A, "fro")**2`.
- Write a function that splits a square matrix into its symmetric and skew-symmetric parts (Exercise 6.66), and test it.
- The command `Q, R = np.linalg.qr(rng.normal(size=(4, 4)))` produces a random orthogonal matrix `Q`. Check that `Q` is orthogonal, that $\det(Q) = \pm 1$ (Corollary 6.93), and that Q preserves lengths: `np.linalg.norm(Q @ x)` equals `np.linalg.norm(x)` for random vectors $\mathbf{x}$, where the length of $\mathbf{x}$ is $\sqrt{x_1^2 + \cdots + x_n^2}$.

P8. Why you should not compute the inverse. The Hilbert matrix H_n is the $n \times n$ matrix with entries $(H_n)_{ij} = \frac{1}{i+j-1}$. It is invertible, but very close to being singular, which makes computations with it very sensitive to rounding errors. For $n = 4, 6, 8, 10, 12$, let `x_true = np.ones(n)` and `b = H @ x_true`, and solve $H_n \mathbf{x} = \mathbf{b}$ in three ways:

```
x1 = np.linalg.solve(H, b)
x2 = np.linalg.inv(H) @ b
x3 = inverse(H) @ b  # your function from P4
```

For each method, print the error $\max_i |x_i - 1|$ and the residual $\max_i |(H_n\mathbf{x} - \mathbf{b})_i|$. What do you observe? Which method keeps the residual smallest? What happens with your own `inverse` for larger n, and why? (Hint: think about the tolerance in `rref`.) Relate your observations to Remark 6.53.

P9. Mini-project: fitting a line to data. Given data points $(x_1, y_1), \ldots, (x_N, y_N)$, we look for the line $y = w_0 + w_1 x$ that fits the data best, in the sense that the sum of squared errors $\sum_i (y_i - w_0 - w_1 x_i)^2$ is as small as possible. This is the simplest example of linear regression. Let X be the $N \times 2$ matrix with rows $(1, x_i)$, and let $\mathbf{y} = (y_1, \ldots, y_N)^\top$. One can show (we do not prove this here) that the best $\mathbf{w} = (w_0, w_1)^\top$ is the solution of the normal equations

$$(X^\top X)\,\mathbf{w} = X^\top \mathbf{y}.$$

- Generate data on a line with some noise:

```
x = np.linspace(0, 1, 50)
y = 2 + 3 * x + rng.normal(0, 0.1, size=50)  # a line plus noise
X = np.column_stack([np.ones_like(x), x])
```

- Compute $\mathbf{w}$ using your own functions from P2 and P4, and check that $\mathbf{w}$ is close to $(2, 3)$. Compare with `np.linalg.lstsq(X, y, rcond=None)[0]`.
- Explain why $X^\top X$ is symmetric. Show that if the columns of X are linearly independent, then $X^\top X$ is invertible. (Hint: if $X^\top X\mathbf{w} = 0$, compute $\mathbf{w}^\top X^\top X \mathbf{w}$ and use Corollary 7.11.)
- Fit a parabola $y = w_0 + w_1 x + w_2 x^2$ to the data $y = 1 - 2x + 4x^2 + \text{noise}$, using the columns $1, x, x^2$. How is this related to the vector space $\mathbb{R}[x]_{\le 2}$?

P10. Power iteration and PageRank. Implement power iteration:

```python
def power_iteration(A, num_iter=100):
    """Approximate an eigenvector for the eigenvalue of largest absolute value."""
    x = rng.normal(size=A.shape[0])
    for _ in range(num_iter):
        x = A @ x
        x = x / np.abs(x).max()  # rescale to avoid overflow
    ...
```

and estimate the eigenvalue from the final vector $\mathbf{x}$, for instance as $(A\mathbf{x})_i / x_i$ for an index i with $x_i \neq 0$.

- Test it on the matrices of Examples 10.7 and 10.8, and compare with `np.linalg.eig`. How does the speed of convergence depend on the ratio $|\lambda_2/\lambda_1|$ of the two largest eigenvalues (in absolute value)?
- What happens for the rotation by $90°$, and for $\operatorname{diag}(1, -1)$? Explain.
- A tiny web has 4 pages, with links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$, $4 \to 3$. Let L be the 4×4 matrix with $L_{ij} = \frac{1}{\text{number of links on page } j}$ if page j links to page i, and 0 otherwise. The PageRank vector is the stationary distribution of $G = 0.85\, L + \frac{0.15}{4}\, \mathbf{1}\mathbf{1}^\top$, where $\mathbf{1}\mathbf{1}^\top$ is the matrix of all ones. Compute it with power iteration, check that it is an eigenvector of G for the eigenvalue 1, and rank the pages.

P11. Diagonalization in floating-point arithmetic. The function `np.linalg.eig(A)` returns the eigenvalues and a matrix `P` whose columns are eigenvectors.

- For a random 5×5 matrix A, check that A is close to PDP^{-1}. (The eigenvalues may be complex.) Compare $PD^{20}P^{-1}$ with `np.linalg.matrix_power(A, 20)`.
- Compute the Fibonacci numbers F_n in two ways: exactly, with Python integers and the recursion, and with Binet's formula (Example 10.33) in floating-point arithmetic, rounding to the nearest integer. For which n does Binet's formula give a wrong answer for the first time, and why?
- Apply `np.linalg.eig` to the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. What do you notice about the two eigenvectors it returns?

P12. Homework. Dual bases and interpolation. Write a function `dual_basis(P)` that returns the rows $r_1, \dots, r_n$ of the dual basis of the columns of an invertible matrix `P` (Proposition 8.7).

- For a random invertible 5×5 matrix P, check that the matrix of all values $\mathbf{r}_i \mathbf{b}_j$ is the identity, and that applying the rows to a vector $\mathbf{x}$ gives its coordinates, i.e., that $\mathbf{x} = \sum_i (\mathbf{r}_i \mathbf{x})\, \mathbf{b}_i$.
- Polynomials of degree at most d are stored by their coefficient vectors in $\mathbb{R}^{d+1}$. For $d+1$ distinct nodes $t_0, \dots, t_d$, the evaluation functionals $p \mapsto p(t_k)$ are the rows of the Vandermonde matrix V with entries $V_{kj} = t_k^j$. Explain why the coefficient vectors of the Lagrange polynomials $L_0, \dots, L_d$ (with $L_j(t_k) = \delta_{jk}$) are the columns of V^{-1}, and compute them for the nodes $-1, 0, 1$; compare with Example 8.9.
- Interpolate $f(t) = \sin(3t)$ at 9 equally spaced nodes in $[-1, 1]$ with the formula $p = \sum_k f(t_k)\, L_k$, and plot f and p. Then repeat with 21 equally spaced nodes on $[-5, 5]$ for $f(t) = \frac{1}{1+t^2}$. What goes wrong near the ends of the interval? (This is Runge's phenomenon.)

P13. Softmax and cosets. Implement the softmax function in two ways:

```
def softmax_naive(z):
    return np.exp(z) / np.exp(z).sum()

def softmax_stable(z):
    z = z - z.max()  # another representative of the same coset
    return np.exp(z) / np.exp(z).sum()
```

- For random vectors $\mathbf{z}$ and constants c, check that `softmax_stable(z + c)` equals `softmax_stable(z)`, i.e., that softmax only depends on the coset $\mathbf{z} + \operatorname{span}\{\mathbf{1}\}$.
- Compare both functions on $\mathbf{z} = (1, 2, 3)$ and on $\mathbf{z} = (1001, 1002, 1003)$. What happens with the naive version, and why does subtracting the maximum fix the problem without changing the result?
- Another canonical representative of the coset is the centered vector $\mathbf{z} - \bar{z}\mathbf{1}$, where $\bar{z}$ is the mean of the entries. Check that it gives the same softmax. Why is subtracting the maximum preferred in practice? (Hint: what is the largest entry after each shift?)

P14. Differentiation and integration as matrices. Store a polynomial $a_0 + a_1 x + \cdots + a_d x^d$ by its coefficient vector `np.array([a0, a1, ..., ad])`.

- Write functions `diff_matrix(d)`, which returns the $d \times (d+1)$ matrix of $D : \mathbb{R}[x]_{\le d} \to \mathbb{R}[x]_{\le d-1}$, $Dp = p'$, and `int_matrix(d)`, which returns the $(d+2) \times (d+1)$ matrix of $J : \mathbb{R}[x]_{\le d} \to \mathbb{R}[x]_{\le d+1}$, $(Jp)(x) = \int_0^x p(t)\,dt$, with respect to the bases of monomials.
- Check that `diff_matrix(d + 1) @ int_matrix(d)` is the identity matrix, and that `int_matrix(d - 1) @ diff_matrix(d)` is not. Which coefficient does the second product lose?
- The midpoint sum $M_N(f) = \frac{1}{N}\sum_{k=1}^{N} f(x_k)$ with $x_k = \frac{k-1/2}{N}$ approximates $\int_0^1 f(x)\,dx$. Explain why M_N is a linear functional on $C[0,1]$, and why it agrees with the integral on all polynomials of degree at most 1. Compute $M_N(f)$ for $f(x) = \sin(\pi x)$ and $N = 10, 100, 1000$, and compare with $\int_0^1 \sin(\pi x)\,dx = \frac{2}{\pi}$. By which factor does the error shrink when N grows by a factor of 10?

P15. Homework. Tensors with NumPy.

- Create the tensor

  ```
  T = np.arange(24).reshape(2, 3, 4)
  ```

  and check that `T[i, j, k]` equals `T.reshape(-1)[12*i + 4*j + k]` for all indices, as in Example 11.10. Compute the unfoldings `T.reshape(2, 12)` and `T.transpose(1, 0, 2).reshape(3, 8)`. What are their ranks?
- Compute all the operations of Table 1 with `np.einsum` for random arrays, and compare with `@`, `np.trace`, `np.outer` and `.T`. For a batch of 100 random 4×4 matrices, stored as an array of shape $100 \times 4 \times 4$, compute all their traces and all the products $A_b B_b$ with a single call of `np.einsum` each, and compare with a Python loop.
- For random matrices A (3×4), B (2×5) and X (4×5), check that `(A @ X @ B.T).reshape(-1)` equals `np.kron(A, B) @ X.reshape(-1)` (Remark 11.17). How many numbers does `np.kron(A, B)` store for $A, B \in \mathbb{R}^{100\times 100}$, and why is it much cheaper to compute $AXB^\top$ directly?

- Write a function that returns the outer product $\mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w}$ of three vectors:

```
def rank_one(u, v, w):
    return np.einsum("i,j,k->ijk", u, v, w)
```

  For random vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$, check that the slices `T[:, :, k]` of `T = rank_one(u, v, w)` are multiples of the matrix $\mathbf{u}\mathbf{v}^\top$ (Example 11.20).

P16. Homework. The LU decomposition and rank-one updates.

```
def lu(A):
    # return L, U with A = L @ U (no row swaps)
    ...

def solve_lu(L, U, b):
    # solve L U x = b by forward and back substitution
    ...
```

- Implement `lu` by the elimination of Theorem 6.106: for each column, store the multipliers in L and replace the rest of the matrix by its Schur complement. Check it on Example 6.107, and check that `L @ U` equals A for random 6×6 matrices.
- Implement `solve_lu`, and compare it with `np.linalg.solve` for random systems. For $A = \begin{pmatrix} 10^{-17} & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{b} = (1, 2)^\top$, the solution is very close to $(1, 1)$. What does your solver return, and why? Swapping the two rows first (partial pivoting) fixes the problem.
- For a random 200×50 data matrix X, compute $(X^\top X)^{-1}$. Then add 100 new random rows one at a time, updating the inverse with the Sherman–Morrison formula of Theorem 6.108, and compare the result with $(X^\top X)^{-1}$ computed from scratch for the final matrix. Compare the running times of the two approaches.

P17. Convolutions and the FFT. Write a function `circulant(c)` that returns the matrix $\operatorname{circ}(\mathbf{c})$ of Definition 10.36.

- For random $\mathbf{c}, \mathbf{x} \in \mathbb{R}^8$, check that the product `circulant(c) @ x` agrees with the inverse DFT `np.fft.ifft(np.fft.fft(c) * np.fft.fft(x)).real` (NumPy's `fft` computes $\hat{\mathbf{x}} = \overline{F}\mathbf{x}$), and that the eigenvalues of `circulant(c)` are the entries of `np.fft.fft(c)`, up to their order.
- Time both ways of computing $\mathbf{c} * \mathbf{x}$ for $n = 2^{12}$ (without counting the construction of the matrix). Which is faster, and by how much?
- Blur a signal $\mathbf{x} \in \mathbb{R}^{101}$ of your choice with the filter $\mathbf{c} = \frac{1}{4}(2, 1, 0, \ldots, 0, 1)^\top$, add noise of size 10^{-3}, and try to recover $\mathbf{x}$ by dividing the DFT of the result by $\hat{\mathbf{c}}$. Plot $|\hat{c}_m|$ and explain why the recovered signal is so noisy. (Lecture 2 explains this with the condition number of $\operatorname{circ}(\mathbf{c})$.)